2023-06

Unleashing Potential: Harnessing Brain-Computer Interfaces to Empower Children with Severe Physical Disabilities

Kelly, Dion Marie

UNIVERSITY OF CALGARY

Unleashing Potential: Harnessing Brain-Computer Interfaces to Empower Children with Severe Physical Disabilities

by

Dion Marie Kelly

A THESIS

SUBMITTED TO THE FACULTY OF GRADUATE STUDIES

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE

DEGREE OF DOCTOR OF PHILOSOPHY

GRADUATE PROGRAM IN NEUROSCIENCE

CALGARY, ALBERTA

JUNE, 2023

Abstract

Independence, inclusion, and full participation are fundamental human rights. Yet, children with significant physical disabilities like quadriplegic cerebral palsy (QCP) encounter numerous barriers to these rights. Despite their cognitive abilities, these children may struggle to communicate, interact with their environment, and participate in activities that other children take for granted. Brain-computer interfaces (BCIs) offer a potential solution. However, pediatric-focused research in BCI literature is strikingly scarce, underscoring the need for targeted research to optimize this technology for children with severe disabilities.

This thesis delves into BCIs tailored for children, specifically those with severe neurological disabilities. We first explored the ability of typically developing children to control BCIs using five different mental control paradigms, refuting earlier assumptions about children's BCI capabilities, and demonstrating that they can competently control BCIs with accuracy similar to adults while highlighting the differential effects of age.

We then examined the impact of gamified calibration on BCI performance, discovering that gamification doesn't benefit all children equally, highlighting the need for personalized approaches. Subsequently, we evaluated a novel Home BCI Program for children with QCP and their families, showcasing its feasibility and benefits such as enhanced convenience and social interaction, while highlighting diverse family needs.

Our fourth study applied user-centered design principles to illustrate BCIs as an empowering tool for children to achieve home-based personalized goals. Lastly, we probed the feasibility of BCI-facilitated online gaming and its potential psychosocial impacts. We provided evidence that BCIs can enhance the quality of life for children with disabilities by enabling participation in simple, yet meaningful activities from which they are typically excluded.

Our research deepens the understanding of pediatric BCIs and underscores their transformative potential for improving the lives of children with disabilities and their families. Moving forward, refining BCI technologies for children and emphasizing collaboration with pediatric end-users and their families is vital.

Preface

This thesis represents original work conducted by myself as the primary researcher responsible for the conception, design, execution, analysis, and composition of the manuscript for the studies presented.

Chapter 2 of this thesis is under review at the *Journal of NeuroEngineering and Rehabilitation* under the title, "Brain-Computer Interfaces for Children: A Comparative Study of Five Common EEG-based Paradigms". I conceptualized the research question and study design, recruited participants, collected, and analyzed all data and composed the manuscript. Zewdie E assisted with concept formation and study design and troubleshooting the technology. Carlson H guided data analysis. Kirton A was the supervisory author and was involved with concept formation and study design and guided data analysis. All authors contributed to manuscript edits.

Chapter 3 of this thesis has been submitted to the *Journal of NeuroEngineering and Rehabilitation* under the title, "Exploring the Impact of Gamification on BCI Performance in Children: The Case for Personalization". I conceptualized the research question and study design, recruited participants, collected, and analyzed all data and composed the manuscript. Irvine B assisted with technology development and troubleshooting and data analysis. Kinney-Lang E assisted with concept formation, technology development, and troubleshooting. Comaduran Marquez D assisted with technology development. Floreani ED assisted with technology development. Kirton A was the supervisory author and was involved with concept formation and study design and guided data analysis. All authors contributed to manuscript edits.

Chapter 4 of this thesis has been submitted to *Nature Communications* under the title, "Integrating Brain-Computer Interfaces into Daily Life at Home for Children with Quadriplegic Cerebral Palsy and Their Families". I conceptualized the research question and study design, recruited participants, collected, and analyzed all data and composed the manuscript. Floreani ED assisted with concept formation, study design, technology development, troubleshooting, participant sessions and data collection. Rowley D assisted with participant sessions and data collection. Kinney-Lang E assisted with concept formation, technology development, and troubleshooting. Keough J assisted with participant sessions. Zewdie E assisted with concept formation, technology development and participant sessions. Jadavji Z assisted with concept formation and technology development. Robu I assisted with technology development. Kirton A was the supervisory author and was involved with concept formation and study design and guided data analysis. All authors contributed to manuscript edits.

Chapter 5 of this thesis has been submitted to *The 2023 Institute of Electrical and Electronics Engineers (IEEE) International Conference on Systems, Man and Cybernetics (SMC)* under the title, “Think BIG: Brain-Computer Interface Goals for Children with Quadriplegic Cerebral Palsy”. I conceptualized the research question and study design, recruited participants, collected, and analyzed all data and composed the manuscript. Rowley D assisted with concept formation, study design, and data collection. Floreani ED assisted with technology development and troubleshooting. Kinney-Lang E assisted technology development and troubleshooting. Robu I assisted with technology development. Kirton A was the supervisory author and was involved with concept formation and study design and guided data analysis. All authors contributed to manuscript edits.

Chapter 6 of this thesis is under review at *Disability and Rehabilitation: Assistive Technology*, under the title, “Separated by distance, united by technology: A pilot study on the psychosocial impact of brain-computer interfaces for online video game play for children with quadriplegic cerebral palsy”. I conceptualized the research question and study design, recruited participants, collected, and analyzed all data and composed the manuscript. Floreani ED assisted with concept formation, study design, technology development, troubleshooting, participant sessions and data analysis. Rowley D assisted with concept formation and participant sessions. Kirton A assisted with concept formation and study design. Kinney-Lang E was the supervisory author and was involved with concept formation, technology development, troubleshooting, and participant sessions. All authors contributed to manuscript edits.

Acknowledgements

I never envisioned myself earning a PhD. Growing up in the small town of Elora, Ontario, the concept of a PhD was a distant one. After high school, my future remained an enigma, an open question mark on the canvas of my life. At the young age of 18, I embarked on a journey to Australia and Southeast Asia, hoping that these experiences would grant me some clarity. They didn't provide a direct answer, but they did instill in me a deep desire for continuous learning and exploration. Now, almost 12 years later, I find myself in an unexpected yet familiar place: Bali, where I am penning the summary of my PhD thesis. This incredible journey, filled with a series of seemingly perfectly timed opportunities, would not have been possible without the generous support and guidance of numerous individuals around me.

First and foremost, I would like to extend my deepest gratitude to my supervisor, Dr. Adam Kirton. His unwavering faith in my abilities and his belief in my potential, even during periods when self-doubt overshadowed my confidence, have been an invaluable source of strength. When I joined his lab nearly five years ago, I had only three months of research experience. Under his guidance, I have evolved into the accomplished researcher I am today. Beyond the professional realm, Dr. Kirton has been an exemplary mentor and role model, profoundly influencing my personal growth over the last five years. For this, I owe him my sincerest thanks.

I also wish to express my profound gratitude to Dr. Ephrem Zewdie. His support extends beyond professional boundaries, blossoming into a cherished friendship. His dependable presence and wisdom have been a constant source of comfort and guidance, for which I am immensely grateful. I would also like to thank my other committee members, Dr. Michael Esser and Dr. Chantel Debert, for their ongoing support and encouragement. Guided by these exceptional leaders, I always felt capable and confident in my abilities.

Every individual in the BCI team and the broader Calgary Pediatric Stroke Project has played an integral role in my journey. This group's remarkable success has led to an impressive expansion, now encompassing numerous staff members and trainees—so many, in fact, that I can scarcely keep count! Among these, I would like to extend special gratitude to my friend and fellow BCI team member, Dr. Zeanna Jadavji. We began our paths as the sole two trainees in the BCI program, during which time she unfailingly served as a source of mentorship. Her empathy and compassion are unparalleled, and she has left an indelible mark on me with her inspiring character.

My sincere appreciation extends to the University of Calgary and the Hotchkiss Brain Institute for fostering an unparalleled environment that has empowered me to dream ambitiously. Their unwavering support for both my PhD studies and my array of diverse extracurricular pursuits has been instrumental in enabling me to reach my personal and professional milestones. This journey would not have been possible without the vital contributions of numerous funding agencies. I am particularly grateful to the Canadian Institutes for Health Research Vanier Canada Graduate Scholarship, the Hotchkiss Brain Institute, the Alberta Children's Hospital Research Institute, and the Indigenous Graduate Scholarships for their generous support.

I extend my heartfelt thanks to all the children and families who have been integral to this research. Over the past five years and countless hours spent together, we have shared a multitude of experiences and significant milestones and they have evolved to an extension of my own family. The candid conversations, profound life discussions, and unending encouragement have been a constant source of inspiration. Each of these families embodies an extraordinary spirit that I have been privileged to encounter in my life. Truly, they are some of the most remarkable individuals I have ever had the pleasure of meeting.

I wish to extend my deepest gratitude to my family, a circle that extends beyond just those immediately related to me. They have consistently encouraged me to believe in the limitless scope of my potential. My heartfelt thanks go to my parents: thank you for always telling me you're proud of me, no matter what I do or don't do. The profound impact of this unwavering belief is beyond words. And to my little sister and best friend, Demi, whose passion, wisdom, and perspective on life have kept me grounded. Thank you for always answering the phone and for being my dependable confidant. You have been an immense source of support to help me navigate through some of the most significant challenges I've had to face throughout this journey.

Dedication

To the inspiring children and resilient families of the BCI program, who have powerfully demonstrated the boundless strength of the human spirit and the invaluable lesson that we should never underestimate anything - be it ourselves, others, or the potential of each day.

Table of Contents

List of Tables

List of Figures

significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * $p<0.05$, ** $p<0.01$, *** $p<0.001$.

Figure 2.4. Feature classification and cortical activation plots generated by the mindBEAGLE® software. Top: ERP graphs showing averaged evoked potentials for each electrode for one session in the AEP paradigm. The vertical red line shows the onset of the stimulus. The green line represents ERPs elicited by target stimuli, while ERPs elicited by non-target stimuli are shown in blue. Significant differences between target and non-target trials are highlighted in green. The brain in the top right corner shows the spatial distribution of maximum peaks of the P300. (A) Participant who achieved 100% accuracy. (B) Participant who achieved 25% accuracy. Bottom: ERD plots for right (top) and left (bottom) motor imagery for electrodes at positions C3 (left) and C4 (right) showing bins that are significantly different from the bandpower during reference the period. The coloured bar above the plots represents the colours associated with relative decreases (red) and increases (blue) in bandpower during the active period. The brains to the right of the plots show the spatial distribution of cortical activations during right (top) and left (bottom) hand motor imagery. (C) Participant who achieved 71% accuracy. (D) Participant who achieved 39% accuracy. ERP = event-related potential; AEP = auditory evoked potential; ERD = event-related desynchronization.

Figure 2.5. Mean change in fatigue before/after completing tasks on each BCI paradigm. Lines over boxplots indicate groups which were significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * $p<0.05$, ** $p<0.01$.

Figure 2.6. Mean motivation score for each BCI paradigm. Higher scores indicate greater levels of overall motivation. Lines over boxplots indicate groups which were significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * $p<0.05$, *** $p<0.001$.

Figure 3.1. Calibration and user task scenes. Top: P300 scenes – standard calibration scene (left), gamified calibration scene (Mole Patrol game, middle), T9 speller scene for spelling task (right). Bottom: MI scenes – standard calibration scene (left), gamified calibration scene (Banana Dash game, middle), cursor control scene for yes/no response task (right).

"brain paintings" using BCI technology, a skill that has earned him global recognition with artworks sold around the world. He showcases his exceptional abilities on his Instagram page @brainpaintbyjohn. John's talent has won him several awards, including the "Make Your Mark" art contest by the Canadian CP Network for World CP Day. Among his many admirers is Olivier Oullier, the former President of Emotiv.

Figure 7.2. Examples of BCI use at home leading to a multitude of previously unimaginable activities. This collage of photos showcases a sample of the variety of activities achieved by Claire, a 7-year-old girl with mixed QCP secondary to HIE, who started in our BCI program at the young age of 5. Beginning with lab-based activities, such as controlling a remote-controlled car, Claire can now manipulate various devices in her environment through the BCI, facilitating participation in a range of activities. These include activities typical for children her age, like painting, as well as those specific to her family, such as their annual pickling tradition.

List of Abbreviations

Abbreviation	Definition
AAC	Augmentative and Alternative Communication
AEP	Auditory Evoked Potential
ALS	Amyotrophic Lateral Sclerosis
ANOVA	Analysis of Variance
API	Application Programming Interface
ASD	Autism Spectrum Disorder
AT	Assistive Technology
BCI	Brain-Computer Interface
BOLD	Blood Oxygen Level Dependent
CAQDAS	Computer Assisted Qualitative Data Analysis Software
CNS	Central Nervous System
COPM	Canadian Occupational Performance Measure
COVID	Coronavirus Disease
CP	Cerebral Palsy
CRPD	Convention on the Rights of Persons with Disabilities
CSP	Common Spatial Pattern
EEG	Electroencephalography
ERD	Event-Related Desynchronization
ERP	Event-Related Potential
ERS	Event-Related Synchronization
GMFCS	Gross Motor Function Classification System
HIE	Hypoxic Ischemic Encephalopathy
ICF	International Classification of Functioning, Disability and Health
ITR	Information Transfer Rate
LDA	Linear Discriminant Analysis
LED	Light Emitting Diode
MEG	Magnetoencephalography
MI	Motor Imagery
MRI	Magnetic Resonance Imaging

NASA-TLX	NASA Task Load Index
NIRS	Near-Infrared Spectroscopy
NV	Non-Verbal
P300	P300 Event-Related Potential
PIADS	Psychosocial Impact of Assistive Devices Scale
PMOT	Pediatric Motivation Scale for Rehabilitation
PODD	Pragmatic Organisation Dynamic Display
QCP	Quadriplegic Cerebral Palsy
QUEST	Quebec User Evaluation of Satisfaction with Assistive Technology
RAM	Random Access Memory
SMR	Sensorimotor Rhythm
SSVEP	Steady-State Visually Evoked Potential
UCD	User-Centered Design
UN	United Nations
UNCRC	United Nations Convention on the Rights of the Child
UNCRPD	United Nations Convention on the Rights of Persons with Disabilities
VTP2	Vibrotactile P300 with 2-tactors
VTP3	Vibrotactile P300 with 3-tactors

Epigraph

The biggest thing I can say to anyone is not to underestimate these kids. She hears everything. She understands everything. When we give them a way that they can show what they know and what they can do–it's a light at the end of the tunnel. The possibilities are endless.

Stephanie Sonnenberg, Mother of Claire Sonnenberg (a young BCI user)

1 Chapter 1 – Background

1.1 Typical Neurodevelopment

In the intricate process of human development, neurodevelopment forms the building blocks that shape an individual's cognitive, emotional, and social abilities. This complex and dynamic process begins in the early stages of prenatal life and continues throughout childhood, adolescence, and into adulthood. It is marked by a series of well-orchestrated events, including neural proliferation, migration, differentiation, and synaptic pruning, which collectively contribute to the formation and maturation of neural networks.[1] These networks, in turn, underpin our capacity to learn, adapt, and interact with the world around us. Disruptions or abnormalities in these processes can lead to severe and lifelong disabilities.

1.1.1 Neuroplasticity

Neuroplasticity, the brain's inherent ability to adapt and reorganize itself in response to experiences, environmental stimuli, and injuries, is a key feature of brain development.[2] This dynamic property facilitates learning, memory consolidation, and skill acquisition throughout an individual's life by enabling neural networks to form new connections, strengthen existing ones, or eliminate less-used pathways. The cellular mechanisms underlying neuroplasticity are complex and metabolically demanding, relying heavily on mitochondria located in presynaptic terminals to supply the energy necessary for these processes.[3]

Neuroplasticity is crucial during critical periods of development, such as infancy and early childhood when the brain is most receptive to change, but also persists throughout adulthood, albeit at a reduced capacity.[2] Although neuroplasticity offers numerous benefits by facilitating adaptation and learning, it can also lead to maladaptive outcomes if factors that guide its activity, including genetics, developmental timing, homeostatic mechanisms, and environmental experiences, diverge from their typical patterns.[2] The enhanced plasticity of the young brain also creates opportunities for learning, including the development of new pathways and networks to potentially overcome or otherwise mitigate the disabilities that result from brain disorders present from early in life.

1.1.2 Critical and sensitive periods

Neurogenesis, synaptogenesis, and synaptic pruning are time-limited processes that represent the foundational elements of neuroplasticity. During these specific time windows, known as critical and sensitive periods, the brain exhibits heightened plasticity and receptiveness to environmental stimuli, allowing for the rapid acquisition of specific abilities and the establishment of neural connections.[4] Critical periods represent finite intervals during which specific experiences must occur for normal development to take place, whereas sensitive periods are characterized by heightened responsiveness to certain stimuli.[2] Both critical and sensitive periods have significant implications for lifelong learning, as the skills and abilities developed during these stages have lasting effects on an individual's cognitive and behavioral capacities. This rapid early neurodevelopment, particularly during the second and third trimesters of pregnancy, not only confers opportunity, but enhanced vulnerability.[5]

1.1.3 Motor and cognitive development

The structural and functional networks formed through neuroplastic processes lead to the behavioural acquisition of age-specific motor and non-motor milestones.[2] The transition from development to motor control function is underpinned by alterations in the anatomy and physiology of the corticospinal system defined by three developmental stages: growth to spinal targets, refinement of spinal terminations, and motor control development.[6] The corticospinal tract is the principal motor pathway for voluntary motor control and the last to reach maturity. Originating in the primary motor cortex (M1), the tract consists of two main parts: the lateral corticospinal tract, which controls movements of the limbs and digits, and the anterior corticospinal tract, responsible for the movements of axial muscles.[7] Pyramidal cells in layer 5 of the M1 (upper motor neurons) project through specific subcortical and brainstem regions to synapse onto lower motor neurons in the spinal cord, which innervate muscle fibers.[6,7] This growth occurs during the prenatal period. Maturity of this system involves neuroplastic changes, including a balance of elimination and growth of projections in the spinal grey matter in response to early experiences during the first two years of life.[6] Motor development refers to the maturation of voluntary movements, such as the ability to hold up one's head, sit without support, crawl, and eventually walk. The voluntary movements are facilitated by a network of structures including M1, supplementary motor areas, premotor cortex, posterior parietal cortex, cingulate motor cortex, basal ganglia, cerebellum, and the spinal cord.[7] Each of these components contributes uniquely to the planning, initiation, coordination, and execution of voluntary movements. The development of coordinated body movements and the manipulation of objects involves the refinement of gross and fine motor skills. This development is largely influenced by the maturation of motor areas in the brain and their connections with sensory areas, allowing for the integration of sensory information into motor responses.

1.2 Early brain injury and cerebral palsy

Injury to the developing brain can occur at any time between gestation to early childhood, the periods of which can be classified as prenatal (before 20 weeks' gestation), perinatal (between 20 weeks fetal life and 28 days postnatal life), or neonatal (from 28 days after birth up to 2 years).[8] The type, timing, and location of the injury can significantly impact any element of neurodevelopment and lifelong function. For example, the perinatal period is characterized by rapid neurodevelopment marked by enhanced neuroplastic processes and high metabolic demand, therefore gray matter areas of the brain, such as the basal ganglia and thalamus, are more vulnerable to injury during this period.[9,10] Certain injuries can selectively affect specific systems while sparing others, leading to different neurodevelopmental disorders. For example, abnormalities in the basal ganglia and thalamus have been linked with both Autism Spectrum Disorder (ASD)[11] and cerebral palsy (CP)[12]. ASD primarily impacts social interaction, communication, and behavior. Its origins are believed to be prenatal, arising from a blend of maternal, genetic, and environmental factors that influence early brain development.[13] This amalgamation of influences can lead to abnormal morphologies in areas of the brain such as the basal ganglia and thalamus.[11] Conversely, damage to the basal ganglia, such as what occurs in some types of CP, can result in severe motor impairments while cognitive functions are spared.

1.2.1 Definition and classification of cerebral palsy

CP is the leading form of lifelong neurological disability, affecting 17 million people worldwide. It is estimated that 1 in 500 babies are born with CP each year, which translates to over 60,000 affected Canadians.[14,15] A clinical descriptive term used to describe a heterogeneous group of non-progressive neurologic disorders characterized by motor impairments, CP is the result of an injury to, or abnormality of, early brain development.[16–18]

The etiology of CP is often multifactorial, although 70-80% of cases are of prenatal origin.[19] CP may be the result of cerebral malformation, hypoxia-ischemia, stroke, infection, metabolic or genetic causes.[16,17] Neuroimaging is an important component of the initial diagnostic evaluation for CP, with magnetic resonance imaging (MRI) being the preferred technique due to its high sensitivity and specificity.[20] Common patterns of MRI abnormalities include premature white matter injury, selective or combined cortical and deep gray matter lesions due to ischemia, and cerebral malformations.[20] Approximately 90% of children with CP have abnormal MRIs, while normal findings are reported in 10% of cases and have been associated with a higher prevalence in dyskinetic, ataxic, and spastic diplegic subtypes.[21,22]

CP can be divided into two main physiologic groups based on the location of brain injury and resultant clinical features. Pyramidal CP (80% of cases) describes conditions in which there is damage to or abnormalities of upper motor neurons that originate in the cortex and descend into the brainstem, resulting in spasticity (increased muscle tone and hyperreflexia).[18] Spastic subtypes are classified based on their topographical distribution: monoplegia (one limb affected), diplegia (lower limbs affected), hemiplegia (one side of the body affected), and quadriplegia (all four limbs affected).[23]

Conversely, extrapyramidal CP describes conditions in which there are signs of damage to subcortical gray matter structures such as the basal ganglia, thalamus and cerebellum.[24] Damage to this tract affects the modulation and regulation of movement. Impairments are characterized by fluctuations in muscle tone and hyperkinetic movements, often involving all four limbs (quadriplegia). Such extrapyramidal types include dyskinetic (15%), which comprises choreoathetotic (irregular, twisting movements) and dystonic types (involuntary postures/movements), and ataxic (5% (uncoordinated movements)).[18,25] Some individuals with CP may demonstrate impairments in multiple categories, which is classified as mixed CP.

To define functional motor abilities in children with CP, the Gross Motor Function Classification System (GMFCS) has been universally adopted. The goal of this scale is to highlight the child's potential for independence, while considering age and developmental milestones.[26] Using a five-level system, the GMFCS indicates the child's motor severity based on their self-initiated movements, particularly mobility. Degree of severity is indicated by a higher classification level: GMFCS level I indicates ability to walk without limitations, GMFCS level II indicates ability to walk with some limitations, GMFCS level III indicates ability to walk with mobility aids (handheld aids indoors/short distances, wheeled mobility for outdoors/longer distances), GMFCS level IV indicates ability to sit supported with self-mobility using a manual or powered wheelchair, and GMFCS level V indicates severe limitations in head and trunk control with no means of independent movement.[27]

1.2.2 Quadriplegic cerebral palsy

Quadriplegic CP (QCP) is the most severe and second most prevalent form of CP, accounting for approximately 35% of CP cases.[28] This is a non-progressive condition, originating from bilateral brain injuries, that manifests as involuntary and uncontrollable movements affecting the entire body, thereby significantly impairing or completely restricting motor function.[29,30] As a result, many children with severe QCP may require full assistance with daily living activities.[31] Importantly, in many children with QCP, cognitive function, including intellectual abilities, is often preserved.[31–34] This means that despite significant physical limitations, these children possess cognitive awareness and capabilities, underscoring the need for effective communication and interaction methods that align with their cognitive abilities. Figure 1.1 shows Dan, a 14 year-old-boy, with dyskinetic QCP secondary to kernicterus, a severe form of neonatal jaundice resulting from untreated high levels of bilirubin resulting in selective bilateral damage to the globus pallidus and subthalamic nuclei of the basal ganglia.[35] Despite his severe physical impairments, he has normal vision and hearing, and is very bright, with a love for Greek mythology (which inspired the name of his chihuahua, Hermes, the son of Zeus and messenger of the gods, known for his speed and cunning).

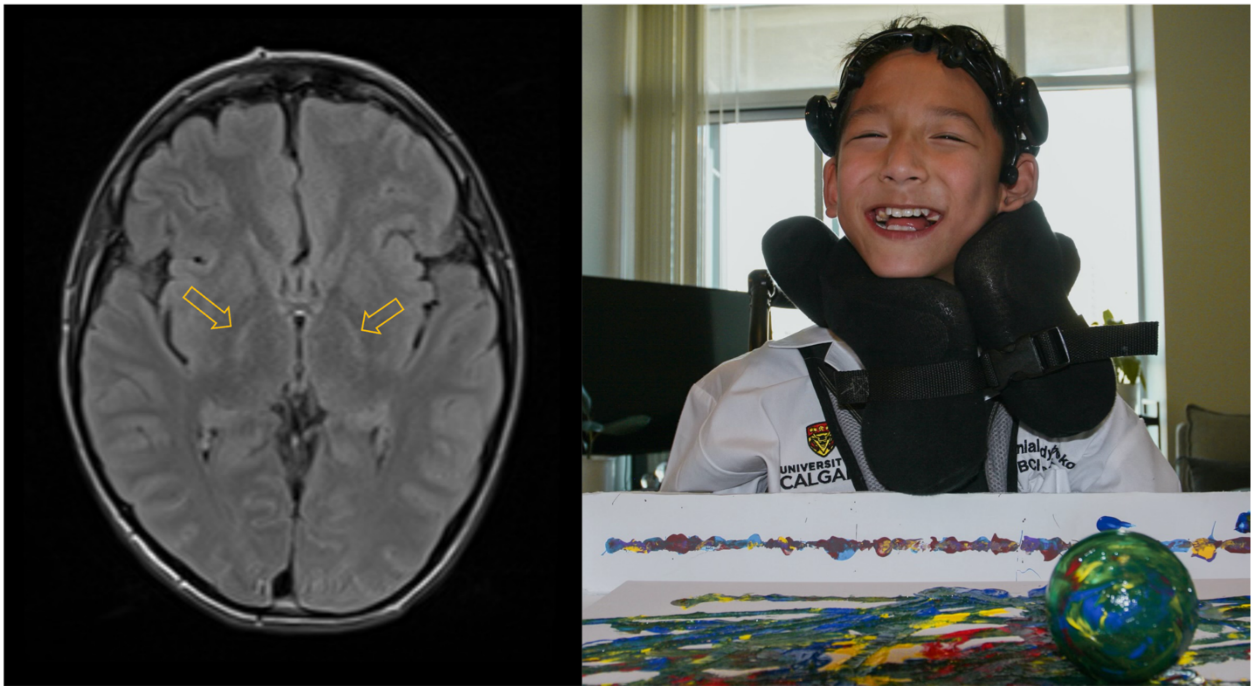


Figure 1.1. Kernicterus case example. Left: Axial T2 FLAIR MRI shows small symmetrical areas of hyperintensity in the globus pallidus bilaterally (arrow). Note that the rest of the brain including the cerebral cortex is entirely intact. Right: Dan, an extremely bright 14-year-old boy with a contagious smile, also lives with dyskinetic QCP.

1.2.3 Functional limitations and impact on daily life: loss of fundamental human rights

Children with severe QCP face significant challenges in participating fully in daily life, which can impact their ability to exercise their fundamental human rights. The World Health Organization's International Classification of Functioning, Disability and Health (ICF) defines participation in the context of nine domains, including learning and applying knowledge, communication, mobility, and social life.[36] Though the severity of impairment has only been shown to be weakly correlated to quality of life, restricted participation is strongly related.[37] Therefore, to improve the quality of life for such children, it's essential to focus on interventions that foster activity and encourage participation in daily life.

The right to equality, the right to freedom of expression, and the right to an adequate standard of living are among the most basic fundamental human rights, as established in the United Nations (UN) Universal Declaration of Human Rights.[38] Further, the UN Convention on the Rights of a Child (UNCRC) recognizes that childhood is a special period in which "children must be allowed to grow, learn, play, develop, and flourish with dignity".[39,40] The UNCRC acknowledges that children with disabilities have the right to enjoy a full and adequate life in environments that ensure dignity, promote independence and foster their participation in society.[40] Importantly, the UNCRC states that implementation of such rights shall be accomplished through "the application of readily available technology".[40] The UN Convention on the Rights of Persons with Disabilities (CRPD) also makes special reference to children and their right to prompt access of assistive technologies for full inclusion and participation in all aspects of life.[41]

1.2.4 Current management and interventions

Although the initial brain injury is static, intrinsic and extrinsic influences may modulate outcomes throughout the lifespan.[33,42] With no strategies for prevention, current management and interventions for QCP focus on improving their quality of life, enhancing functional abilities, and minimizing complications. As cerebral palsy encompasses a wide range of motor and cognitive impairments, the management approach is typically multidisciplinary and tailored to each child's specific needs[17]. Key components of the management plan for children with severe CP include: physical, occupational, and speech therapy; pharmacological and surgical treatment for the management of symptoms such as spasticity, seizures, and pain; mechanical aids and assistive technology.[17] Early intervention and ongoing support are imperative to achieving the best possible outcomes for these children and their families.

1.2.5 Assistive technologies for children with quadriplegic cerebral palsy

For children with severe QCP, unless effective interventions are provided, the negative effects of their impairments are exacerbated by missed opportunities for interaction and learning.[43]

Assistive technologies, defined as "any item, piece of equipment, software program, or product system that is used to increase, maintain, or improve the functional capabilities of persons with disabilities," play a crucial role in enhancing the quality of life for children with QCP.[44] These technologies, which can range from specialized wheelchairs to communication boards, are designed to promote independence, mobility, communication, and participation in everyday activities.

1.2.5.1 Alternate access technologies

Alternate access technologies provide alternative ways for children with disabilities to access and interact with assistive technology. One such technology is the switch, which can be activated by various physical actions to improve access for individuals with limited motor skills. There are different types of switches, including mechanical, touch, sensor/proximity, pneumatic (e.g. sip and puff), and voice/sound.[45] Mechanical switches, the most common type, resemble large buttons and can be conveniently placed on a wheelchair for easy access, typically with a finger press. Head tracking devices, another form of alternate access technology, are particularly useful for those who have severe physical limitations but retain head control. They work by tracking the user's head movements and translating them into cursor controls on a screen.[45] This allows the user to navigate digital interfaces, select icons, or input text, simply by moving their head.

Eye tracking devices represent a further advancement in this field. They operate by following the gaze of the user, enabling them to control a cursor or make selections on a screen using only their eyes.[45] This technology can be invaluable for individuals who may have very limited physical mobility but retain control and movement of their eyes.

1.2.5.2 Augmentative and alternative communication (AAC) technologies

AAC technologies offer opportunities for children with physical and communicative impairments to participate in communication.[43] AAC technologies range from low- to high-tech systems, including printer communication boards with pictures or symbols (low-tech), dynamic display communication boards (mid-tech), and speech-generating devices that produce synthetic or digitalized speech (high-tech).[46] With printed communication boards, users communicate by pointing to or gazing at pictures or symbols on a board. Dynamic display communication boards and speech-generating devices can be accessed using methods such as eye gaze, allowing users to select words, phrases, or symbols that the device then communicates.

Despite their promise, these technologies require the assistance of others or a degree of controlled movement for effective access. Eye gaze technology, while a significant step forward, presents calibration and usage challenges for children with severe QCP, whose involuntary and uncontrolled movements can interfere with the system's ability to accurately detect and track eye movements.[47]

Current assistive technologies may have limited benefit for children with severe CP due to their strong dependence on sufficient motor control and the assistance of others. A variety of other challenges, including high costs, technical complexity, inadequate training and support, incompatibility with other technologies, and the potential for stigmatization, all contribute to high abandonment rates, and further complicate the use of these technologies. There is an urgent need for innovative new neurotechnologies to facilitate function and independence in children with QCP.

1.3 Brain-computer interfaces

Brain-computer interfaces (BCIs) represent a promising solution by which children with QCP might be able to better communicate, interact with their environment, and better realize their own independence.[48] By circumventing peripheral nerves and muscles, BCIs can function in a closed-loop system to allow for real-time mental control of an external device, such as a computer cursor or a communication device.[48] This can be achieved through sampling and translation of brain signals into useable commands.[49] Though specific components of BCI systems can differ, the general framework of all BCIs are similar: brain signals are acquired, amplified and processed to extract specific features, which are then translated by an online classification algorithm into commands to control an output device.[49]

1.3.1 Signal acquisition: Invasive systems

Invasive BCIs involve surgical implantation of the signal acquisition device. These can be placed either intracortically or extracortically. Due to their proximity to the brain, such techniques allow for strong signal quality, wide frequency range detection, and high-resolution topography.[48] An innovative addition to this field is the Stentrode™, which serves as an electrode similar to a cardiovascular stent. These electrodes are strategically placed in the cerebral vasculature, demonstrating promising experimental results.[50] A distinct advantage of the Stentrode™ is that it doesn't necessitate a craniotomy. Instead, the electrodes are guided to the vicinity of the targeted brain region through the venous vessels, where they record the activity of nearby neurons from its inner wall.[51] This method capitalizes on established practices and expertise from vascular stenting surgery, thereby gaining considerable interest in clinical applications.[51]

Intracortical recording devices, such as single electrodes or electrode arrays, require implantation of electrodes directly into the cortex of the brain, usually the motor cortex.[52] Through such devices, some patients have been able to control computer cursors[53] and robotic arms[54,55], as well as communicate[56]. However, they are limited by their higher risks including tissue damage and infection, inflexibility once the electrodes have been implanted, and poor feasibility for their use outside of a controlled environment.[48,57]

Electrocorticography (ECoG), on the other hand, mitigates some of these risks with its less invasive approach. In ECoG, strips or grids of electrodes are positioned on the brain's surface, an extracortical placement that balances risk and benefit.[58] Since its clinical introduction in the early 1960s for treating epilepsy patients, ECoG has proven advantageous for pinpointing precise signal locations.[59] For use with BCI, most ECoG arrays have been implanted in the motor cortex[60], while implantation in the dorsolateral prefrontal cortex has also been used to facilitate cognitive control.[61] ECoG systems have made significant strides in the field of BCI, showcasing impressive capabilities such as facilitating rapid communication[62], and exoskeleton control for enhanced mobility[63]. Importantly, their potential for at-home use marks a critical advancement[64,65], making assistive technology more accessible and convenient for individuals in their daily lives. However, a major limiting factor in widespread adoption of such BCI systems by patients is the requirement for surgery. To date, no child has ever been implanted with an invasive BCI system. Despite the potential benefits that invasive BCIs offer, the risks and ethical considerations surrounding surgical implantation in children are substantial. For example, there remains uncertainty about the potential shift or damage an implanted electrode grid could cause as children undergo growth and development.[66] Given these limitations and the invasiveness associated with ECoG and other invasive BCI techniques, it's essential to explore non-invasive alternatives, especially when considering applications for children.

1.3.2 Signal acquisition: Non-invasive systems

Non-invasive BCIs record brain signals from outside of the scalp and measure electrophysiological, magnetic, and/or metabolic brain signals.[48] Among these methods are electroencephalography (EEG), which provides a direct measure of the brain’s electrical activity, and others like magnetoencephalography (MEG), functional near-infrared spectroscopy (fNIRS), and magnetic resonance imaging (fMRI) that provide indirect measures of brain activity. MEG measures small magnetic fields produced by the electrical currents of the brain, while fNIRS and fMRI measure the blood oxygen level dependent (BOLD) response, which is indicative of brain activity.[57,67] Due to its portability, low cost and high temporal resolution, EEG is the most widely used recording method for non-invasive BCI control.[57]

1.3.2.1 EEG fundamentals

To fully grasp the potential and intricacies of BCIs, it is essential to first understand the fundamentals of EEG, as it forms the basis of capturing and interpreting the brain's electrical signals that BCIs rely on for operation. The extent to which potentials originating in the cortex can be measured at the scalp level has been attributed to five primary factors, as outlined by Gevins and Smith (2006): the number of active neurons, the spatial orientation of the cortical neurons, the synchronicity of the neuronal activity, the depth of the cortical source, and the conductive properties of different brain, skull, and scalp tissues.[68]

1.3.2.1.1 Signal origin

It is estimated that the human brain comprises over 100 billion neurons, each with over 10,000 connections, forming an intricate network enabling various bodily functions and cognitive processes.[69] These extensive interconnections can be further subdivided into subnetworks of cells, including parallel pyramidal cells, which are primarily responsible for the generation of EEG signals.[69] The EEG signal arises from the synchronized synaptic activity of these cells. This activity creates an extracellular voltage around the neural dendrites, referred to as a dipole, which has a positive region (source) and a negative region (sink).[70] The dipoles of multiple neurons in a subnetwork sum together and are measurable as a single dipole, with magnitudes reflecting the number of neurons contributing to it.[70]

1.3.2.1.2 Signal propagation

Signals are propagated within the brain through volume conduction, driven by the basic principles of electrical charge, where like charges repel each other, creating a wave of charged ions that travels through the extracellular space.[70] However, this flow of ions can be impeded by electrical and physical barriers, such as myelin-coated nerve tracts. Propagation also depends on the size of the dipoles, with larger dipoles travelling further than smaller ones.[70]

1.3.2.1.3 Signal measurement

The electrical signal created by neural firing must traverse multiple layers - from the brain, through the dura, skull, and scalp, to an external electrode for measurement. Volume conduction propagates the brain's signal through extracellular fluid, however once an insulating layer is encountered, propagation becomes a function of capacitive conduction.[70] Capacitance occurs when two pools of charges are separated by an insulating layer, preventing the charges from mixing, and creating a charge difference across the insulating layer.[70] When measuring EEG, the gap between the skin surface and the electrode needs to be bridged by a conductive connection.[69] Different types of electrodes utilize different conductive substances. For example, gel-based electrodes, considered the gold standard, utilize a highly conductive gel, while more recently-developed dry electrodes utilize a metal alloy or conductive rubber that relies on perspiration.[71]

1.3.2.2 EEG Signal Processing

EEG signal processing is a fundamental part of BCIs and typically involves three main steps: preprocessing, feature extraction, and classification. First, raw EEG data, which often contains a significant amount of noise, undergoes signal filtering and artifact rejection to improve the signal to noise ratio and to isolate the frequency bands of interest. This is crucial in enhancing the quality and reliability of the EEG data. Next, feature extraction is applied to the filtered signals, which is the process of distilling meaningful, quantifiable characteristics from the EEG data. These features may include time-frequency domain characteristics, or event-related potential measures, among others. The goal here is to represent the EEG data in a way that accurately captures the underlying brain activity while reducing the complexity of the data. Finally, the extracted features are input into a classifier, which is a machine learning algorithm trained to distinguish between different cognitive states or tasks. The classifier assigns labels to the data, effectively translating brain signals into commands for the BCI. Together, these steps create a robust method for translating the complex electrical activity of the brain into meaningful and actionable information.

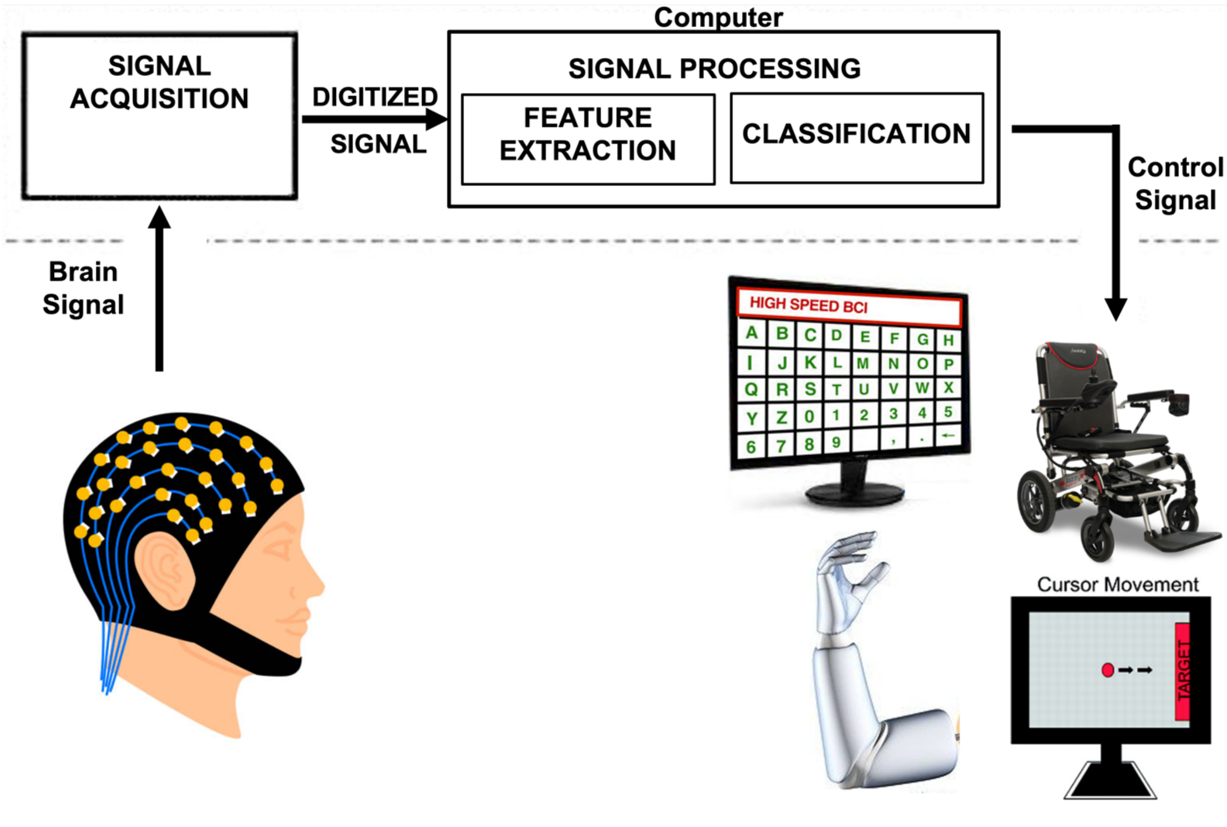


Figure 1.2. Mechanism of BCI control. To control a BCI, a user encodes their intent in brain signals, which are translated by the BCI into a control signal for an external device. Acquired brain signals are transmitted to a computer for processing, where specific features are extracted and translated by a machine learning algorithm (classification) to produce a control signal for control of external devices, including spellers, power mobility chairs, neuroprosthetic limbs, and computer cursors.

1.3.3 Mental control signals for EEG-BCI

Mental control signals used to control non-invasive BCIs can be produced in response to external stimuli (evoked signals) or they can be generated voluntarily by the user (spontaneous signals).[57,72] Common evoked signals are event-related potentials (ERPs) and steady state visual evoked potentials (SSVEPs). ERPs are variations in EEG signal amplitudes that correspond to specific time points following the onset of an event or stimulus.[73]

The P300, which is a positive peak in the ERP time-locked to 220-500 milliseconds after the presentation of a new surprising stimulus, is the most studied ERP component for BCI control.[73] The P300 occurs over the centroparietal area and can be elicited through the "oddball paradigm", where a target stimulus is presented infrequently among frequently presented nontarget stimuli. Such stimuli can be visual, auditory, or somatosensory.[73] SSVEPs are evoked by repetitive visual stimuli, such as LED lights, that are flickering at the same time but at different frequencies (between 6 and 30 Hz). The evoked brain signals occur over the occipital cortex and correspond to the unique frequency that the user is focused on.[57] The advantages to using evoked signals for BCI control are that they involve little to no training for the user to achieve efficient control. However, since they require extensive focus and attention, such paradigms are often fatiguing.[73]

The most preferred endogenous signals are the sensorimotor rhythms (SMRs), which are oscillations in brain activity in the mu (8-13 Hz) and beta (13-30 Hz) frequency bands.[57] These signals can be modulated through the imagination of a movement without physically moving—a phenomenon known as motor imagery. This causes event-related desynchronizations and synchronizations of the SMRs, which occur over the sensorimotor cortex.[73] Explicit motor imagery can occur from three possible perspectives: kinesthetic (feeling the movement), internal visual (visualizing the movement in first person), and external visual (visualizing the movement in third person).[74] Though motor imagery is the most commonly used task for SMR-BCI control, drawbacks include poor reliability and long training periods for efficient control. More importantly, such a mental control task may not be intuitive for individuals with severe motor impairments, for whom BCIs are targeted.[75] Recent research has shown that, similar to motor imagery, alpha rhythms (8-13 Hz) can be modulated through alternative mental imagery tasks.[75,76]

1.3.4 Applications for EEG-BCI

BCI applications have been designed to "replace, restore, enhance, supplement, or improve" the interactions between the brain and its environment.[77] Clinical uses have focused on two main areas of application: direct control of assistive technologies and neurorehabilitation.[48,72] The goal of assistive BCIs is to enable communication, movement control, environmental control and mobility for patients with severe neurological disabilities, where the threshold for sufficient control is defined as a Cohen's kappa score of 0.4 or an accuracy of 70%.[48,78,79] Unfortunately, less than 10% of BCI studies have included patients and the majority of studies have been in healthy adult controls, thus the transferability of results to clinical populations remains relatively undetermined.[80,81] Awareness of BCI technology by health care professionals caring for persons with severe disability is poor yet they endorse enormous potential once educated.[82]

Clinical studies evaluating BCI communication have mainly comprised case studies in adult patients with amyotrophic lateral sclerosis (ALS) in a "locked-in" state, which is characterized by a loss of all neuromuscular control apart from eye movements and preserved consciousness.[72] Other common adult populations include brainstem stroke[83] and cervical spinal cord injury.[84] A common feature to all these adult clinical populations is an intact cerebral cortex following normal development, creating highly predictable anatomical and functional brain organization. Studies in adults with such severe motor impairments have shown that most can achieve successful control of P300-based systems for communication and entertainment, including gaming and art.[83,85–89] In patients with locked-in syndrome due to brainstem stroke, SSVEP-based BCIs have been successful for communication with better accuracies than P300-based spellers.[90] SMR-BCIs have been successfully used by motor-impaired patients in cursor control for gaming and communication applications.[91,92] A few studies have evaluated the use of BCI in adults with severe quadriplegic CP (spastic and dystonic), with results demonstrating that more than half of individuals can achieve a statistically significant level of accuracy.[93,94] However, this is not indicative of sufficient control and more training would likely be warranted for efficient performance.

1.3.5 BCIs for children

Children have been almost entirely neglected from BCI research, with over 10,000 studies on BCIs and < 2% focused on pediatrics. Invasive BCI has been evaluated in six pediatric patients undergoing ECoG monitoring during surgery for epilepsy, where a cursor control task was performed using motor imagery.[95] After only 11 minutes of training, these children demonstrated a mean accuracy of 81% which was similar to adult performance, demonstrating that decoding signals from the cortex of children is feasible for BCI control.[66,95] A few studies have compared SSVEP-BCI in typically developing children compared to adults, yielding accuracies ranging from 50-79% for children aged 6-11 and 75-79% for adults.[96,97]

A systematic review recently conducted by Orlandi et al. identified just twelve studies that focused on pediatric BCIs.[66] This limited research may reflect the unique complexities and ethical issues associated with conducting studies on vulnerable groups, such as children. Moreover, the inherent challenges of assessing these systems are magnified when dealing with the continually evolving landscape of a child's brain[98], a scenario that becomes even more intricate when injury is involved. While adults with acquired brain injury may still exhibit predictable EEG responses, children with congenital injury may have atypical neurodevelopment and neurophysiology. Though adult BCI studies show promise, their optimized algorithms and established tasks might not directly translate to children. Age-related differences in mental control signals, which are crucial for BCI operation, may hinder the effectiveness of algorithms optimized for adults. For instance, children's abilities to perform motor imagery—essential for many BCI applications—develop alongside cognitive processes that underpin motor representations, typically maturing between the ages of 7 and 12.[74] While adults studies have demonstrated kinesthetic motor imagery to be most effective in producing detectable EEG changes[99], pediatric studies have shown that children have the greatest ability to use external visual imagery[74]. Moreover, research involving children with unilateral hemiplegic and spastic diplegic CP indicates that explicit imagery abilities might be preserved despite their conditions.[100]

Different mental tasks must also be considered for children. While movement-related tasks may be intuitive for adult BCI users, children may find them abstract and difficult to understand, particularly for children who have never had functional movement.[100] For tasks based visual stimuli, such as SSVEP and P300, caution must be practiced as children with disabilities may have undiagnosed photosensitivity or visual impairments.[101] When selecting mental tasks for children, it is essential that they are engaging, easy to comprehend and require minimal effort.[66]

Other considerations include differences in head size and shape, attention, cognition, and motivational needs as well as different underlying neurological conditions.[66,102]

Based on their findings, Orlandi and colleagues proposed a list of potential requirements for clinical translation of pediatric BCI, including collecting more qualitative data that considers aspects such as BCI experience, fatigue, and workload; developing child-friendly protocols, utilizing comfortable headsets; and including gamified activities to increase engagement in BCI paradigms.[66]

We therefore initiated a clinical pediatric BCI program to provide children with severe neurological disabilities unique opportunities to try a suite of BCI systems to explore possible new avenues for their independence.[103] Studies from our lab have shown that typically developing children, those with perinatal stroke, and those with QCP can successfully perform SMR-BCI tasks using the Emotiv EPOC headset (Emotiv, USA) .[103–105] In tasks involving driving a remote-controlled car and cursor control, older children demonstrated performance comparable to adult studies, with higher performance in the car task compared to the cursor task.[104] Children with severe QCP have demonstrated successful BCI control in activities such as playing computer games, driving a toy car, and controlling a Sphero SPRK+ robot (Sphero, USA).[103] Some children with QCP were also able to successfully spell using a P300 spelling board, however most encountered challenges due to difficulties maintaining engagement and attention during the task.

The increased lifespan of children with QCP[106], often spanning decades, coupled with their heightened plasticity and capacity for learning, underscores the considerable benefits that could stem from early life application of BCIs. Recognizing and addressing the distinct elements that influence pediatric BCI performance is crucial if we aim to harness the transformative potential of BCIs for children with disabilities. Therefore, to achieve this goal, it becomes imperative to adapt and refine the rapidly evolving BCI paradigms, initially designed for the relatively static adult brain, to suit the dynamic, highly adaptable, and developing brains of children.

1.4 Optimizing BCIs for Children

Such studies demonstrate an important aspect of BCI use in children: children are not "small adults". Therefore, BCIs for adults are not designed, let alone optimized, for children. Ideally, an interface should have universal usability. However, that is not the case for BCIs: adult studies have shown that 10 - 50% of users cannot achieve sufficient control.[107,108] Even with cutting edge signal processing and machine-learning classification algorithms, significant intra- and inter-user variability remain a major problem for the realization of reliable BCI performance.[107–110] To bridge this reliability gap, Kübler and colleagues identified four aspects that might determine BCI control: individual characteristics of the BCI user, characteristics of the BCI, type of feedback and instruction, and the application controlled by the BCI.[111] Although no previous studies have considered these issues for children, they are considered here with a pediatric perspective.

1.4.1 Individual characteristics of pediatric BCI users

The unsuccessful efforts in resolving the reliability issue at the hardware and software level have led the BCI research community to appreciate one of the most important aspects of BCI control: the individual characteristics of the user. Such individual characteristics, and two predictors of BCI performance that have been studied thus far, include physiological and neuropsychological factors. Establishing predictors of BCI performance could reduce the amount of time and effort spent on determining a suitable BCI paradigm for the user, thereby bypassing arduous training of a suboptimal control system. This is especially important for children, who's limited attention may drive frustration and reduce motivation as time and effort are invested into a system with unsatisfactory outcomes. These considerations are equally important for parents, as failure of sufficient BCI control could lead to feelings of discouragement and despair.[78] Further, known predictors could inform the design of optimized training protocols and BCI-controlled applications by utilizing a user-centred approach.[109,110] The advancement of BCI for severely disabled children demands attentive exploration and understanding of their needs and requirements, as well as the restrictions that accompany their impairments.[85,110]

1.4.2 Characteristics of BCIs for children

Enhancements to BCI systems can be directed towards both hardware (such as headsets and electrodes) and software (including processing and classification methodologies). At present, there is a lack of headsets specifically designed for children. Given their heightened sensitivity to discomfort, it's imperative that pediatric headsets be not only comfortable but also easy and quick to set up.[66] Furthermore, to accommodate the unique head shapes of children and ensure optimal contact between the electrodes and the scalp, these headsets must offer customization options. In addition to hardware, there's a notable absence of guidelines for processing brain signals in children, underscoring another area for improvement in pediatric BCI technology.[112]

1.4.3 Child-friendly instruction and feedback

A major challenge with BCI use is the requirement for subject-specific training, which is often tedious and unengaging for the user, but necessary to improve efficiency.[113,114] This is especially true for children, who's limited attention and motivation pose pressing challenges for the implementation of BCI. Games present clear objectives with short-term goals that confer rewards and act as external motivators to provide a seamless sense of advancement.[115] Several studies have shown that the addition of scoring systems, prizes, and awards to tasks, a process known as "gamification", can increase motivation, attention and task performance in children[115,116] including those with neurological disabilities [117–120].

BCI control is an acquired skill; performance depends on both the user and the system.[121] As with any skill, performance can be improved with frequent practice. Efficient BCI control involves the cooperation of the user, who learns to convey their intent through the generation of recognizable brain signals; and the BCI system, which adapts (via machine learning) to recognize and translate the signal features to accomplish the user's intent.[121] Improvements in BCI performance can therefore be achieved through modifications to either the user-training or the system. However, most studies have focused on enhancing the components of BCI systems, while largely neglecting the user-training component of BCI control, particularly in children where factors are almost certainly different.[121]

1.4.4 Child-friendly BCI applications

Most studies in children have focused on communication and mobility applications of BCI, based off assumptions from adult studies of what children with disabilities would want to use BCIs for.[66] Children may have different priorities and goals compared to adults, thus applications that may be favoured in adult populations may be different in pediatric populations. For example, more entertaining application such as games and art are likely more appealing compared to communication applications.[103]

1.5 Integrating BCIs into the lives of end-users

A major limitation with BCI research to date is that nearly all studies have been conducted in a controlled environment, thereby neglecting the true environment of the target end-user.[122] This presents a crucial translational gap. Problems that may arise in realistic end-user environments, such as increased noise and distraction, could lead to substantially different EEG signals and, consequently, the failure of such lab-based BCI systems.[123] For example, when BCI was transferred to the home environment of users with ALS their accuracy on a spelling task dropped by 13%, demonstrating significantly worse performance than when they completed the same task in a controlled environment managed by BCI experts.[89] Many of the current BCI systems are impractical for prolonged independent use by end-users.[123] There is thus a great need for translational studies that evaluate BCIs in a real-life context.

Zickler et al. proposed a User-Centre Design (UCD) to bridge the translational gap.[87] The UCD promotes BCI research and development as a holistic user experience, rather than focusing on single aspects of design, such as improving signal processing or classification algorithms.[85] It is based upon six principles, including early and active involvement of end-users (and their families); guidance and fine-tuning driven by user-centered evaluation; and continuous interaction between the BCI team and end-users.[85] Four stages for the implementation of such principles are: identification and understanding the context of use and potential impact; understanding user requirements; designing solutions to meet user requirements; and evaluating usability of the design.[85] The three facets of usability that should be evaluated in the UCD approach are effectiveness (accuracy), efficiency (information transfer rate (ITR) and subjective workload), and user satisfaction (with general aspects of the device and overall satisfaction).[85]

Caregivers play an essential role in the integration of BCI into the home environment.[47] Therefore, the first aspect of home-based BCI use must focus on the experiences of people without technical expertise who will be regularly setting up these systems.[124] In studies of home-based BCI use, the most common feedback from caregivers concerned the need for more technical training on BCI systems, more reliable connections between the headset and computer, and easy-to-clean, preconfigured caps/headsets.[105] Time commitments for BCI setup, training, and cleanup have been reported by caregivers to contribute to attrition of home-based BCI.[105,106] Equally important, Kleih and Kübler have advised that the mental and emotional state of caregivers must also be considered in the use of BCI at home, as the occurrence of frustration and dissatisfaction could hinder successful BCI use.[51]

In UCD, the development of assistive technologies revolves around the target end-user, specifically their needs and requirements. For example, in a study of adults with severe neuromotor impairment, communication applications were found to be more effective (accuracies > 80%) and efficient (higher ITR) than entertainment applications, however users indicated that only the entertainment applications were feasible for use in daily life.[85] Other common needs that have emerged from end-user feedback of home-based systems are reliability and speed of the system, headset aesthetic (wireless, easier to adjust), and the need for quicker/more engaging calibration phases.[85,87,124,125,127,128] Nevertheless, the impact of home-based BCI has demonstrated benefits for quality of life for patients with severe motor impairments.[86,89,128]

1.6 Family centred-research and patient engagement

Patient-centred care has gained widespread acceptance in healthcare, representing a paradigm shift towards recognizing and incorporating the invaluable insights and experiences of patients and their families into the research process. This shift ensures that healthcare services, treatments, and policies are truly responsive to the needs and preferences of those they directly affect, leading to improved patient satisfaction, engagement, and outcomes.[129] BCI research stands to greatly benefit from this approach. By involving patients and their families in the design and implementation of BCI technologies, we can enhance usability and acceptance of these systems, ultimately contributing to better quality of life for users.

Individualized approaches are standard practice for current management and interventions for children with QCP. This is because each child's condition, abilities, goals, and personal circumstances are unique, and these differences can significantly impact their process and outcomes. Similarly, this personalized approach should be integral to the implementation of BCI, ensuring these systems meet the specific needs of each child.

In addition to these child-specific considerations, family plays an integral role in pediatric care. Parents or caregivers are often the primary advocates for their children and can provide critical insights into the child's behaviours, preferences, and responses to interventions. They are also typically responsible for supporting the child in carrying out therapeutic activities at home, and their ability to do so can significantly influence the child's progress.[130] Therefore, it's crucial to engage families in the research and development process of BCIs for children, to gain a holistic understanding of their needs, values, and capabilities, and to involve them in goal-setting and decision-making.

Introducing such innovative technologies to families requires sensitive communication and couching of expectations. Clear, honest information about the capabilities, limitations, and potential risks of BCI should be provided, ensuring that families are prepared for this new experience. The promise of BCI must be balanced with the understanding that, like all technologies, its application is a tool with both benefits and limitations rather than a cure-all solution. This underscores the importance of coupling technological advancements with deeply rooted principles of patient and family engagement to ensure optimal outcomes.

1.6.1 Mixed methods approach for patient-centred research

Mixed methods research, combining both qualitative and quantitative approaches, is increasingly being utilized in patient engagement and family-centered research. This complementary use of methodologies allows for a more comprehensive understanding of the intricate aspects of patients' experiences and healthcare outcomes.[131] Qualitative methods, such as interviews or focus groups, delve into the intricate narratives of individuals, capturing rich, context-specific insights into their experiences, needs, and challenges, which is particularly crucial when engaging with families with complex needs. Conversely, quantitative methods provide the ability to measure and compare variables, identify patterns, and generalize findings to larger populations, contributing to the establishment of evidence-based practices. When integrated, these methods can yield a deeper understanding of patients' experiences and health outcomes, leading to the development of more effective, personalized interventions. Engaging families with complex needs in this mixed methods research often involves additional considerations, such as creating an open and supportive environment, ensuring clear and comprehensible communication, and recognizing and addressing potential barriers to their involvement, in order to foster authentic engagement and valuable contributions.

1.7 Rationale & Aims

EEG-based BCIs are safe, painless, and adaptable to home-based use with modern systems advancing rapidly toward improved performance and clinical utility. While there has been an explosion in recent BCI development, nearly all studies have been in adults with gross neglect of pediatric populations.[66] However, children with severe neurological disabilities stand to benefit immensely from BCI technology, given their heightened plasticity, capacity for learning, and increased lifespan. The current BCI research landscape, consisting predominantly of studies conducted on healthy adult controls, makes it challenging to determine the transferability of results to clinical populations. Studies by our lab demonstrated that healthy school-aged children and those with hemiplegic CP have the ability control simple EEG-based BCI systems with minimal training and performance comparable to adults.[104,132] However, further research with children with QCP and their families is required to evaluate the transferability of results.

It is estimated that more than 13,000 Canadians might benefit from BCI technologies, including children.[82] However, a lack of awareness and knowledge of BCIs in both clinicians and patients presents a barrier to access.[82,133] Clinical research can fill a fundamental gap that exists between technology development and the patients and families who could benefit from these new tools.[134] Our recently established clinical pediatric BCI program has provided children with QCP with unique opportunities to try a suite of BCI systems and demonstrated the feasibility and positive impact of the program for enhancing independence, inclusion, and recognition for these children and their families.[103,133] A major hurdle to advancing this progress is that there is no established baseline for how well children can perform across the variety of mental control signals and tasks offered by currently existing BCI systems, none of which have been designed with children in mind. Translational studies evaluating BCIs in real-life contexts can bridge the gap between controlled laboratory environments and the daily experiences of end-users. User-centered design principles, involving the active participation of children and their families, can guide the development of BCI systems that meet their specific needs and requirements, turning the promise of this technology into a reality to transform the lives of thousands of children and their families.

In chapter 2, we conducted a cross-sectional study comparing BCI performance in typically developing children across five paradigms well-established in adults.

Aim: To establish baseline measures of performance for typically developing children across a suite of non-invasive BCI systems and tasks.
Hypothesis: Most typically developing children can achieve basic BCI competency on all modern systems comparable to that seen in adults.

In chapter 3, we conducted a cross-over study to explore the impact of gamification on BCI performance in visual P300 and motor imagery-based paradigms in typically developing children.
Aim: To optimize the performance of BCI systems in typically developing children by integrating gamification techniques.
Hypothesis: Integrating gamified elements into BCI calibration paradigms will increase classification accuracy.

In chapter 4, we established a novel patient- and family-centred pilot Home BCI Program for children with QCP and their families.
Aim 1: To evaluate the feasibility of home BCI use.
Hypothesis 1: A home-based BCI program is feasible and beneficial for children with QCP and their families.

Aim 2: To better understand how BCIs can be effectively implemented and integrated into the daily lives of children with QCP and their families.
Hypothesis 2: Effective implementation and integration of BCI into the daily lives of children with QCP and their families can increase the children's independence and enhance their overall quality of life.

In chapter 5, we utilized a collaborative goal-setting approach to investigate the goals of children using BCI in their homes.
Aim: To investigate the feasibility and usability of home-based BCIs for children with quadriplegic CP to achieve their personal goals.
Hypothesis: Home-based BCIs are feasible and usable for children with QCP to achieve their personal goals.

In chapter 6, we conducted a pilot case study to explore the feasibility and psychosocial impact of enabling online multiplayer BCI video game play for children with QCP.
Aim 1: To demonstrate the feasibility of online BCI-facilitated for children with QCP.
Hypothesis 1: Online multiplayer BCI gaming is feasible for children with QCP.
Aim 2: To explore the potential psychosocial impacts of using BCI to enable multiplayer online gaming.
Hypothesis 2: Online multiplayer BCI gaming will have a positive psychosocial impact on children with QCP.

Chapter 2: Brain-Computer Interfaces for Children: A Comparative Study of Five Common EEG-based Paradigms

Dion Kelly, Ephrem Zewdie, Helen Carlson, Adam Kirton

2 Chapter 2

2.1 Abstract

Quadriplegic cerebral palsy (QCP), the most severe form of cerebral palsy (CP), affects millions of individuals worldwide. Children with QCP often have intact cognitive function but face challenges in communication or interaction with their environments, which may result in a condition similar to "locked-in syndrome." Brain-computer interfaces (BCIs) hold potential to help, but pediatric BCI research has been limited. This study aimed to establish baseline performance of common BCI paradigms in typically developing children to support applications in children with disabilities.

In this cross-sectional study, performance on five BCI paradigms, including visual (P300), auditory (AEP), and vibro-tactile (VTP2 and VTP3) event-related potentials, and sensorimotor rhythm (SMR) modulation using motor imagery, was evaluated in thirty school-age children using tasks with predefined goals. Two commercially available EEG-based BCI systems, Mindbeagle® and intendiX®, were used. The primary outcome was online classification accuracy. Potential factors affecting performance, including age, sex, motivation, tolerability, and fatigue, were also explored.

We found that most children were able to demonstrate competency on multiple BCI paradigms with favorable tolerability and no serious adverse events. Mean accuracy across all paradigms was 77.03%, with 73% achieving BCI competency. Performance on P300-based paradigms was better than the SMR paradigm, with the highest performance observed in the VTP2 paradigm (89.48%), and the lowest in the SMR paradigm (55.68%). Significant differences in accuracy and fatigue were observed across the paradigms, with the visual P300 spelling paradigm showing the highest motivation and lowest fatigue. Age was correlated only with AEP BCI performance, while no other factors appeared to influence performance across paradigms.

We conclude that evoked potential BCI paradigms are generally effective in children as young as 6 years of age in a laboratory setting for potentially meaningful tasks, such as communication, recreation, and computer operation. The research contributes to the limited knowledge on non-invasive BCI performance in children and offers insights into factors affecting performance. More research is needed to understand how these BCI paradigms can be

optimized for children and implemented in real-world environments and as assistive technology for youth with disabilities.

2.2 Introduction

Cerebral palsy (CP) is the most common physical disability in childhood and the leading form of lifelong disability, affecting over 85,000 Canadians and millions worldwide.[135] Quadriplegic CP (QCP) is a prevalent and most severe form of CP, accounting for approximately 35% of all cases.[28] It is a nonprogressive disorder characterized by impaired movement of the whole body, often resulting in limited or no motor function.[24] Many common causes of QCP selectively injure the motor system, leaving other neural systems relatively intact including cognition. Many children with QCP have an intact cerebral cortex and a substantial number have normal MRIs despite severe motor impairments.[136] Such cognitive motor dissociation is increasingly recognized and includes many youth with high capacity, some of whom have average or above-average intellectual abilities.[31–34] With limited means to communicate or interact with their environments, children who are cognitively aware and highly capable may experience a condition similar to "locked-in syndrome."[31]

The rights to equality, freedom of expression, and an adequate standard of living are among the most basic fundamental human rights, as established in the United Nations (UN) Universal Declaration of Human Rights.[38] Further, the United Nations Conventions on the Rights of a Child (UNCRC) and Persons with Disabilities (UNCRPD) acknowledge that children with disabilities have the right to enjoy a full and meaningful life in environments that respect dignity, promote independence, and foster their participation in society.[40,41] Importantly, the UNCRC and UNCRPD state that implementation of such rights shall be accomplished through "the application of readily available technology".[40] Traditional assistive technologies employing kinetic methods, such as joysticks and mechanical switches, are often impossible for children with severe QCP to use. Eye-gaze methods may address some difficulties[137] but dyskinetic body and eye movements and common comorbidities such as central visual impairment limit efficacy for many users .[47] Consequently, children with severe QCP currently have very limited options available to realize their fundamental human rights.[40]

Brain-computer interfaces (BCIs) are a potential solution to improve the lives of children with severe physical disabilities. By utilizing electroencephalography (EEG) and other non-invasive means, BCIs can directly capture the user's real-time brain activity converting it into intentional actions without requiring any physical movement.[138] This process involves sampling, amplifying, and processing intentional brain signals to extract specific features, which are then translated into commands using a classification algorithm.[139] These intentional brain signals can either be evoked by external stimuli or generated voluntarily by the user.[57] Common signals used for command-following include event-related potentials (ERPs) and modulation of sensorimotor rhythms (SMRs).[57,73] ERPs are variations in the EEG signal that correspond to specific time points after a stimulus onset, while SMRs can be modulated through imagined movement or motor imagery. The P300, a positive deflection in the EEG signal occurring approximately 300 ms after a stimulus presentation, is the most studied ERP feature for BCI control.[73] Motor imagery induces event-related desynchronizations (ERDs) and synchronizations (ERSs) of SMRs over the sensorimotor cortex.[73] The number and performance of BCI paradigms have been steadily increasing, but research has primarily focused on adults.

In contrast, pediatric BCI research has been relatively non-existent to date. A recent review found that in the last 14 years, there have only been eight studies published examining non-invasive BCI use in children.[140] No studies have investigated real-time P300-BCI use and only three studies have investigated motor imagery for SMR-BCI in children[104,141,142], including two from our group, with all using a simple commercially available BCI system (Emotiv EPOC). Clinical research is essential to bridge this fundamental gap between rapidly advancing technology development and the pediatric patients who stand to benefit.[82]

The aim of this study was to establish baseline levels of BCI performance in typically developing children across multiple BCI paradigms well-established in adult populations to provide foundational knowledge necessary for future applications in children with neurological disabilities. We hypothesized that the majority of typically developing children can attain adult-level BCI proficiency, characterized by an average classification accuracy of 70% or higher[143], across various BCI paradigms.

2.3 Methods

2.3.1 Participants

wWe conducted a cross-sectional study, with participants recruited from the community through the Healthy Infants and Children's Clinical Research Program, a database of typically developing children whose families had volunteered to participate in medical research.[144] Recruitment occurred between January and July 2021. Inclusion criteria were typical neurodevelopment with no neurological conditions, age 6 to 18 years, and informed consent/assent. Methods were approved by the Conjoint Health Research Ethics Board at the University of Calgary (REB18-2033). Participants received a $25 gift card for volunteering.

2.3.2 Experimental Setup

All testing occurred in the BCI Lab at the Alberta Children’s Hospital. Two commercially available EEG-based BCI systems, Mindbeagle® and intendiX® [145,146] (g.tec, Austria) were employed to evaluate five BCI paradigms well-established in adults: visual, auditory and vibro-tactile P300 event-related potentials (with two and three mechano-vibrators), and SMR modulation using motor imagery (Fig. 2.1). Each paradigm consisted of tasks with predefined goals and lasted between 8-20 minutes, with optional 5-minute breaks in between to reduce mental fatigue. The primary outcome was online classification accuracy. Secondary outcomes were potential factors affecting performance, including age, sex, motivation, tolerability, and fatigue.

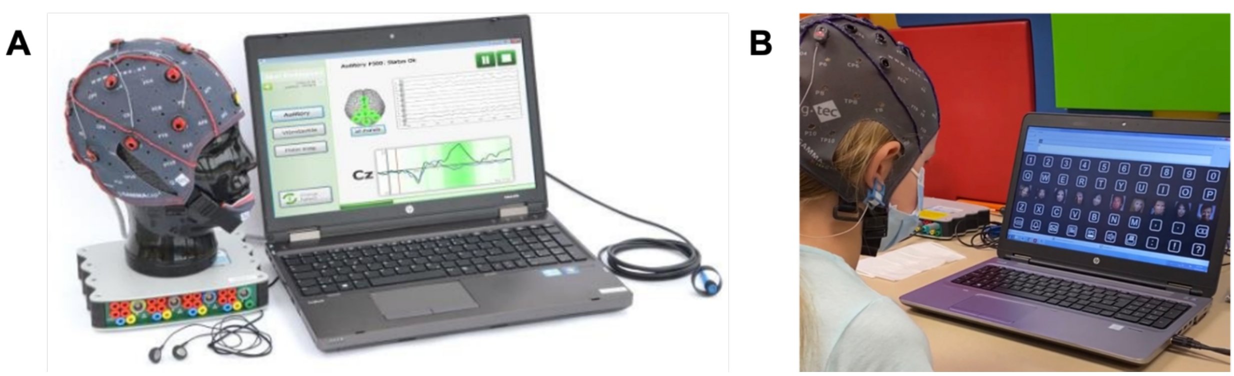


Fig. 2.1. The g.tec BCI systems. (A) The mindBEAGLE® system (photo retrieved from https://www.gtec.at/product-configurator/mindbeagle). (B) A participant completing the spelling task on the intendiX® P300-based speller.

Participants were comfortably seated at a table and were instructed to remain still and relaxed with their eyes open. Participants were positioned such that they could not see the computer monitor to mitigate potential distractions. Therefore, participants did not receive real-time feedback. Once a run was completed, the participant was shown the result page produced by mindBEAGLE software to inform them of their performance.

EEG voltage signals were acquired from the scalp using the g.SCARABEO active gel-based electrode system and amplified using the g.USBamp. Participants were fitted with an EEG cap sized for their head circumference, with electrodes placed according to the international 10-20 system. Electrode configurations differed depending on which BCI paradigm was being evaluated. For both systems, a ground electrode was positioned at FPz and a reference electrode was clipped onto the right ear lobe.

Paradigm 1: Communication board spelling using visual P300 event-related potentials (P300)

In the visual P300 paradigm, participants were asked to spell words using the intendiX® row/column spelling system at various difficulty levels corresponding to flash rates (fewer flashes = greater difficulty). The intendiX® speller board presents 50 cells arranged in a 5 x 10 matrix, displaying letters from A to Z, numbers from 0 to 9, and special characters.[147] Cells flash in rows and columns, which means that only one row or column of characters changes during each flash to display a famous celebrity's face in each flashing cell.[148] Flashes last 100 ms, with an inter-stimulus interval of 75 ms. EEG signals were sampled from 8 electrodes positioned at Fz, Cz, P3, Pz, P4, PO7, Oz, and PO8.

Participants were instructed to direct their attention to letters on the spelling board. They began by training a classifier to recognize their unique EEG data by silently counting in their heads the number of times a target letter flashed, while ignoring other flashes.[149] Participants were asked to attend to each letter of the word "WATER" in sequential order to spell the word in copy-spelling mode at 15 flashes per row/column, as recommended by the program.[149] In copy-spelling mode, the target letters are inputted into the application and displayed at the top of the screen so that the user pays attention to only one specified target letter for the specified period of time (number of flashes), before directing their attention to the next target letter. Training was completed offline with no feedback and lasted approximately 5 minutes.

After training was complete, the participant's classifier was loaded into the intendiX® program, and they were instructed to free-spell the five-letter word "SNACK" beginning at 10 flashes per row/column and decreasing by increments of 2 flashes for each subsequent run down to a minimum of one flash, for a total of 6 runs. This word was chosen because five letters is the average word length in the English language and "snack" is a grade 1 spelling word. For free-spelling, the participant was instructed to pay attention to and to silently count the number of times each target letter flashed, until a letter was selected and displayed providing real-time feedback. The participant was instructed to start paying attention to the next letter in the word once a selection was made, regardless of whether the selection was correct or incorrect. The number of correctly selected characters was recorded and divided by the total number of target letters to generate an accuracy score at each flash rate.

Paradigm 2: Identifying target stimuli in the oddball paradigm using auditory event-related potentials (AEP)

Performance on the AEP was assessed using the mindBEAGLE® system, which is a complete BCI system designed for use in a clinical setting to assess cognitive abilities and conscious awareness in patients with disorders of consciousness, as well as provide a potential means of binary communication.[145] EEG signals were sampled from 8 electrodes positioned over the centroparietal region at FCz, C3, Cz, C4, CP1, CPz, CP2, and Pz.

The task employed an auditory oddball paradigm, where deviant target stimuli (high tones) were presented infrequently (12.5%) among frequently presented non-target stimuli (low tones, 87.5%). Participants were given headphones that were connected to the Audio Trigger Adapter Box. First, the task was explained and the auditory stimuli that were presented during each run were played to ensure understanding and adequate volume levels. Participants were instructed to silently count in their head the number of times they heard the target high tones (1000 Hz beep), while ignoring the nontarget low tones (500 Hz beep). One run consisted of 120 trials (15 target; 105 non-target stimuli), lasting approximately 7 minutes. To mitigate mental fatigue and to promote attention, the run was broken into four segments and paused approximately every 1.5 minutes. Participants completed three runs in which accuracy scores were averaged to obtain a mean accuracy for each participant.

Paradigm 3: Identification of a target stimulus in the oddball paradigm using 2-tactor vibro-tactile P300 event-related potentials (VTP2)

Performance in the VTP2 paradigm was assessed using the mindBEAGLE® system, with EEG signals sampled from 8 electrodes positioned over the centroparietal region at FCz, C3, Cz, C4, CP1, CPz, CP2, and Pz.

Participants wore headphones while vibro-tactile stimulators (mechanovibrators) connected to the g.STIMbox were secured onto each wrist. Similar to the AEP paradigm, the VTP2 assessment employed an oddball paradigm where non-target stimuli were presented frequently (87.5%) relative to target stimuli (12.5%). The task was explained and the vibrotactile stimulations were demonstrated. Participants were instructed to silently count the number of times they felt the target vibration on their left wrist, while ignoring distractor vibrations on their right wrist. Brown noise was played to prevent auditory potentials from the sound of the mechanovibrators. One run consisted of 120 trials (15 target; 105 non-target stimuli), lasting approximately 2.5 minutes. Participants completed three runs in which accuracy scores were averaged to obtain a mean accuracy for each participant.

Paradigm 4: Identification of a target stimulus in the oddball paradigm using 3-tactor vibro-tactile P300 event-related potentials (VTP3)

Methods for the VTP3 paradigm were the same as the VTP2 paradigm (explained above), with the following differences: stimulators were fixed onto each wrist, with an additional 'distractor' stimulator fixed to the right ankle; non-target stimuli were presented on the right ankle (75%) and target stimuli were presented on the left and right wrists (12.5% for each wrist). If a participant achieved >60% or higher classification accuracy, the threshold recommended by g.tec medical engineering[151], they were assessed in 'communication mode' using that classifier. In 'communication mode', participants were asked to confirm the answers to ten questions with known answers. They then answered the questions again by paying attention to stimuli on the appropriate wrist (left for 'yes'; right for 'no') for approximately 75 seconds before an answer was chosen by the program. If the 'distractor' was chosen, the response circle would remain in between the two answers reflecting an indeterminate response.[151] Chosen answers were compared against known answers to generate an accuracy score for communication mode.

Paradigm 5: Sensorimotor rhythm modulation (SMR) using hand motor imagery

Participants were evaluated in the SMR paradigm using the mindBEAGLE® system, with 16 electrodes positioned over the sensorimotor cortex at FC3, FCz, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1 CPz, CP2, CP4 and Pz. Additional electrodes allowed the BCI system to accurately detect and classify the spatially distributed brain patterns, providing better spatial resolution to differentiate between various imagined movements and their locations. Participants were given headphones that were connected to the Audio Trigger Adapter Box. First, the task was explained and the verbal cues that were presented during the motor imagery task were played to ensure understanding. Participants were then instructed to imagine seeing themselves or someone else performing left- and right-hand movements (opening and closing) from a third person perspective (external visual imagery). Each trial consisted of 8 seconds of imagined movement, with 1-2 seconds between trials. One run consisted of 60 imagined movements (60 trials) per hand presented in randomized order, lasting 9 minutes. Approximately every 3 minutes, the assessment was paused to check-in with the participant and give them an optional 1-minute break. Participants completed one run to generate a classifier.

Requirements for assessment in 'communication mode' were the same as in the AEP paradigm (explained above), with the following differences: to answer questions, participants were required to imagine left-hand movement for 'yes' and right-hand movement for 'no' for 8 seconds before an answer was determined by the program. At the end of the run, a circle would either move over 'yes' or 'no'. Chosen answers were compared against known answers to generate an accuracy score for communication mode.

2.3.3 Signal processing and classification

All data recording, processing, and real-time feedback was managed by mindBEAGLE® and intendiX® software. Data was sampled at 256 Hz and band-pass filtered between 0.1-30 Hz to extract signal features while the participant performed the tasks relevant to each paradigm. A notch filter was applied at 60 Hz to reduce power line interference. The digitized data was then transmitted to a computer for processing and classification. For P300-based paradigms, the mindBEAGLE software extracted ERPs from data recorded 100 ms before and 600 ms after stimulus presentation. A linear discriminant analysis (LDA) classifier was then used to classify ERPs as target or non-target stimuli and calculate a classification accuracy ranging from 0% to

100%.[151] For the SMR paradigm, ERDs and ERSs around 10-12 Hz were extracted from data recorded 3s before and 5s after presentation of auditory cues. A common spatial patterns (CSP) classifier was used for feature extraction to enhance the separability of the motor imagery classes (left vs right) by focusing on the most relevant spatial information from the electrodes. This established weights for each electrode, representing their relative contributions to classification. Subsequently, an LDA classifier was applied to the extracted features to distinguish between left vs right hand motor imagery, producing a classification accuracy between 0% and 100%.[151] In addition to reporting the median classification accuracy for each run, the results page presented by mindBEAGLE also reports how many trials were influenced by artifact (i.e. physiological or external sources of noise that contaminate the EEG signal).

2.3.4 Randomization

The five paradigms were divided into two sessions. The SMR and AEP paradigms were kept separate as we hypothesized that these would be the most fatiguing. The other paradigms were grouped together based on ease of setup, since AEP, VTP2, and VTP3 all utilized the same electrode montages. Participants were randomly assigned to one of two groups using an online randomization tool (www.randomlists.com). Group 1 was assessed on paradigms 1 and 2 on the first day, followed by paradigms 3, 4, and 5 on the second day. Group 2 was the opposite. Paradigm order was also randomized within groups to mitigate within session learning effects and fatigue.

2.3.5 Questionnaires

Handedness was assessed at baseline using a child-adapted Edinburgh Handedness inventory.[152] Factors potentially affecting performance were collected at the start of each session. These included age, sex, vision problems/corrected vision, concussion history, previous BCI experience, and state including measures of mood, fatigue, sleep, exercise, in addition to 24-hour estimates of caffeine, sugar, alcohol, and drug consumption. Participants were also asked about hobbies and sports, including duration, and estimated daily practice. A motivation questionnaire, adapted from the Pediatric Motivation Scale (PMOT) for rehabilitation[153], was administered at the end of each paradigm. The PMOT was developed for children as an event-based measure of motivation (rather than motivation as a trait) following activities. It contains 21 items which are divided into six subscales to evaluate subjective feelings of effort/importance, relatedness, autonomy, interest/enjoyment, competence, value/usefulness and open-ended items. Children responded to five items taken from the interest/enjoyment, effort/importance, and competence subscales using a 6-point ordinal face scale, where 1 was "not true at all" and 6 was "definitely true". To promote valid responses throughout the scale, some items were framed negatively and were reverse scored. Children reported their perceived fatigue on a scale of 1 to 10 at the end of each paradigm as well as at the end of each session. Change in fatigue was calculated by subtracting the reported fatigue recorded after each paradigm from the reported fatigue recorded immediately before each paradigm. Tolerability was assessed at the end of each session including ranking their experience among seven common childhood events.[154]

Behavioural observations, including potential signs of fatigue, boredom, and engagement, were systematically noted during the study. The researcher (DK) monitored participants' facial expressions, body language, and verbal cues to identify these signs. The frequency and duration of these behaviors were recorded and analyzed to understand their relationship with BCI performance and user experience. The average number of artifacts for each run for each paradigm were also reported as a sign of boredom/lack of engagement from restlessness.

2.3.6 Statistical Analysis

Statistical analyses were performed using SPSS Statistics for Macintosh version 28.0 (IBM, USA) and Jamovi version 1.6.23 (Jamovi, 2021). Graphs were created using GraphPad Prism version 9.0.0 for Macintosh, GraphPad Software, San Diego, California USA, www.graphpad.com. Accuracy scores were averaged to obtain a mean score for each paradigm. For questionnaire data, responses were transformed into corresponding numerical values. Outcomes were tested for normality using the Shapiro-Wilk test. Parametric or nonparametric tests were used as appropriate. Differences in dependent variables between the five BCI paradigms, including performance, fatigue, and motivation were compared using the Friedman two-way analysis of variance (ANOVA) by ranks followed by pairwise comparisons (Dunn's Multiple Comparison's test). For pairwise comparisons that showed significant differences, the choice of the statistical test to further examine these differences depended on the distribution of the differences between paired observations. If the differences were normally distributed, a parametric paired samples t-test was used. If the differences were not normally distributed, the nonparametric Wilcoxon Signed-Rank Test was applied. Resulting effect sizes are reported as Cohen's d (parametric) or rank biserial correlations (non-parametric, r_rb). A parametric paired samples t-test was performed to evaluate differences in tolerability between sessions. Independent samples Mann-Whitney U and Kruskal-Wallis tests were performed to evaluate the effects of sex, age group, handedness, and assessment order on BCI performance in each paradigm. Spearman's correlations were conducted to evaluate potential factors affecting performance, including age, motivation, tolerability, fatigue, and mood. Partial correlations were completed to control for factors found to have a significant effect on BCI performance.

2.4 Results

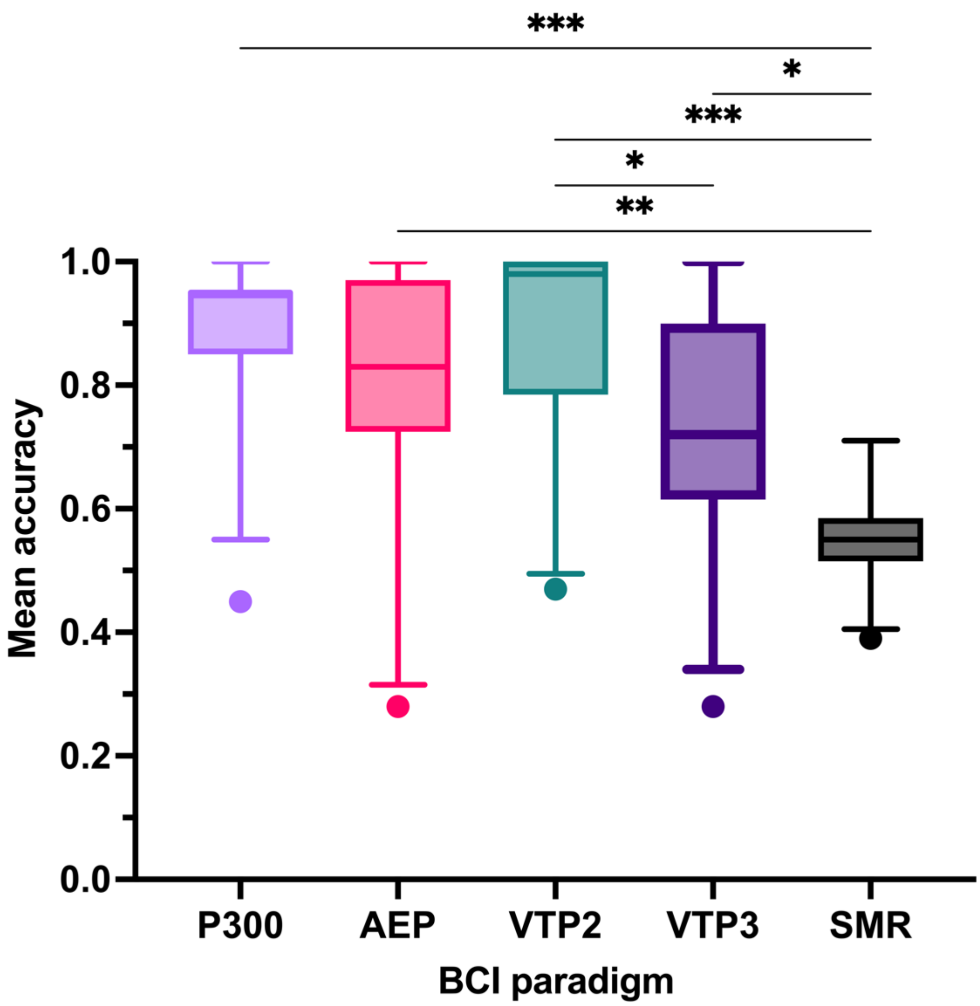
2.4.1 Participants

The final population consisted of thirty typically developing children (53% female, mean [SD] 11.3 [3.4] years, range 6–16, 93% right-handed). The recruitment rate, determined as the ratio of participants who were successfully enrolled in the study to the total number of individuals who were approached and met the eligibility criteria, was approximately 35%. There were no serious adverse events. Sessions averaged 50-75 minutes with the longest being 120 minutes. Two participants dropped out after the second run of the AEP paradigm, with one citing extreme fatigue and the other reporting a headache. One of these participants also did not complete the VTP2 paradigm.

2.4.2 BCI Performance

Mean accuracy scores for each of the five BCI paradigms are summarized in Figure 2. Mean accuracy across all BCI paradigms was 77.03±1.7%, with a range from 59-93%. Overall, 73% of participants achieved BCI competency, defined as a classification accuracy of 70% or higher [143], in paradigms involving evoked potentials with a mean accuracy of 82.43±2.0% and a range from 58-99% (Table 1). Only 2/30 participants achieved competency in the SMR paradigm. The highest performance was achieved in the VTP2 paradigm (89.48%), while the lowest was observed in the SMR paradigm (55.68%). Accordingly, significant differences in average accuracy among the paradigms were observed ($\chi2(4) = 46.1$, $N = 29$, $p < 0.001$). Post-hoc Dunn's multiple comparisons tests were conducted, and the results showed significant differences between the following pairs of BCI paradigms (adjusted p-values are reported): P300

> SMR ($p < 0.001$); AEP > SMR ($p = 0.002$); VTP2 > SMR ($p < 0.001$); VTP3 > SMR ($p = 0.01$); and VTP2 > VTP3 ($p = 0.02$). SMR performance was significantly lower than P300 (paired samples t-test: $t_{29} = 12.22$, Cohen's d = 2.231), AEP (Wilcoxon Signed-Rank Test: W = 426.0, r_rb = 0.832), VTP2 (Wilcoxon Signed-Rank Test: W = 432.0, r_rb = 0.986), and VTP3 (paired samples t-test: $t_{29} = 4.92$, Cohen's d = 0.898) paradigms. Similarly, VTP3 performance was significantly lower than VTP2 performance (paired samples t-test: $t_{28} = 3.93$, Cohen's d = 0.730).



P300 = visual P300; AEP = auditory evoked potential; VTP2 = vibrotactile with 2 tactors; VTP3 = vibrotactile with 3 tactors; SMR = sensorimotor rhythm

Figure 2.2. Mean classification accuracies for all BCI paradigms. Mean accuracies for the P300 paradigm reflect mean accuracies averaged for all trials at flash rates demonstrating competency (accuracy > 70%; flash rates 4-10). Data is shown as a boxplot of the accuracies averaged over

all runs for each BCI paradigm, where the box corresponds to the interquartile range (25th - 75th percentile), the line inside the box denotes the median (50th percentile), the whiskers extend down to the 5th percentile and up to the 95th percentile, and the dots beyond the whiskers are outliers. Lines over boxplots indicate groups which were significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * $p<0.05$, ** $p<0.01$, *** $p<0.001$.

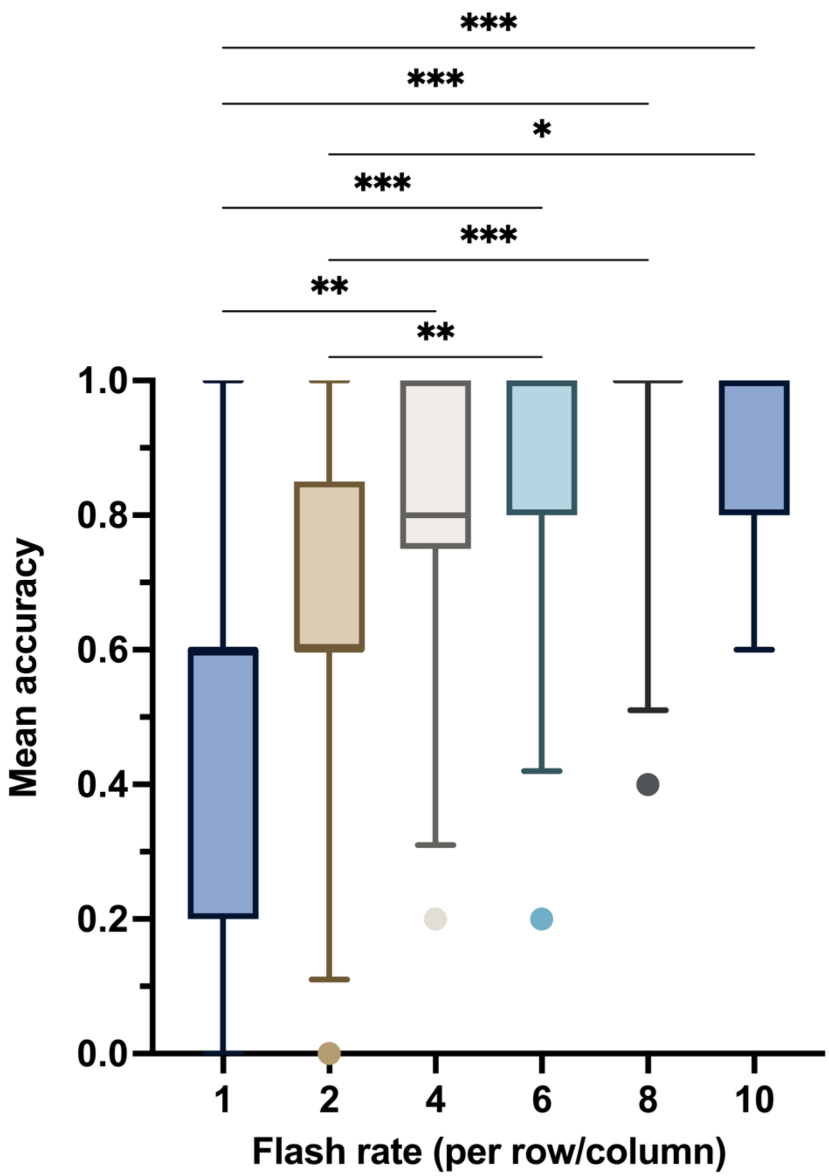
Table 2.1. Frequency of participants who obtained a certain classification accuracy in each BCI paradigm evaluated using the mindBEAGLE system.

Mean classification accuracy (%)	VTP2 (n=29)	VTP3 (n=30)	AEP (n=30)	SMR (n=30)
100	14	3	7	0
90-99	6	6	5	0
80-89	2	4	6	0
70-79	2	4	6	2
60-69	3	8	0	4
50-59	1	2	1	20
< 50	1	3	5	4

Visual P300 Paradigm

Figure 2.3 demonstrates BCI performance on the visual P300 spelling paradigm, as indicated by mean accuracy score at various flash rates. Scores ranged from 43% to 97% in a dose-dependent manner with the highest accuracy observed at 8 flashes per row/column (93.33±2.8%; range 40-100) and the lowest at 1 flash (48.67±5.05%; range 0-100). Mean accuracy at 10 flashes was 89.33±2.7% (range 60-100); 6 flashes was 87.33±3.7% (range 20-100); 4 flashes was 80.67 ± 4.1% (range 20-100); and two flashes was 66.0±5.0%; (range 0-100). Differences in average accuracy across the six flash rates were observed ($\chi2(5) = 66.8$, N = 30, $p < 0.001$). Average accuracy was higher at flash rates of 4 or more flashes per row/column, with significant differences between the following pairs of flash rates: 4 vs. 1 ($p = 0.003$); 6 vs. 1 ($p < 0.001$); 6 vs. 2 ($p = 0.007$); 8 vs. 1 ($p < 0.001$); 8 vs. 2 ($p < 0.001$); 10 vs. 1 ($p < 0.001$); 10 vs. 2

($p = 0.01$). The overall mean accuracy for flash rates ranging from 4 to 10, which had average accuracies above 70%, was 87.67±2.4%. However, there was variability among participants (ranging from 45 to 100) across this range. Table 2 shows the frequency of participants who achieved specific classification accuracies at different flash rates. Twenty-eight of the thirty participants (93%) were able to attain 100% accuracy in at least one of the flash rate conditions, with only one unable to achieve 80% accuracy in one condition. Three participants reported forgetting/mixing up the letter arrangement after a mistake was made. One participant was visibly fatigued, almost falling asleep during the 5-minute training period, and had the lowest performance.



P300 = visual P300; AEP = auditory evoked potential; VTP2 = vibrotactile with 2 tactors; VTP3 = vibrotactile with 3 tactors; SMR = sensorimotor rhythm

Figure 2.3. Mean classification accuracies for the visual P300 spelling paradigm at various flash rates. Accuracy reflects the number of correctly selected characters divided by the number of target characters for the word "SNACK". Lines over boxplots indicate groups which were significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * p<0.05, ** p<0.01, *** p<0.001.

Table 2.2. Frequency of participants who obtained a certain classification accuracy at each flash rate using the intendiX P300 spelling system.

Mean classification accuracy (%)	10 flashes	8 flashes	6 flashes	4 flashes	2 flashes	1 flash
100	18	24	19	13	7	2
80	8	3	5	10	6	4
60	4	2	5	3	11	10
40	0	1	0	3	2	6
20	0	0	1	1	3	5
0	0	0	0	0	1	3

AEP Paradigm

Overall mean accuracy for the auditory paradigm was 79.37±3.9% (range 28-100). Figure 2.4 shows the ERP classification and spatial distribution of maximum P300 amplitudes generated by the mindBEAGLE® software for a high-accuracy participant and a low-accuracy participant. Twenty-seven of the thirty participants (90%) were able to achieve 70% or greater accuracy in at least one run, with eighteen (60%) attaining 100% accuracy in at least one run. Thirteen (43%) showed signs of boredom, including restlessness (finger tapping, swiveling chair, fidgeting with wires, looking around the room) and fatigue (yawning, dozing off, struggling to keep eyes open). This was supported by a significant increase in the average number of artifacts between the first and last run (22.0±6.8, 25.77±7.5, and 39.54±14.1; $p<0.01$). Two participants developed a headache during the AEP paradigm.

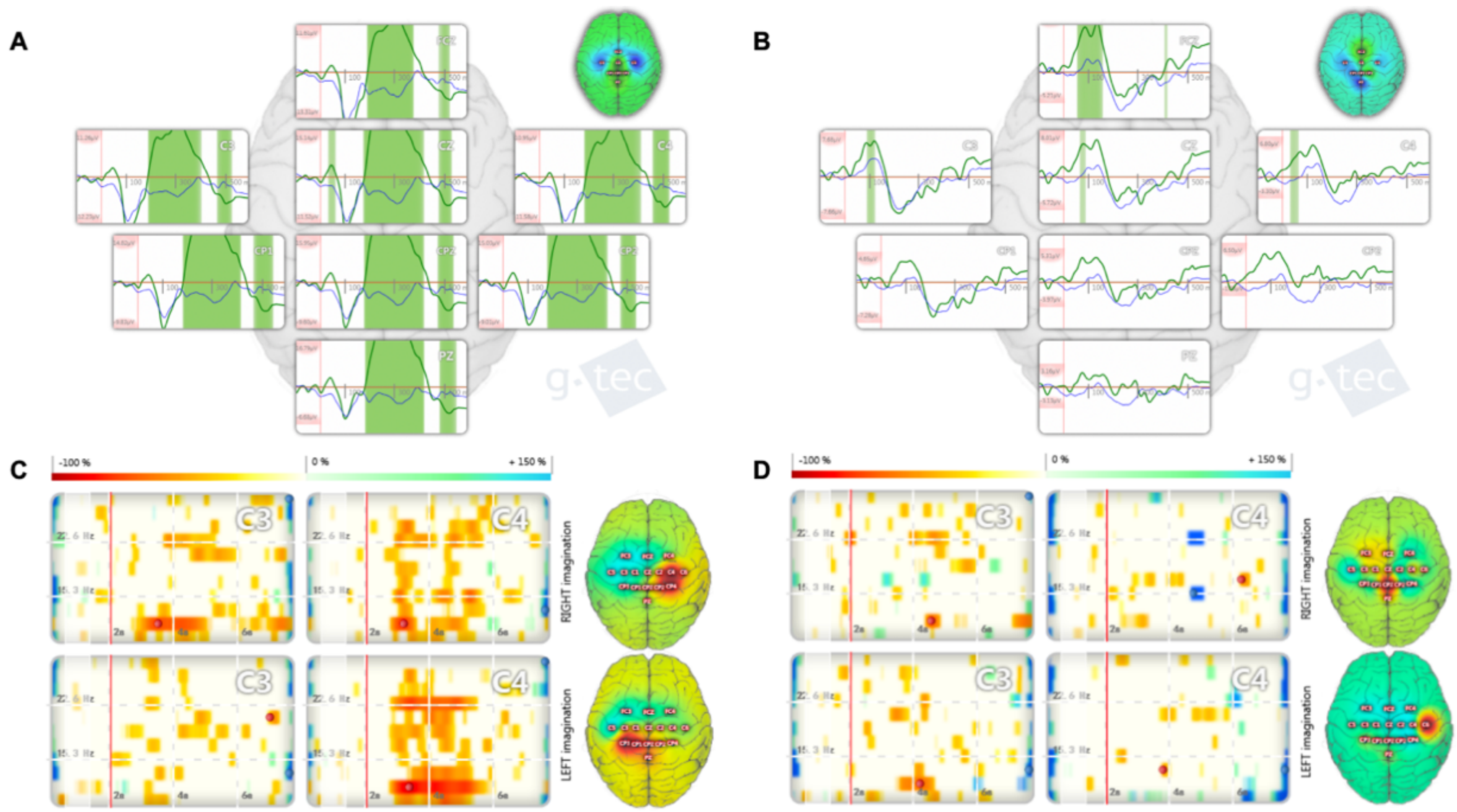


Figure 2.4. Feature classification and cortical activation plots generated by the mindBEAGLE® software. Top: ERP graphs showing averaged evoked potentials for each electrode for one session in the AEP paradigm. The vertical red line shows the onset of the stimulus. The green line represents ERPs elicited by target stimuli, while ERPs elicited by non-target stimuli are shown in blue. Significant differences between target and non-target trials are highlighted in green. The brain in the top right corner shows the spatial distribution of maximum peaks of the P300. (A) Participant who achieved 100% accuracy. (B) Participant who achieved 25% accuracy. Bottom: ERD plots for right (top) and left (bottom) motor imagery for electrodes at positions C3 (left) and C4 (right) showing bins that are significantly different from the bandpower during reference the period. The coloured bar above the plots represents the colours associated with relative decreases (red) and increases (blue) in bandpower during the active period. The brains to the right of the plots show the spatial distribution of cortical activations during right (top) and left (bottom) hand motor imagery. (C) Participant who achieved 71% accuracy. (D) Participant who achieved 39% accuracy. ERP = event-related potential; AEP = auditory evoked potential; ERD = event-related desynchronization.

VTP2 Paradigm

Twenty-nine participants completed the VTP2 paradigm. Mean classification accuracy was 89.48±3.0% (range 47-100). Twenty-two (76%) were able to achieve 100% accuracy in at least one run, with all participants (100%) achieving at least 80% accuracy in at least one run. The average number of artifacts did not significantly change across runs.

VTP3 Paradigm

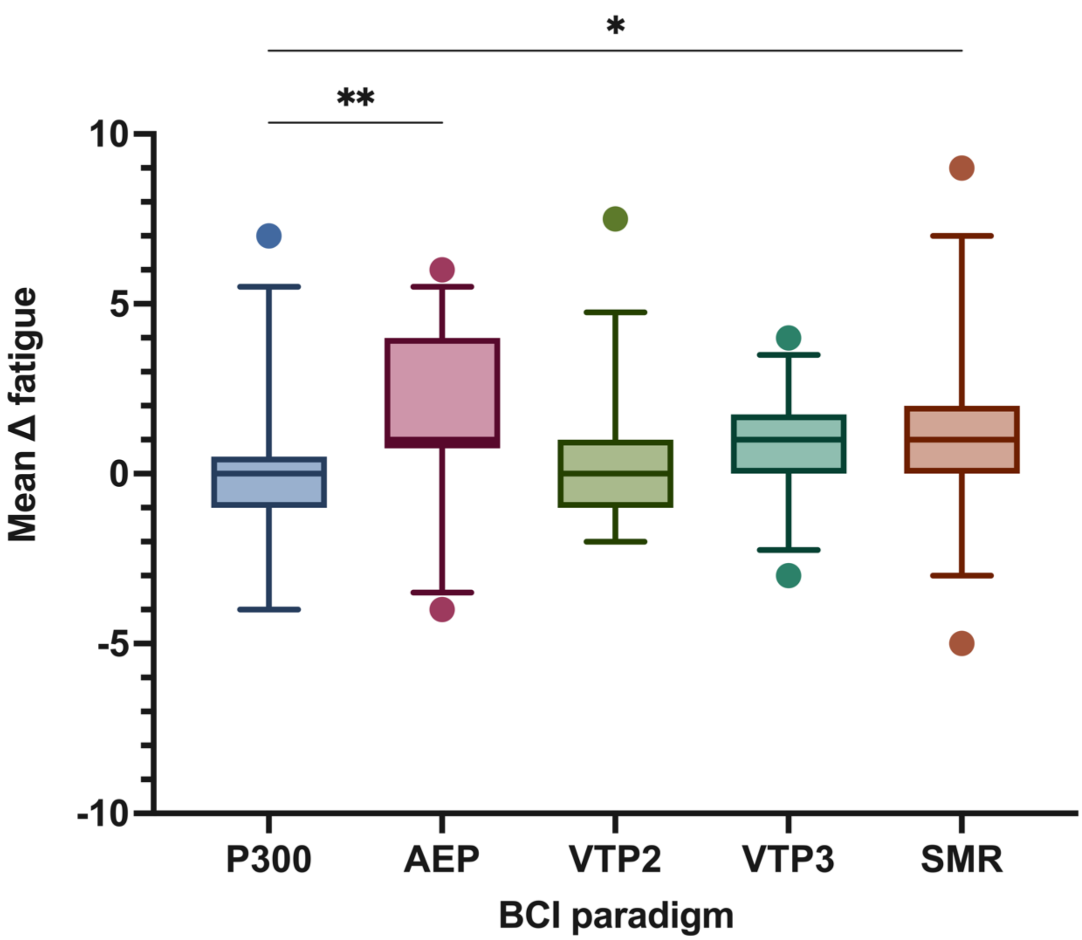
Thirty participants were assessed in the VTP3 paradigm. Overall mean classification accuracy was 73.37±3.5% (range 28-100). Fifteen participants (50%) achieved 100% accuracy in at least one run, with twenty-eight (93%) achieving at least 70% accuracy in at least one run. Average number of artifacts did not significantly change across runs.
Twenty-eight (93%) participants attained the classification accuracy threshold of 60%[151] in at least one run and were assessed in 'communication mode' by responding to ten questions with pre-confirmed answers. On average, the classifier selected the correct answer 36% of the time (SD 0.37; range 1-8) while the incorrect answer was selected in 17% of cases (SD 0.22; range 0-4). The most common selection was an indeterminate response (M=46.8%; SD 0.4; range 0-7).

SMR Paradigm

Thirty participants were assessed in the SMR paradigm. All participants completed one run in assessment mode to generate a classifier based on a generic data template. Overall mean classification accuracy was 55.68±1.5% (range 39-71). Figure 4 shows the ERD plots and cortical activation plots generated by the mindBEAGLE® software for a high-accuracy participant and a low-accuracy participant. Two of the thirty participants (6.67%) were able to achieve 70% accuracy, while most participants (80%) generated a classifier with an accuracy less than 60%. Six participants (20%) attained the classification accuracy threshold of 61%[151] and were assessed in 'communication mode' by responding to ten questions with pre-confirmed answers. On average, the classifier selected the correct answer 72% of the time (SD 0.60; range 6-9) while the incorrect answer was selected in 28% of cases (SD 0.60; range 1-4).

Fatiguability

Average change in fatigue for all BCI paradigms was 0.69 (SD±0.18; range -1.60 to 3.50) (Fig. 2.5). The greatest change in fatigue was observed in the AEP paradigm (1.58±0.4; range -4.0 to 6.0), followed by the SMR paradigm (1.15±0.4; range -5.0 to 9.0). Significant differences in average fatigue were found across the five BCI paradigms ($\chi2(4) = 20.6$, N = 29, $p < 0.001$). The only negative change in fatigue, indicating that participants were less fatigued after completing the task, was observed in the visual P300 spelling paradigm (-0.18 ± 0.4; range -4.0 to 7.0). This differed significantly from fatigue scores for AEP (paired samples t-test: $t_{29} = 4.24$, $p < 0.01$) and SMR (Wilcoxon Signed-Rank Test: W = 248, $p < 0.05$) paradigms, with large effect sizes (Cohen's d = 0.78 and r_rb = 0.65, respectively). The change in fatigue for the VTP2 and VTP3 paradigms were 0.16±0.4 (range -2.0 to 7.5) and 0.68 ± 0.3 (range -3.0 to 4.0), respectively.



P300 = visual P300; AEP = auditory evoked potential; VTP2 = vibrotactile with 2 tactors; VTP3 = vibrotactile with 3 tactors; SMR = sensorimotor rhythm

Figure 2.5. Mean change in fatigue before/after completing tasks on each BCI paradigm. Lines over boxplots indicate groups which were significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * $p<0.05$, ** $p<0.01$.

Motivation

Mean motivation across all BCI paradigms was 4.71 (SD±0.13; range 3.12-5.88) out of 6.0 (Fig. 2.6). Significant differences were observed across the five BCI paradigms ($\chi2(4) = 48.9$, $N = 29$, $p < 0.001$). The highest motivation was in the visual P300 spelling paradigm with an average score of 5.30±0.11 out of 6.0 (range 4.0-6.0). This was higher than motivation scores for the VTP3 (Wilcoxon Signed-Rank Test: $W = 374$, $p < 0.001$, r_rb = 0.840), AEP (paired samples t-test: $t_{29} = 7.35$, $p < 0.001$, Cohen's d = 1.34), and SMR paradigms (paired samples t-test: $t_{29} = 7.24$, $p < 0.001$, Cohen's d = 1.32). Similarly, mean motivation score for the VTP2 paradigm (4.88±0.1; range 2.8-6.0) was higher than the AEP (paired samples t-test: $t_{28} = 4.00$, $p < 0.05$, Cohen's d = 0.744) and SMR paradigm (paired samples t-test: $t_{28} = 3.73$, $p < 0.05$, Cohen's d = 0.693) scores. The average score for the VTP3 paradigm was 4.60±0.2 (range 2.0-6.0). The lowest motivation scores were seen in the AEP and SMR paradigms, with both having average scores of 4.38±0.16.

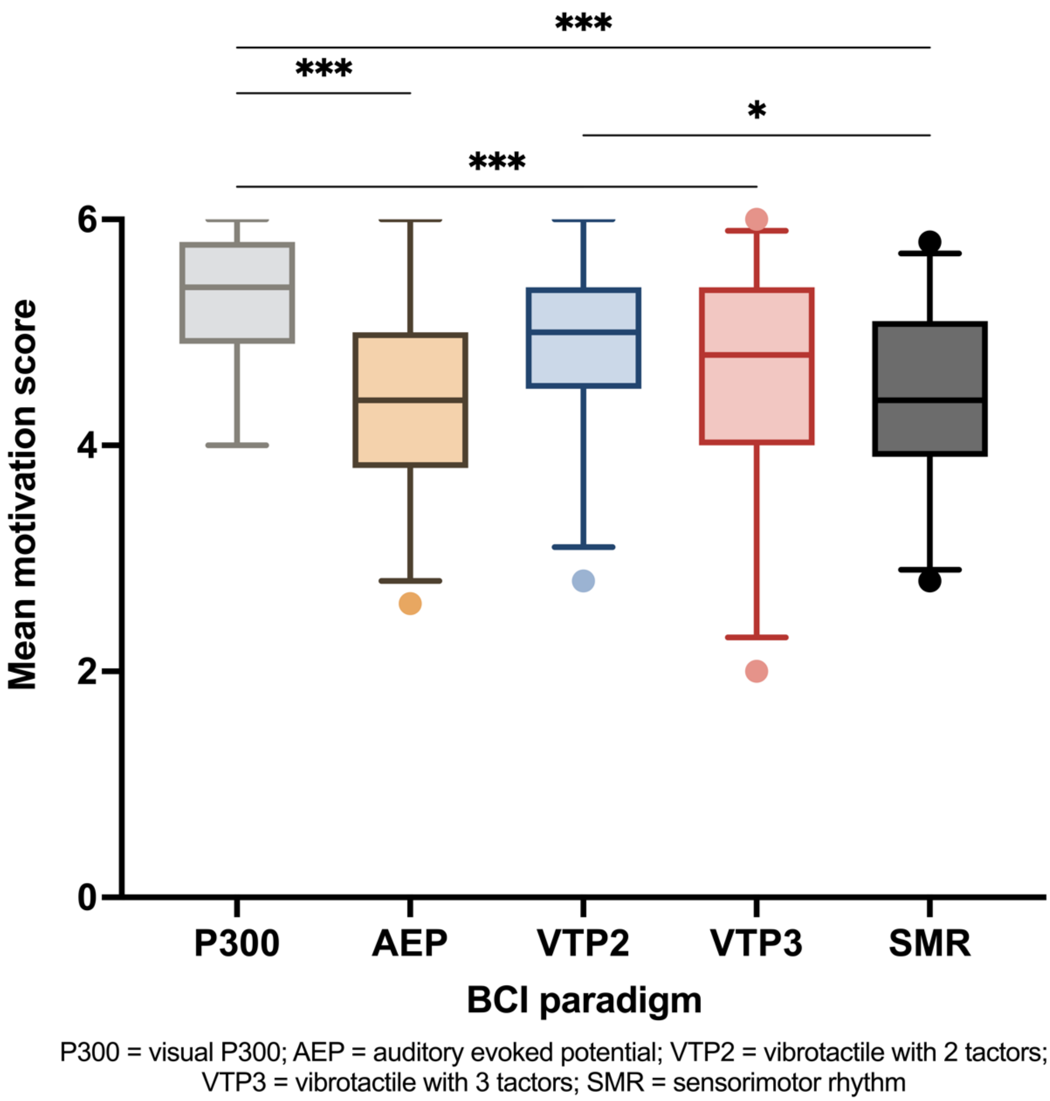


Figure 2.6 Mean motivation score for each BCI paradigm. Higher scores indicate greater levels of overall motivation. Lines over boxplots indicate groups which were significantly different (Friedman Test and Dunn's Multiple Comparison's Tests). * $p<0.05$, *** $p<0.001$.

Tolerability

Average tolerability for all BCI paradigms was 4.67 (SD±0.13, range 3.3-6.0). Tolerability scores for sessions with the SMR/P300 paradigms were higher as compared to the AEP/VTP2/VTP3 sessions (4.78 versus 4.55, $t_{29}=2.43$, $p<0.05$). When ranking their BCI experience among seven common childhood experiences, mean tolerability rank score was 3.67±0.25 with average rankings falling between watching TV (2.75±0.2) and a long car ride (5.22±0.3).

2.4.3 Factors affecting performance

We observed a significant correlation between age and performance on the AEP paradigm ($r_s(28) = 0.56$, $p < 0.001$). Therefore, age was controlled for in subsequent analyses of all other factors affecting AEP performance. Partial correlations revealed that average accuracy on the AEP paradigm was positively correlated with change in fatigue for the AEP paradigm ($r_s(28) = 0.385$, $p < 0.05$). When participants were stratified into age groups (young (6-8 years), middle (9-11 years), young teens (12-14 years), and teens (15-17 years)), a difference in AEP accuracy was seen ($H(3)=11.86$, $p=0.008$) with teen groups achieving higher accuracies. Additionally, we found that children who play musical instruments performed better overall on the visual P300 paradigm ($U=19.00$, $p=0.04$). When evaluating effects of order, we found that children who completed the VTP2 paradigm before the VTP3 paradigm performed worse on the VTP3 paradigm ($U = 139$, $p = 0.045$). Similarly, children who completed the AEP paradigm before the VTP2 paradigm performed worse on the VTP2 paradigm ($U = 139.5$, $p = 0.040$). No associations were observed between sex, handedness, concussion history, motivation, tolerability, video game play, or sports and BCI performance.

2.5 Discussion

This study investigated the performance of thirty typically developing children in five different BCI paradigms. We showed that the P300-based paradigms, including visual, vibrotactile (VTP2 and VTP3), and auditory evoked potentials, demonstrated significantly better performance than the SMR paradigm, with accuracies similar to adult-level BCI competency. Our results suggest that P300-based paradigms, particularly the VTP2 paradigm, may be more suitable for use with children, as they demonstrate higher performance levels compared to the SMR paradigm. This information can be valuable for researchers and clinicians when selecting BCI paradigms for pediatric populations.

Non-invasive BCIs have potential as a rehabilitation and assistive technology for children, yet studies in this area are limited.[46,112] Orlandi and colleagues' 2021 review found only 12 studies involving BCI use in children, with eight using non-invasive BCI.[140] Sample sizes for pediatric studies were small, with half involving less than 10 participants. Typically developing children participated in single sessions lasting 45-60 minutes, while children with disabilities participated in sessions lasting 15-30 minutes. Our study involved 30 participants and two sessions with considerably longer durations, ranging from 50-75 minutes on average. Additionally, our study investigated five BCI paradigms, whereas all previously reported studies only investigated one. The average age of participants in all studies was 13.3±3.2 years, with only four participants under the age of nine. In contrast, our sample size was more than triple that of previous studies, and the mean age of participants in our study was 11.3±3.4 years, with eight participants under the age of nine. The 12 reported studies had a lower percentage of female pediatric participants (42%) than our study, which had a more balanced sample with 53% female participants. This study significantly contributes to the growing body of knowledge on BCI performance in pediatric populations by providing a more comprehensive evaluation of various paradigms, including a larger and more diverse sample of children. Our results encourage further research on BCI applications in pediatric populations, with a focus on exploring the effectiveness of BCI technologies across a broader age range and various use cases.

Studies on non-invasive BCI performance in typically developing children have evaluated evoked potentials and motor-related activities. Evoked potential BCI paradigms are the most successful in adults due to high achievable accuracies with short training time.[73] In children, studies have investigated steady state visual evoked potentials[155,156], motion-onset visual evoked potential[157,158], and visually-evoked P300 potentials.[159] Our study focused only on visual P300. Only one other group has evaluated P300 BCIs in children, but there were many methodological differences between the studies.[159] Our stimuli was presented in a 5x10 matrix using a row/column paradigm for a spelling task, while Vařeka and colleagues utilized a paradigm in which a single number from 1-9 was displayed in sequence in random order for a “Guess The Number” task. Our paradigm also used the ‘face speller’ approach, which has been shown in adults to be optimal in improving P300-BCI performance compared to traditional ‘black and white’ conditions.[148] They did not assess online BCI performance and only reported offline accuracy for single trials, which was approximately 63% using the same classification algorithm as was used in this study (LDA). Our study demonstrated significantly higher classification accuracies compared to Vařeka et al.'s study, with their offline single-trial classification accuracy being similar to our online classification accuracy at 2 flashes per row-column. Evaluating online performance can provide more meaningful insights into the real-world utility of BCIs for children. While Vařeka et al. used only three electrodes at midline positions Fz, Cz, and Pz to reduce setup time, we utilized 8 electrodes over the centroparietal regions. However, our setup time was still minimal, taking only about 5 minutes, and did not cause any discomfort to the participants. These findings suggest that using electrode montages with up to 8 electrodes, matrices with up to 50 characters, and row/column stimulus presentation paradigms with faces and flash rates as low as 3 flashes per row/column can be effective for utility-driven tasks such as spelling in children as young as 6 years old. Additionally, such features are optimal for P300 use by children, as opposed to single character stimuli presentation, which would take too long to spell a word. Given these methodological differences, we highlight the importance of adopting efficient, engaging, and age appropriate BCI paradigms for children. By focusing on visual P300 paradigms, employing engaging task designs, assessing online BCI performance, optimizing electrode montages, and tailoring stimulus presentation methods, future pediatric BCI research can contribute to the development of more effective and user-friendly BCI technologies for children. This, in turn, will enhance the potential of BCIs as powerful tools for rehabilitation and assistive technology for children with disabilities.

In light of the various methodological approaches and their implications, it is clear that there is no one-size-fits-all solution for pediatric BCI research. The variety of methodologies in pediatric BCI research, including our own study, makes direct comparisons challenging. Despite this, they signify progress in adopting and optimizing pediatric BCI studies. To fully optimize BCI in this field, future research should prioritize personalized approaches, considering factors like user physiology, anatomy, psychology, hardware, software, instructions, feedback, and control output.[111,140]

Moving forward, it is essential to strike a balance between controlled experimental conditions and real-world settings to ensure the practical applicability of pediatric BCI research. Another important distinction between this study and the study by Vařeka et al. was the data acquisition environment. In our study, data was collected in a controlled laboratory setting with only the researcher and child (sometimes one family member) present in a quiet room. In contrast, the study by Vařeka et al. collected data from participants in a noisy school setting, which likely contributed to hindered performance, in addition to the aforementioned factors. It is worth noting that the environment in the Vařeka study is more realistic for real-life end-user scenarios compared to the controlled setting in our study. Conducting studies in realistic end-user environments is crucial for clinical translation, which is a limitation of our present study.

As we explore the factors influencing BCI performance in children, it is important to compare other types of paradigms, such as SMR, and examine how these findings align with or differ from related research. Two participants (6.67%) in our study achieved the 70% threshold for BCI competency in the SMR paradigm, while six (26%) were able to achieve the 61% threshold established by g.tec for communication in this paradigm. Only one other study has reported pediatric SMR BCI performance[104], using Cohen's kappa as the metric. The threshold for BCI competency corresponds to a kappa score of 0.4. Zhang et al. (2019) reported an average Cohen's kappa score of 0.46, with 42% of their participants achieving a kappa score of 0.4 or higher. Differences in BCI performance between the present study and the study by Zhang et al. (2019) may be explained by several factors, particularly the characteristics of the BCI systems themselves. Zhang et al. used the wireless, inexpensive, commercially available Emotiv EPOC, whereas we used the wired, clinical-grade g.tec BCI system. The two systems differ in the number, placement, and types of EEG electrodes, as well as amplification, impedance, sampling rate and resolution, size and weight.[160] Moreover, the signal processing and classification methods differ as well, with Emotiv using Probabilistic Neural Network and Radial Basis Function to classify ERDs, while the g.tec mindBEAGLE software uses CSP and LDA.[104] The BCI tasks themselves also differed significantly between the two studies, with Zhang et al. evaluating BCI performance in controlling a remote-controlled car and cursor control using two different thought strategies: goal-oriented thought and motor imagery. Their study found that BCI performance was significantly higher for the remote-controlled car task and the goal-oriented strategy. The motor-imagery based cursor control task had the poorest performance, resulting in an average Cohen's kappa score of 0.32, which falls below the threshold for BCI competency. Similarly, our motor imagery task had poor performance with an accuracy score below the BCI competency threshold. The lack of feedback or output for the user during the 9-minute motor imagery task could have contributed to this poor performance. These results suggest BCI performance in children is highly dependent on the design of the task, feedback, and output, rather than solely on the grade of the BCI system. Both research-grade and consumer-grade systems can yield successful results if the task is engaging and relevant for the child. The SMR paradigm requires sustained attention, concentration, and motivation, and, not surprisingly, appears to be more challenging for children. The presence of movement artifacts suggests that children may lose focus and attention during unengaging BCI tasks. This underlines the need to develop more engaging tasks and provide adequate feedback to maintain children's motivation and concentration and improve performance in SMR-based BCI applications.

By comparing our findings to adult BCI studies, we emphasize the importance of tailoring BCI research for children while drawing insights from adult research. Guger et al. tested the same intendiX® P300 spelling paradigm on 17 healthy adults with the same methodology as the present study.[148] Accuracy remained high (above 90%) until 2 flashes, but decreased to around 80%, and 60% at one flash, which is similar to our findings. Based on our results at 4 and 2 flashes, we predict that performance at 3 flashes would be around 73%. Our study suggests that children can use this P300-BCI speller with flash rates as low as 3 per row/column, which is considered reasonable in modern P300 BCIs.[148] The mindBEAGLE system was validated in 10 healthy adults and produced similar results to our study in healthy children.[161] Chatelle et al. 2018 reported mean classification accuracies of 93% for AEP, 97.5% for VTP2, 74.5% for VTP3, and 56.4% for SMR. The better AEP BCI performance by adults in Chatelle et al. 2018 is not surprising, given our findings of a positive correlation between age and AEP BCI performance. Mean classification accuracies for the VTP3 and SMR paradigms were similar between adults and children, with adults achieving only 1% better accuracies in both paradigms. However, only 80% of adults achieved the communication threshold for the VTP3 paradigm, while 93% of children in our study achieved the required threshold. Similarly, 20% of adults reached the threshold for communication mode in the SMR, which was the same for pediatric participants in our study. These results demonstrate that typically developing children can achieve the same level of BCI competency as healthy adults. Nonetheless, Chatelle et al. 2018 concluded that the accuracy of the paradigms should be improved, and the duration of the protocol should be shortened before the system is ready for clinical implementation. This conclusion is particularly relevant for children.

BCI performance variability is a major challenge in research, but there is no consensus on the specific factors responsible for it. We investigated potential factors and found no associations with BCI performance, except for age in the AEP paradigm. This is expected, as previous studies have demonstrated age-related differences in EEG responses to auditory stimuli, including full maturation of auditory evoked potentials at 16 years of age.[162] In the study by Vařeka et al. (2020), a notable finding was the significant variability of P300 components in pediatric EEG signals.[159] Surprisingly, this association was not present in other P300 paradigms, even though the P300 is a well-known feature associated with executive functions such as attention and inhibition, and has been used to study functional brain development over adolescence.[163] Additionally, we were surprised to not find an association between age and BCI performance on the SMR BCI paradigm. This is unexpected given that children's abilities to perform motor imagery develop in parallel with cognitive processes underlying motor representations between the ages of 7 and 12.[74] Previous research has shown that this progressive improvement in motor imagery performance is similar to adults after 10 years of age.[164] In contrast, Zhang et al. (2019) found a significant correlation between age and SMR BCI performance. The findings suggest that there is significant variability in pediatric BCI performance, and while age appears to be a factor in some paradigms such as the AEP, it is not consistently associated with performance in other paradigms such as the P300 and SMR, highlighting the need for further research to better understand and address the challenges of BCI performance variability in pediatric populations.

The differences in performance between the SMR and P300-based paradigms could potentially be attributed to the varying levels of engagement these paradigms offer to the young users. The SMR paradigm, which requires sustained attention, concentration, and motivation, seems to pose a challenge for children, possibly due to its lack of immediate feedback or output. This draws attention to the necessity of creating more engaging tasks and providing adequate feedback to keep children motivated and concentrated. Regardless of the system grade, whether research-grade or consumer-grade, we may attain successful results if we ensure that the BCI tasks are relevant and engaging for children. In contrast, in the P300 paradigm, we observed a high accuracy level, similar to that found in adult studies, even with flash rates as low as 3 per row/column. This suggests that P300 paradigms could be more conducive to children's BCI performance, making it a promising avenue for further exploration. The threshold of 70% for BCI competency is meaningful since it provides a benchmark for assessing performance. However, given the varied performance in different tasks and the overall lower scores in tasks involving motor imagery, it suggests that more factors, such as task design, the type of feedback provided, and individual children's attention span and motivation, can play a significant role in determining successful BCI use.

Our findings have practical implications for BCI use in children, where studies are limited. We have provided the first comprehensive description of baseline BCI performance in typically developing children across different age groups and BCI paradigms. Additionally, our study has demonstrated the validity of using two commercial EEG-BCI systems in typically developing children, showing that they are capable of controlling advanced BCIs with accuracy similar to that of adults. These promising findings suggest that BCI technology has the potential to be a valuable tool for millions of children with disabilities worldwide.

This study also lays a solid foundation for future pediatric BCI research, providing considerable opportunities to improve BCI performance in children by modifying various parameters to make them more child-friendly and engaging. However, to realize the clinical implementation of BCI technology in pediatric patients and real-world settings, future research should further validate these systems.

2.5.1 Limitations

The modest sample size in our study was informed but also a potential limitation. We recruited 30 participants, which may have reduced our ability to detect more subtle differences, though notably this sample is larger than most in the literature. Our sample was also restricted to a particular age group (6-18) and geographic region (Calgary, Alberta), which may have introduced sampling biases. Future research with larger and more diverse samples is necessary to confirm or refute our findings. The limitations of our study are further associated with the controlled laboratory environment in which our data was collected. Therefore, it is necessary to exercise caution when generalizing the findings to other settings.

Various factors such as noise, lighting, the presence of people, and familiarity with the environment could potentially impact the data quality and participant performance. Thus, the generalizability of the results should be interpreted with caution, and future research should aim to address these limitations by collecting data in more diverse settings that are reflective of a true end-user environment.

2.6 Conclusion

We demonstrated the effectiveness of using evoked potential BCI paradigms in children, as young as 6 years old, for utility-driven tasks such as spelling in a controlled laboratory setting. While conducting studies in realistic end-user environments is crucial for clinical translation, this study adds to the limited body of knowledge on non-invasive BCI performance in children. It also provides important insights into factors that may affect BCI performance in this population.

Further research is necessary to explore the potential of BCIs as rehabilitation and assistive technology for children with disabilities.

Chapter 3: Exploring the Impact of Gamification on BCI Performance in Children: The Case for Personalization

Dion Kelly, Brian Irvine, Eli Kinney-Lang, Daniel Comaduran Marquez, Erica D. Floreani, Adam Kirton

3 Chapter 3

3.1 Abstract

A major challenge with BCI use is the requirement for subject-specific calibration, which is often tedious and unengaging, but necessary to improve performance. This is especially true for children, whose limited attention and motivation may restrict the duration of endurable calibration periods. Games present clear objectives with short-term goals that confer rewards and act as external motivators to provide a sense of advancement. Several studies have shown that the addition of scoring systems, prizes, and awards to tasks, a process known as "gamification", can increase motivation, attention, and task performance in children. This randomized, prospective, cross-over study aimed to address this challenge by comparing the effects of gamified versus non-gamified calibration environments on classification accuracy and BCI performance on utility-driven tasks.

Thirty-two typically developing children (14 female, mean age 11.9 years, range 5.8-17.9) attended two sessions lasting between 1.5-2 hours, to perform two standard paradigms: spelling using visual P300 event-related potentials (P300) and cursor control using sensorimotor rhythm (SMR) modulation, following gamified and non-gamified calibration. Gamified paradigms incorporated elements of game design, such as meaningful stories, quests, points and sounds. The primary outcome was BCI performance, which included performance of the classification model and online accuracy. Motivation, tolerability, and mental workload (NASA-TLX) were evaluated following each paradigm.

For the P300 paradigm, mean classification accuracy was similar after gamified (96.81±3.5%) and non-gamified (96.52±2.4%) calibration. Mean classification accuracy for the SMR paradigm was 61.81±13.4% with gamification and 59.84±11.4% without gamification (n.s.). Mean online accuracy for SMR cursor control was 63.2% for both conditions. For the P300 spelling task, online performance was significantly lower following gamified training ($p < 0.01$). Motivation and tolerability scores across all four paradigms were high. There were no significant differences found between classification accuracy, online BCI performance, motivation, tolerability, or perceived mental workload.

Our results reinforce the ability of typical children to control advanced BCI systems with performance comparable to adults. Gamified calibration environments may not enhance BCI classification and performance in children though the gamified environments utilized in this

study may not have been engaging enough. To our knowledge, this is the first study that investigates the effects of gamified calibration paradigms on classification accuracy and BCI performance in children. Future research on optimizing BCI training paradigms for children is necessary to facilitate implementation of BCI in pediatric populations.

3.2 Introduction

Brain-computer interfaces (BCIs) have gained significant attention in recent years due to their potential to transform the lives of individuals with significant disabilities, enabling them to interact with their surroundings and communicate.[72] BCIs function by detecting and translating brain signals into actionable commands, bypassing the need for physical interaction. Despite the promising advancements in BCI technology, a major challenge remains in the need for subject-specific calibration. This training is essential to ensure that the BCI system is tailored to the unique brain patterns and cognitive processes of each user, enabling efficient and accurate command execution. However, calibration can be a lengthy and tedious process, often requiring users to engage in repetitive tasks that may lead to decreased motivation and attention.[109] This challenge is further exacerbated in pediatric populations, who typically have shorter attention spans and may find traditional BCI training protocols unengaging. As such, addressing the challenges of calibration is critical in optimizing BCI use and unlocking its full potential for various populations, including children who have been neglected from BCI research.

Implementing BCIs for children presents a unique set of challenges. Firstly, the pediatric brain is constantly undergoing development, with evolving cortical organization and changing neurophysiology.[165] Much of this occurs in the face of major brain injuries early in life in target clinical populations such as youth with quadriplegic cerebral palsy. This complex neural plasticity complicates the process of designing and adapting BCI paradigms, as the algorithms and techniques optimized for adult brains may not be directly transferable to children.[140] Additionally, children often have shorter attention spans and may require more engaging and motivating tasks to maintain their focus throughout the BCI training sessions. Thus, child friendly BCI paradigms must be developed that effectively capture and sustain user interest. Furthermore, children with disabilities who could immensely benefit from BCIs, may experience unique cognitive or sensory impairments that require specialized approaches to BCI design and training.[166] Addressing these challenges is crucial to harness the potential of BCI technology to enhance the lives of children with disabilities and enabling them to achieve a greater degree of autonomy and independence.[167]

Gamification represents one promising solution to improve BCI systems for pediatric users by making the training process more engaging and enjoyable. By incorporating elements of game design, such as points, quests, and meaningful stories, gamification can help maintain interest and motivation throughout training sessions.[168] This increased engagement can potentially lead to better focus and sustained attention, which are crucial for effective BCI use. Moreover, gamification can create a sense of progression and achievement, further enhancing the intrinsic motivation of the child.[168] By catering to the unique needs of children, gamified BCI systems can potentially reduce the barriers associated with traditional BCI training paradigms and improve overall performance. As a result, gamification could play an important role in bridging the gap between the potential benefits of BCI technology and its successful implementation for pediatric populations, particularly for those with motor impairments or communication difficulties.

We have shown that both typically developing children[104,169] and children with cerebral palsy[103,132,170] can learn to control EEG-based BCI systems using standard calibration paradigms. The aim of this study was to investigate the effect of gamified calibration on BCI performance on two common BCI control paradigms, the P300 event-related potential and sensorimotor rhythms (SMR), in typically developing children. We hypothesized that gamified calibration would lead to improved performance in an online BCI classification task.

3.3 Methods

3.3.1 Participants

To ensure that our study was adequately powered to detect the effect of gamified calibration on BCI performance in children, we conducted an a priori power analysis using G*Power software (version 3.1.9.4). Based on pilot study results[171], we estimated an effect size of $f \geq 0.4$ for the interaction between calibration condition (gamified vs. non-gamified) and paradigm (P300 and SMR) in a 2x2 factorial within-subjects design. We set the desired statistical power ($1-\beta$) at 0.80 and the significance level (α) at 0.05 for a two-tailed test. The power analysis indicated that a total sample size of $N = 31$ participants would be required.

Participants were recruited from the community via a database of typically developing children whose families have volunteered to participate in medical research (Healthy Infants and

Children's Clinical Research Program).[144] Inclusion criteria were typical neurodevelopment with no neurological conditions, age 6 to 18 years, and informed consent/assent. Methods were approved by the Conjoint Health Research Ethics Board at the University of Calgary (REB21-1883). Participants received a $25 gift card for volunteering.

3.3.2 Protocol

We conducted a prospective, randomized, cross-over study to investigate the effects of gamified calibration in comparison to non-gamified calibration on two different paradigms: P300 and SMR. We employed a 2x2 factorial within-subjects design, with the factors being the paradigm (P300 vs. SMR) and the calibration condition (gamified vs. non-gamified). To ensure a balanced allocation of participants across the different conditions and to control for potential order effects, we utilized a Latin square design. Participants were randomly assigned to one of four conditions using an online randomization tool (www.randomlists.com). This resulted in four groups of eight participants each (total N = 32), with each group experiencing a unique sequence of the four combinations of paradigms and conditions. Participants attended two sessions lasting approximately 1.5 hours each. The primary outcome was classification accuracy, with secondary outcomes including precision and recall scores of the classification model from calibration, and the online accuracy achieved in subsequent BCI tasks.

Each visit had a similar protocol: 10-minute set up with a preliminary assessment questionnaire to record potential factors affecting performance (concussion, vision problems/corrected vision, mood, previous night's sleep, tiredness, exercise, food, alcohol/drug consumption including caffeine, and hobbies/sports), and the adapted Edinburgh handedness inventory.[152] At the start of the first visit, participants also completed three tests from the CNS Vital Signs computerized neurocognitive test battery to evaluation attention and working memory: the Stroop Test, the Shifting Attention Test, and the 4-part Continuous Performance Test.[172] At the start of the second session, participants completed two N-back tasks (1-back and 2-back) to evaluate working memory and visuospatial memory. The results from these neurocognitive tests will be reported in a separate paper.

Before beginning the BCI paradigms, the participants were instructed to sit still, with feet flat on the floor and hands/arms either resting on their lap or on the table in front of them. Participants were then evaluated on two classifier training environments for each paradigm: one gamified and one non-gamified. Afterward, they were assessed on their BCI performance on a "utility driven" BCI task: spelling for the P300 paradigm and cursor control for the SMR paradigm. After each paradigm, motivation, tolerability, and workload were evaluated with a series of questions.

The performance of the classification model was evaluated by calculating the metric "accuracy", which represents the proportion of correct predictions among all predictions made. However, to provide a comprehensive assessment of the model's performance and ensure a balanced evaluation, we also reported precision and recall values alongside accuracy. Precision measured how many of the positive predictions made by the model were correct relative to the total number of positive predictions and serves as a measure of the model's ability to correctly identify positive instances without generating false positives. Recall measured how many of the true positive values in the test data were identified by the model relative to the total number of true values and helps us assess the model's sensitivity and its effectiveness in detecting all positive cases.

3.3.3 Experimental Setup and BCI System

A research-grade EEG-based BCI system was utilized for signal acquisition and processing (g.tec medical engineering GmbH, Schiedlberg Austria). Gamified and non-gamified classifier training paradigms were generated in-house.

EEG signals were acquired using the gSCARABEO active wet electrodes and amplified by the gUSB amplifier (g.tec medical engineering GmbH, Austria). Montages for SMR and P300 paradigms are specified in Table 3.1. All signals were sampled at 256Hz. Before initiating the BCI session, impedance was assessed to be <5 kΩ for optimal signal quality and to minimize artifacts.

Table 3.1. Channel Configurations for P300 and SMR BCI Paradigms.

Paradigm	# Channels	10-20 Channel Locations
P300	8	Fz, Cz, P3, Pz, P4, PO7, Oz, PO8
MI	16	FC3, FCz, FC4, C5, C3, C1, CZ, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, Pz

3.3.4 BCI Paradigms

One standard calibration scene, one gamified calibration scene, and one task scene were developed for each P300 and MI (Fig. 3.1). For the gamified calibration scenes, we incorporated game elements such as a storyline and quest, scoring system, and audio-visual features like characters, background music, and sound effects associated with points.

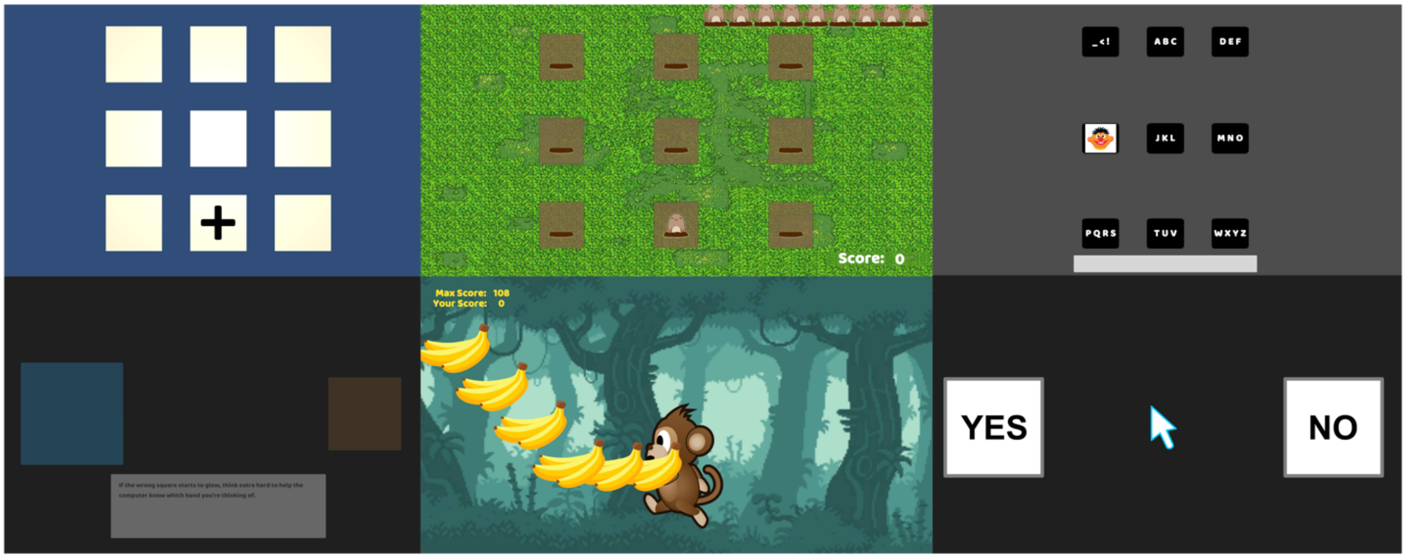


Figure 3.1. Calibration and user task scenes. Top: P300 scenes – standard calibration scene (left), gamified calibration scene (Mole Patrol game, middle), T9 speller scene for spelling task (right). Bottom: MI scenes – standard calibration scene (left), gamified calibration scene (Banana Dash game, middle), cursor control scene for yes/no response task (right).

P300 Paradigm

Design

A pseudo-random single character flashing design was chosen for the P300 paradigm due to its numerous reported advantages over the traditional row-column design, including mitigating adjacency-distraction effects and double-flashing errors.[173] A 3x3 matrix (9 potential targets) was used for both calibration conditions and the spelling task, chosen to minimize total eye movement and user fatigue.[173]

The familiar face paradigm is known to elicit stronger ERP responses in adults, leading to faster target selection and higher accuracy, in addition to increased potential for enhanced user engagement.[174] These factors are crucial for optimizing BCIs for children with limited attention spans. Rezeika et al. (2018) suggested that using culturally well-known faces can lead to high and consistent effects across individuals[173]; thus, we incorporated popular children's cartoon characters. Cells flashed one at a time, which means that only one key changes during each flash to display a cartoon character's face in each flashing cell. Flashes lasted 100ms, with an inter-stimulus interval of 75ms.

Calibration

In the non-gamified calibration task, a target square was identified with a "+". Users were instructed to pay attention to that square and silently count the number of times a character flashed, until the word "done" appeared on the screen, signaling the end of the training.
For the gamified P300-based BCI training, titled "Mole Patrol" (similar to whack-a-mole), participants were instructed to use their brain power to "whack" moles appearing from different holes. To achieve this, they had to silently count the number of times the mole appeared in the target hole, accompanied by a distinct sound, while ignoring other cartoon characters who were also attempting to catch the mole. Participants were guided through the game interface, with the number of moles "whacked" and the number of remaining moles displayed in the top left corner, and the participant's score shown in the bottom right corner.

After completing a run, participants were required to enter the number of times the mole appeared in its hole. Correctly counting the mole's appearances earned them 100 points. The farther they were from the correct count, the fewer points they received for each mole. The highest possible score was 900 (9 runs x 100 possible points per run).

For both paradigms, each run consisted of 9 trials with a random number of flashes ranging from 10-15, which represented the number of times the mole emerged from its hole or the number of "flashes". The total calibration time was approximately 5 minutes. Once the classification model was trained, a confusion matrix was provided alongside percentages for the performance of the model, including the accuracy, precision, and recall.

Spelling Task

Using the classifier that was trained in the calibration phase, participants then completed a free spelling task using a two-stage T9 speller. Participants were required to write a five-letter word beginning at 10 flashes per key and decreasing by increments of two flashes for each subsequent word until two flashes per key, and a final word at 1 flash per key. Target words were different for each flash rate to mitigate potential learning effects. Thus, participants spelled a total of seven different five letter words. They were instructed to employ the same method as in the calibration task, by focusing on the key containing their target letter and silently counting the number of times the cartoon character flashed on the key. Each five-letter word required ten actions to spell - two actions (selecting a key and confirming the choice) for each of the five letters.

The accuracy of the task was calculated at each flash rate as the ratio of the correct number of selections (correctly chosen and confirmed keys) to the total number of selections (ten actions per each five-letter word).

Participants were tested at 10 and 8 flashes. If their online accuracy dropped below 40% on two subsequent attempts, the task was discontinued to avoid causing frustration and disappointment.

SMR Paradigm

Design

For the SMR task, a binary class paradigm of left- and right-hand motor imagery was chosen. The training protocol was based off the Graz training paradigm, which is the standard training approach for motor imagery-based BCI[175]. It involves the imagined movement of specific motor tasks, such as moving the left or right hand, while EEG signals are recorded and processed. After the first pair of imagined actions, real-time feedback was provided to users based on a classifier trained on the completed sets of imagined actions. The duration of the training protocol was 5.67 minutes with the following parameters: window length=1.5s; number of training windows=6; pause before training=2s; number of training selections=20; pause after training=2s; train break=4s.

Calibration

During the non-gamified calibration task, participants were instructed to concentrate on two gray squares presented on the screen. They were asked to imagine opening and closing the hand corresponding to the side with the larger square. The participants received feedback after one trial per side. The feedback was provided by making the square corresponding to the classified side more opaque. This real-time feedback encouraged participants to think more intently about the imagined hand movement and to continue imagining the movement until instructed to relax.

The gamified task, titled "Banana Dash," aimed to engage participants in teaching baby Moe, a virtual monkey character, how to catch falling bananas using their imagined hand movements. Participants were instructed to transfer their brain power to baby Moe by imagining grabbing the bananas with their left or right hand, depending on which side the bananas were falling. The task was designed to resemble a learning process, with Moe initially observing the participant's imagined actions before attempting to catch the bananas himself.

During the first two trials, Moe did not run towards the bananas, as he was simply observing the participant's thought patterns. After these initial trials, Moe would attempt to catch the bananas based on the participant's imagined hand movements. If Moe started moving in the wrong direction, participants were instructed to concentrate more intensely on imagining the correct hand movement to guide Moe towards the bananas.

Participants were encouraged to remain focused on consistently imagining the hand movements throughout the task and to keep imagining until Moe signaled them to relax. The objective of the game was to help Moe catch as many bananas as possible to achieve the maximum score. The gamified task aimed to maintain participant engagement and motivation while collecting motor imagery-based BCI data.

Cursor Control Task

The online task for the SMR paradigm was cursor control on a one-dimensional plane (horizontal) to answer yes/no questions by imagining left-hand and right-hand movements. The cursor control task entailed responding to ten yes/no questions with pre-confirmed answers. Participants were instructed to imagine left-hand or right-hand movements, which moved the cursor in the respective direction to indicate a 'yes' or 'no' response. Accuracy was calculated by dividing the number of correct responses by the total number of questions.

Signal Processing and Classification

The classification steps for both P300 and SMR were identical in the standard and gamified calibration processes.

P300 Processing

After each visual stimulus the following 600ms of EEG data were recorded. Each 600ms epoch was filtered with a 5th order Butterworth bandpass filter with a passband of 0.1-15 Hz. Once the stimulus presentation for a single round was finished, the epochs for each object were ensemble-averaged, yielding one 600ms averaged epoch per object. These epochs were then saved with binary labels indicating whether the respective object was the target or not. This was repeated for each of the nine rounds of flashing. These epochs were then used to estimate the XDawn filtered ERP covariance matrices. ERP covariance matrices were mapped to their respective tangent space representations. This resulted in nine feature sets with the target label and 72 feature sets with the non-target label. To overcome this class imbalance, oversampling was used to oversample feature sets of the target class to match the non-target class. Feature sets were used to train a shrinkage linear discriminant analysis (sLDA) classifier using 5-fold cross validation. Reported classification accuracy did not include the oversampled feature sets used for training. This XDawn and sLDA were selected based on recommendations from Lotte et al.[176] for state-of-the-art BCI pipelines with small datasets. For the online spelling task, the trained classifier was used to evaluate each object's posterior probability of belonging to the target class. The object with the greatest posterior probability was selected as the user's chosen target.

SMR Processing

Each 9-second imagined action was segmented into 6 epochs of 1.5 seconds each. These epochs were filtered using a 5th order Butterworth bandpass filter with a passband range of 5-30 Hz. The covariance matrix was then calculated for each epoch. The covariance matrices were subsequently mapped to their corresponding tangent space representations, producing a feature set with a length equal to the number of channels. Logistic regression was employed for classification of the feature sets, following an approach similar to Barachant et al., but substituting logistic regression in place of LDA.[177]

An iterative classification approach was chosen over a static one to provide real-time feedback. This feedback allowed users to adjust their mental strategies and enhanced the learning experience by offering more opportunities for improvement. A classifier was trained after the first two imagined actions, and updated after every subsequent pair of imagined actions, utilizing all available training data. Finally, a classifier was trained using all calibration data once it became available.

3.3.4.1 Questionnaires

Motivation was assessed using a shortened version of the Pediatric Motivation Scale (PMOT), which measures motivation as an event-based state from a child's perspective.[153] Workload was assessed using the child-adapted NASA Task Load Index (NASA-TLX), which measures subjective mental workload.[178] The PMOT was developed for children as an event-based measure of motivation (rather than motivation as a trait) following activities. It contains 21 items which are divided into six subscales to evaluate subjective feelings of effort/importance, relatedness, autonomy, interest/enjoyment, competence, value/usefulness and open-ended items. Children responded to five items from the interest/enjoyment, effort/importance, and competence subscales using a 6-point ordinal face scale, where 1 was "not true at all" and 6 was "definitely true". Tolerability was assessed at the end of each session with a questionnaire using the same 6-point ordinal face scale. To promote valid responses throughout the scale, some items were framed negatively and were reverse scored. Children reported their perceived fatigue on a scale of 1 to 10 at the start of each session and at the end of each paradigm. To quantify the level of fatigue attributed to each paradigm, change in fatigue was calculated by subtracting the reported fatigue recorded after each paradigm from the reported fatigue recorded immediately before each paradigm.

3.3.4.2 Data Analysis

Statistical analyses were performed using SPSS Statistics version 28.0 (IBM, USA), and GraphPad Prism version 9.0.0 (GraphPad Software, USA) was utilized for further statistical evaluations and the generation of graphs. Accuracy scores were averaged to obtain a mean score for each paradigm. Responses from questionnaires were transformed into corresponding numerical values. Outcomes were tested for normality using the Shapiro-Wilk test and parametric or nonparametric tests were used as appropriate.

To analyze the differences in accuracy across various flash rates for the P300 paradigm, we conducted a Friedman two-way analysis of variance (ANOVA) by ranks followed by pairwise comparisons using Dunn's Multiple Comparison's test. To report the overall online BCI accuracy for the P300 spelling paradigm, we computed the average accuracy score for runs within the stable range of flash rates where accuracy was not significantly impacted. Performance, encompassing classification model metrics (accuracy, precision, and recall) and online accuracy for utility-driven tasks, was compared between gamified and non-gamified calibration using Wilcoxon signed-rank tests. Differences in factors affecting performance, including motivation, tolerability, workload, and fatigue were analyzed using Wilcoxon sign-rank tests. Participants were stratified into four distinct age groups for the purpose of evaluating differences between ages. These groups were defined as follows: 'young' (6-8 years), 'middle' (9-11 years), 'young teens' (12-14 years), and 'teens' (15-17 years). A Kruskal-Wallis test was subsequently performed to evaluate potential differences in accuracy across these age groups. Bonferroni-Dunn corrections were applied to control for Type I errors in multiple comparisons.

A proficiency threshold for the SMR paradigm was set at 74%, which is the upper confidence limit of chance results for a 2-class classification model with 10 trials/class at α=5%.[151,179] This means that a classification accuracy higher than 74% would be considered statistically significant, indicating that the BCI performance is better than random.

3.4 Results

3.4.1 Population

The final population consisted of thirty-two typically developing children (14 female, mean 11.9 ± 3.9 years, range 5.8–17.9, 91% right-handed). The recruitment success rate was approximately 27%. There were no serious adverse events. Sessions averaged 60-90 minutes with the longest being 120 minutes. One participant dropped out after completing their first session. The participant who withdrew became discouraged and upset after comparing their performance to that of their sibling and chose not to continue with the study. As a result, the final sample size was reduced to 31. For the P300 spelling task, one participant did not meet the threshold for task continuation (accuracy >40%) in the first two runs (10 flashes and 8 flashes) and had a low recall score for their classification model. Given their low recall score, it was unlikely that their performance would improve in subsequent more difficult runs with lower flash rates. To prevent further frustration, the task was discontinued for this participant after the first two runs. Consequently, their data was not included in the calculation of the overall accuracy for the P300 spelling task.

3.4.2 Effect of gamification on BCI performance

3.4.2.1 Classification models

Mean performance metrics for the classification models (accuracy, precision, recall) are summarized in Table 3.2. The Wilcoxon signed-rank tests revealed no significant differences in classification model performance between the gamified and non-gamified calibration conditions for either paradigm for all three metrics: P300 accuracy ($W = -50.0$, $p = 0.58$); P300 precision ($W = 17.0$, $p = 0.17$); P300 recall ($W = -67.0$, $p = 0.43$); SMR accuracy ($W = 110.0$, $p = 0.24$); SMR precision ($W = 121.0$, $p = 0.20$); SMR recall ($W = 38.0$, $p = 0.67$).
In the gamified SMR calibration, 7 participants (22.58%) reached the 74% accuracy threshold, compared to 6 participants (18.75%) in the non-gamified calibration.

Table 3.2. Classification model performance metrics (accuracy, precision, and recall) for each paradigm (P300 and SMR) for each condition (non-gamified and gamified).

Paradigm	Metric	Mean	Std. Deviation	Range
P300 (non-gamified)	Accuracy	96.81%	3.5%	89-100%
	Precision	94.61%	14.3%	50-100%
	Recall	72.42%	28.2%	11-100%
P300 (gamified)	Accuracy	96.52%	2.4%	93-100%
	Precision	99.00%	3.9%	83-100%
	Recall	68.81%	20.8%	33-100%
SMR (non-gamified)	Accuracy	59.84%	11.4%	44-91%
	Precision	59.61%	11.3%	45-92%
	Recall	59.61%	12.9%	40-90%
SMR (gamified)	Accuracy	61.81%	13.4%	43-96%
	Precision	61.94%	13.6%	42-98%
	Recall	60.32%	15.1%	40-95%

3.4.2.2 Effect of gamified calibration on subsequent performance on utility-driven tasks

P300 Spelling

Figure 3.2 illustrates the mean online accuracy in the P300 spelling task across various flash rates, following both non-gamified and gamified calibration. After non-gamified calibration, online accuracy scores ranged from 82.19±17.7% at 10 flashes (n=32, range: 30-100%) to 38.89±19.1% at 1 flash (n=27, range: 0-90%). Conversely, after gamified calibration, scores varied from 74.84±24.9% at 10 flashes (n=31, range: 10-100%) to 42.27±21.1% at 1 flash (n=22, range: 10-90%). Significant differences in mean accuracy were observed at different flash rates, both after non-gamified ($\chi^2(6)=94.62$, n=27, $p<0.0001$) and gamified calibration ($\chi^2(6)=66.61$, n=22, $p<0.0001$). The flash rate's reduction, which amplified task difficulty, led to significant accuracy differences, especially noticeable when the flash rate fell below 4 (Table 3.3).

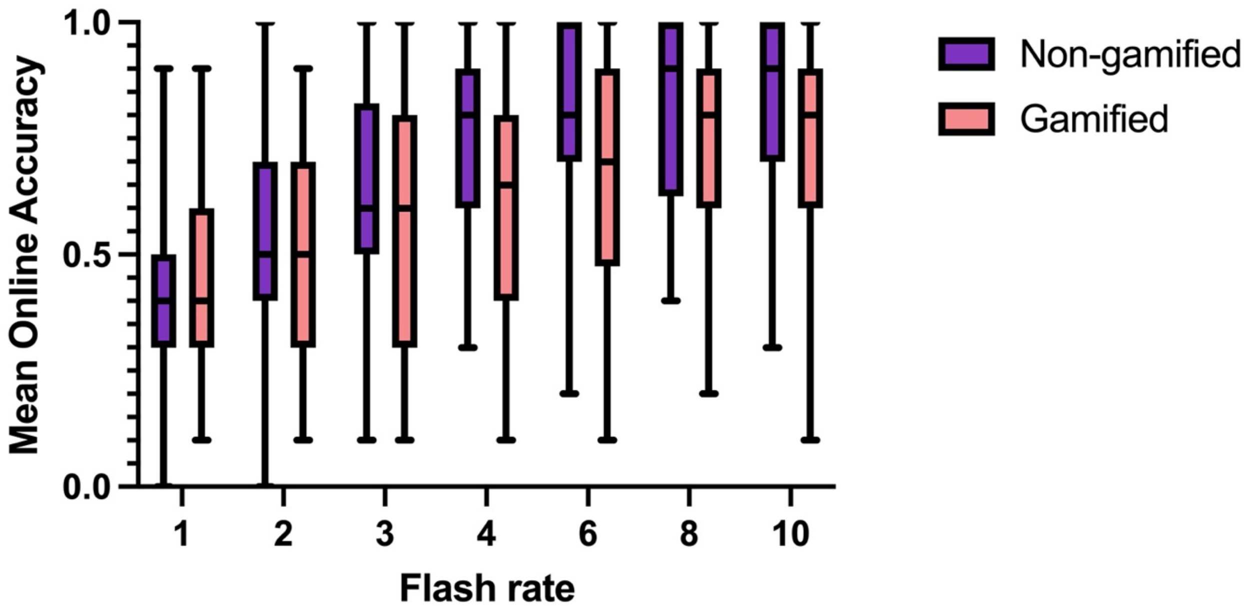


Figure 3.2. Mean online accuracy for P300 spelling at different flash rates following non-gamified and gamified calibration.

Table 3.3. Dunn's Multiple Comparison's test of accuracy between different flash rates for non-gamified and gamified conditions.

Flash rate	**Non-gamified**	**Gamified**
1 vs 3	p=0.034	n.s.
1 vs 4	p<0.001	p<0.001
1 vs 6	p<0.001	p<0.001
1 vs 8	p<0.001	p<0.001

1 vs 10	p<0.001	p<0.001
2 vs 4	p=0.018	p=0.020
2 vs 6	p<0.001	p<0.001
2 vs 8	p<0.001	p<0.001
2 vs 10	p<0.001	p<0.001
3 vs 6	p=0.018	n.s.
3 vs 8	p=0.009	p=0.022
3 vs 10	p=0.004	p=0.007

Table 3.4 summarizes the mean online classification accuracy for both BCI tasks, P300 spelling and SMR cursor control, following gamified and non-gamified calibration. Mean online classification accuracy for the P300 spelling following non-gamified calibration group was 80.47±17.1%, with a range from 45-100%, while accuracy following gamified calibration was 71.47±19.9%, with a range from 18-100%. This median decrease in accuracy of 10% for the gamified condition was significantly lower compared to the non-gamified condition (W=-238.0, adjusted p=0.017). Mean online classification accuracy for SMR cursor control follow non-gamified training was 63.23±16.0%, with a range from 40-100%, while mean accuracy following gamified calibration was 63.23±18.9%, with a range from 30-100%, which was not significantly different (W=6.000, p=0.90) (Fig. 3.3).

Table 3.4. Mean online accuracy for each utility driven task (P300 spelling and SMR cursor control) using the classification model generated in each condition (standard and gamified).

Task (calibration condition)	**Mean online classification accuracy**	**Std. Deviation**	**Range**
P300 Spelling			
Non-gamified	80.47%	17.1%	45-100%
Gamified	71.47%	19.9%	18-100%
SMR Cursor Control			
Non-gamified	63.23%	16.0%	40-100%
Gamified	63.23%	18.9%	30-100%

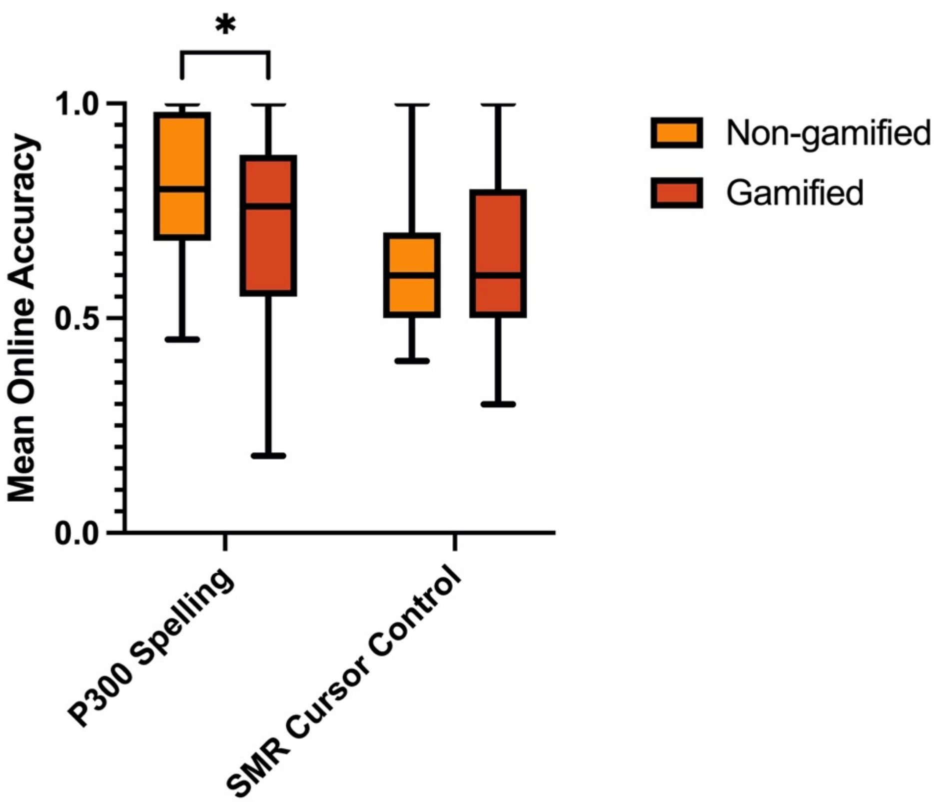


Figure 3.3. Mean Online Accuracy for P300 Spelling and SMR Cursor Control Paradigms following Non-gamified and Gamified Calibration.

3.4.3 Factors Affecting Performance

Age

No significant differences in accuracy were found for non-gamified calibration in the SMR paradigm, or for gamified calibration in either paradigm. Conversely, differences in accuracy were observed between age groups for non-gamified P300 calibration ($H(3)=9.35$, $p=0.025$), with teens achieving significantly higher accuracy compared to young children ($p=0.026$).

Motivation

Motivation scores for the P300 paradigms were 5.03±0.8 (range: 3.22-6.00) and 4.91±0.7 (range: 3.44-6.00) for non-gamified and gamified conditions, respectively. For the SMR paradigms, scores were 5.02±0.6 (range: 3.11-6.00) and 4.96±0.6 (range: 3.33-6.00), for non-gamified and gamified conditions, respectively. No significant differences in motivation scores were found between the paradigms.

Tolerability

Tolerability scores for the P300 paradigms were 4.71±1.1 (range: 2.00-6.00) and 4.73±0.8 (range: 2.67-6.00) for non-gamified and gamified conditions, respectively. For the SMR paradigms, scores were 4.83±0.7 (2.83-6.00) and 4.66±0.9 (2.33-6.00) for non-gamified and gamified conditions, respectively. No significant differences in tolerability scores were found between the paradigms.

Mental Workload (NASA-TLX)

NASA-TLX scores for the P300 paradigms were 32.02±17.0 (1.67-65.83) and 33.84±15.9 (7.50-59.17) for non-gamified and gamified conditions, respectively. For the SMR paradigms, scores were 36.55±13.6 (13.33-70.83) and 39.32±15.4 (10.00-73.33) for non-gamified and gamified conditions, respectively. No significant differences in NASA-TLX scores were found between the paradigms.

Fatigue

The average change in fatigue for the P300 paradigm was 0.48±1.9 (range -4.0 to 6.0) for non-gamified calibration and 0.16±1.4 (range -3.0 to 6.0) for gamified calibration. For the SMR paradigm, the change in fatigue was 0.52±1.9 (range -2.0 to 5.0) for non-gamified calibration and 0.20±1.11 (range -2.0 to 3.0) for gamified calibration. There were no significant differences in change in fatigue found between calibration conditions for either paradigm.

3.5 Discussion

This study investigated the effect of gamified calibration on BCI performance in typically developing children on two well-known paradigms: P300 and SMR. The only significant effect of gamification was found on performance in the P300 spelling task, where gamified calibration had a possible negative effect on performance. Age-related differences in accuracy were found only in non-gamified P300 calibration, with teens performing better than younger children.

Overall, our results suggest that gamification may not be beneficial for everyone and an ongoing need to define personalized approaches in BCI calibration.

The lack of serious adverse events and the duration of sessions indicates that the paradigms (both gamified and non-gamified) were generally well-tolerated by our young participants. However, the withdrawal of one participant due to discouragement elucidates many important considerations for introduction of BCIs to naive users, particularly children. A BCI, like many technologies, requires time to learn and adapt to.[121] A new user may not perform well initially, leading to discouragement. This emphasizes one of the greatest ethical challenges in BCI deployment: the need to manage expectations, and ensure that users, particularly children, understand that it's normal to struggle in the beginning.[180,181] This also highlights the potential psychological effects of using a BCI, such as frustration or feelings of inadequacy, which can be detrimental to BCI performance and adoption.[182] It's important to consider these effects when designing and introducing BCIs for children where research experience remains limited. This may also reflect on participant selection and preparation processes. Improved screening and education methods may be beneficial to ensure that participants are prepared for the challenges of using a BCI and understand that initial struggles are part of the learning process.

There is a notable lack of literature on BCI performance in children, making the findings of this study particularly informative regarding BCI applications in children. We demonstrated that the classification model performance in the P300 paradigm was consistently high in both non-gamified and gamified conditions, with mean accuracies exceeding 96%, indicating stable classification accuracy across paradigms. Additionally, gamification appeared to improve precision, reaching 99% (SD=3.9%) in the gamified condition, while reducing variability compared to the non-gamified condition (SD=14.3%). Recall rates were also relatively high, at 72.42% (SD=28.2%) for non-gamified and 68.81% (SD=20.8%) for gamified conditions. To date, only one other study has investigated visual P300 BCI classification in children, comparing the performance of various classification models.[159] That study found similar classification accuracies across all classifiers, ranging from 62-64%, with precision and recall rates between 61.5-63.5% and 62-66%, respectively. In contrast, our study exhibited substantially higher classification model performance. This suggests that our approach might offer advantages in experimental design, paradigms, or data processing techniques. Moreover, our results demonstrate the reliability of P300 BCIs as a tool for BCI control in children, irrespective of the presence of gamified elements. The high performance of the classification model in our study implies that P300 BCIs could be effectively employed in various applications for children, including assistive technology, rehabilitation, and gaming. These encouraging results may inspire the development of child-friendly P300 BCI systems that address the specific needs and preferences of younger users.

The performance of the classification model in the SMR paradigm was noticeably lower than in the P300 paradigm, with mean accuracies around 60% for both the standard and gamified conditions. This suggests that the SMR paradigm may be more challenging for BCI control, again regardless of whether calibration is gamified or not. This observation is consistent with previous research in adults, which has documented that P300-based BCIs generally outperform SMR-based BCIs.[183] Our prior study evaluating P300- and SMR-BCI performance in children mirrored this finding.[169] In the gamified SMR calibration, 7 participants (22.58%) reached the 74% accuracy threshold, compared to 6 (18.75%) in the non-gamified calibration. This suggests that gamification may have a slight (if any) positive impact on achieving the proficiency threshold. Compared to our previous study, six participants (20%) attained the threshold (mean accuracy was 55%), demonstrating that the SMR calibration methods employed in the current study neither improved nor diminished performance.[169] It's important to note that children in this study only had two sessions using the SMR paradigm, with the total time for each session—including both calibration and the cursor control task—averaging around 10 minutes. Given the complex nature of motor imagery, proficiency is not typically achieved immediately; it's an acquired skill that usually requires repeated practice to gain mastery.[139] It is also important to note that compared to the previous study, which utilized the g.tec mindBEAGLE system[151], the current study had fewer trials (10 vs. 30 per class). However, our trials were broken up into segments of shorter durations (20 segments of six 1.5s-epochs vs 60 segments of 8s) with longer duration of breaks between segments (8s vs 2s). Therefore, the SMR paradigm in this study had 180s of imagined movement with 160s of breaks, whereas the previous study had 480s of imagined movement with 90s breaks, which contributed to a longer calibration period (9.5 mins compared to 5.67 mins). The modified calibration approach, with longer breaks between segments and a reduced overall calibration period, did not negatively impact performance, and thus may be more suitable for children, potentially reducing fatigue and maintaining engagement during the calibration process. These results highlight the importance of tailoring BCI paradigms to the specific needs of children, to improve usability and effectiveness. Further research is required to identify more effective strategies to enhance BCI proficiency among children.

We did not find significant differences in performance (accuracy, precision, recall) between the gamified and non-gamified calibration conditions for either the P300 or SMR paradigms. It is worth noting, however, that the high mean classification accuracy of 96% for both non-gamified and gamified P300 calibration paradigms suggests that there is limited room for improvement, making it challenging to discern any substantial benefits of the gamified approach. This ceiling effect may have masked the true potential of gamification in enhancing BCI performance, as the already high baseline performance in the P300 paradigm might overshadow any possible advantages. Our findings of no effect of gamification on SMR calibration align with the results from Castro-Cros et al. (2020), who found no effect of gamification on classification accuracy in an MI-BCI rehabilitation application.[184] There are several important factors that may have contributed to this outcome, including individual interests and the design of the gamified paradigms. The effectiveness of gamification can vary depending on individual preferences and interests.[168] The gamified elements might not have been appealing or motivating enough, thus not impacting performance. There are several psychological mechanisms through which gamification affects motivation, including personal relevance and competence.[168] In the present study, the gamified environments were designed by the researchers without input from the target population. This may have led to a mismatch between the participants' preferences and the implemented game elements, diminishing the motivational effects of gamification. Further, gamified tasks should offer challenges that are well-matched to the participants' abilities. If the tasks were too easy or too difficult, it could have led to boredom or frustration, respectively, and negatively impacted their sense of competence. By involving the target users in the design process and tailoring gamified elements to individual interests, BCI developers and researchers can ensure that challenges are well-matched and meaningful elements are incorporated into calibration paradigms, which may enhance users' motivation and engagement with the tasks.[168] While a high accuracy of the classification model during calibration is generally desirable, it does not guarantee high accuracy during online BCI control. Factors such as signal variability, user fatigue, and the complexity of the BCI task can all impact the accuracy of online BCI control. This was observed in the P300 spelling task, with online accuracies falling approximately 15% and 20% with classifiers generated from non-gamified and gamified calibration, respectively. What was more surprising was the lower online accuracy for gamified classifiers compared to non-gamified classifiers. This difference may be attributable to the differences in the design of the calibration tasks. Specifically, the presence of auditory stimuli in the gamified paradigm likely had a significant impact on the brain signals produced by the participants. In the Mole Patrol game, the characters made sounds when they popped up on the screen, with the mole making a distinct sound. These auditory stimuli likely elicited an auditory P300 in the participants' brain signals, which differs from visual P300 response in terms of distribution, latency, and amplitude.[1] However, during the utility-driven spelling task, there were no sounds present, and therefore, the auditory P300 component would not be expected. This discrepancy between the presence of an auditory P300 component in the gamified calibration and its absence in the utility-driven task could have led to classifiers that were less effective in the latter context, resulting in a lower online accuracy for the gamified classifiers. In order to clarify the observed differences in online accuracy between gamified and non-gamified classifiers, we plan to conduct a detailed EEG analysis, which will be reported in a separate paper. However, it's important to note that this does not necessarily undermine the potential of multimodal paradigms. In fact, these results could be interpreted as a call for more comprehensive training and calibration procedures that fully utilize the advantages of multimodal stimuli to optimize classifier performance. Combining different sensory modalities, could prove to be more effective for pediatric BCI applications, warranting further investigation in this direction. These findings highlight the importance of considering the context in which BCIs are used and calibrated, as these factors can significantly impact the system's performance.

No significant age-related differences were found in the gamified P300 calibration, while such differences were observed in the non-gamified P300 calibration, which suggests a possible influence of the gamification on age-related performance. Gamification could have made the task more engaging or enjoyable for younger children, improving their attention and thus their accuracy. This approach could potentially diminish the observed performance gap between younger and older children that was evident in the non-gamified version. This gap was similarly observed in our previous study where typically developing children used standard adult-oriented P300-BCI systems.[169] Alternatively, it's possible that the gamified version was less engaging or even distracting for older children, decreasing their accuracy and thus bringing their performance closer to that of the younger children. However, this disparity could also be attributed to age-related differences in the neural and cognitive processes underlying the P300 response, including attentional capacity.[1,185] Future research could explore whether the age-related performance differences observed in the non-gamified P300 calibration could be mitigated through tailored gamification designs suitable for each age group.

Gamification did not significantly affect motivation, tolerability, or mental workload as measured by the NASA-TLX in either the P300 or SMR paradigm. Participants reported similar motivation and tolerability scores for both gamified and non-gamified conditions across paradigms, indicating comparable levels of engagement and comfort. The mental workload, as indicated by the NASA-TLX scores, was also consistent regardless of gamification, suggesting that the cognitive demand of the tasks remains stable regardless of the presence of game elements. This indicates that gamification, while potentially offering other benefits, does not appear to have strong effects on these specific aspects of user experience.

This study offers valuable insights into BCI use and development in children, revealing that gamification may not be universally beneficial as we anticipated. Our findings emphasize the importance of understanding individual variability in BCI performance and the need for personalized approaches in BCI design and training. Furthermore, we demonstrate that most children can effectively operate a P300-based BCI with high classification and online accuracies, while many can achieve above-chance classification accuracies for SMR BCIs. These insights emphasize the importance of ongoing research and development in the realm of BCIs for children. By concentrating on strategies to optimize BCI performance, such as gamification, researchers and developers can create more effective and accessible technologies to assist children with diverse needs, including those with motor impairments or communication challenges. In our clinical program, gaming has emerged as a popular choice for children.[103,186] It's important to note that for children who may not have access to traditional video games due to their disabilities, any form of gaming, even if it's simple or adapted, can be a rewarding experience and a significant motivator for their BCI training. Therefore, developing and incorporating games that are accessible and enjoyable for all children should remain a key focus in the development of pediatric BCI systems and training programs.

Several important limitations may have influenced our findings. The presence of sounds in the gamified P300 paradigm, for instance, could have introduced an auditory P300 component that was not present in the utility-driven task, potentially leading to a discrepancy in brain signal patterns between the two contexts. This might explain the lower online accuracy observed for gamified classifiers and highlights the importance of ensuring consistency in experimental conditions. The ceiling effect observed in the P300 paradigm may have concealed the true impact of gamification on BCI performance. Since the baseline performance was already high, the gamified paradigm might not have been able to demonstrate significant improvements, which could undermine the potential benefits of gamification. This suggests the need for more sensitive measures or alternative paradigms that allow for a clearer assessment of gamification effects. In light of these limitations, future studies should consider maintaining consistency in experimental conditions, exploring alternative paradigms to overcome the ceiling effect, and increasing sample sizes for a more comprehensive understanding of individual differences. Engaging the target population in the design process to create personalized gamified elements may also enhance the effectiveness of gamification in BCI calibration, ultimately leading to improved performance and user experience.

3.6 Conclusion

This study contributes to the understanding of the impact of gamification on BCI performance in children and highlights the need for personalized approaches in BCI calibration. We demonstrated that while gamification may enhance BCI performance in some individuals, it may not be beneficial for everyone, indicating that a one-size-fits-all approach to BCI calibration is insufficient. By analyzing the effects of age, motivation, tolerability, and workload on BCI performance, this study has identified potential avenues for the development of targeted strategies to optimize performance in diverse user groups, including children. Future research is necessary to investigate individual variability and the need for tailored BCI design and training.

Chapter 4: Integrating Brain-Computer Interfaces into Daily Life at Home for Children with Quadriplegic Cerebral Palsy and Their Families

Dion Kelly, Erica Danielle Floreani, Danette Rowley, Eli Kinney-Lang, Joanna Keough, Ephrem Zewdie, Zeanna Jadavji, Ion Robu & Adam Kirton

4 Chapter 4

4.1 Abstract

Children with quadriplegic cerebral palsy (QCP) are often trapped in their bodies, unable to access current assistive technologies and denied their basic human rights. Brain-computer interfaces (BCIs) are an emerging technology that represent a promising solution to foster interaction and participation for children with QCP. However, BCI research has been confined to adults in controlled laboratory environments, representing critical translational gaps. Our goal was to assess the feasibility of home-based BCI use and its integration into the daily lives of children with QCP and their families. Eight children with severe QCP participated in a novel Home BCI Program over periods of up to 26 months including 242 sessions lasting up to 136 minutes. Outcomes included the experiences and opinions of families and aides (n=11) focusing on four aspects: adoption, preferences and perceptions, technical considerations, and integration with other assistive technologies. Consistent benefits included flexibility, convenience, and increased opportunities for social interaction. Challenges involved technical problems, setup, and scheduling. Confidence in BCI setup improved over time, and hybrid programs combining home and lab sessions were preferred by caregivers. Understanding diverse family needs, flexible scheduling, and tailored support may foster more successful, sustainable home BCI experiences. Improved customization, continuous support, and stronger collaborations amongst stakeholders may effectively cater to specific capabilities, requirements, and preferences for integrating BCIs into the daily lives of children with QCP and their families.

4.2 Introduction

Quadriplegic cerebral palsy (QCP) is the most severe and second most prevalent form of cerebral palsy in children. Children with QCP are often cognitively capable but unable to communicate or interact with their environments due to severely impaired movements that affect their entire body.[24] This often makes engaging in simple everyday activities impossible, infringing on their basic human rights, including their right to freedom of expression and participation.[38] This is underscored by the United Nations Conventions on the Rights of the Child (UNCRC) and the Rights of Persons with Disabilities (UNCRPD), both of which emphasize the importance of equal rights and opportunities for all individuals and the urgent need to focus on technological advances for affected children.[40,41] While the extent of impairment has been found to have variable associations with quality of life, limited participation has a strong association.[37]

Assistive technologies (AT) such as eye-gaze devices, head-tracking, and mechanical or proximity switches can enable access and participation for people with disabilities. However, existing technologies rely heavily on accurate and consistent motor control, and may consequently be impossible for children with QCP to operate.[46] There is thus an urgent need for novel assistive technologies to enhance independence, participation, and overall quality of life for children with QCP.

Brain-computer interface (BCI) technologies are emerging as a promising solution to empower children with QCP and enable their participation in daily life activities. BCIs bypass the need for motor control by decoding intent directly from the user's neurophysiological activity.[77] This activity can be acquired non-invasively, the most common modality being electroencephalography (EEG). BCIs are calibrated as users learn to modulate their brain signals in specific ways to encode their intent while the system adapts to recognize their unique signal patterns through the training of machine learning algorithms.[77] Once the algorithms are trained, they can translate the user's brain signals into useable commands in real time to control external devices such as computer cursors or communication aids.

The majority of BCI studies have been conducted in adults in controlled laboratory environments, with near-total neglect of children despite pressing need and clear potential to use BCIs as a lifelong assistive technology.[140,187] While our research group has demonstrated the ability of both typically developing children[104,188] and children with cerebral palsy[103,132,133,170,186] to successfully control BCI systems, no studies have investigated home-based BCI use for pediatric end-users. This presents a critical translational gap. Problems that may arise in realistic end-user environments, such as increased noise and distraction, could lead to challenges in reliability and control and, consequently, the failure of the BCI system to support end-user goals of participation and inclusion.[123]

Despite over 10,000 BCI research studies, <0.3% have investigated BCI use at home, and none have explored home use by children. We introduce a patient- and family-centred pilot Home BCI Program for children with QCP, which aims to evaluate the feasibility of home BCI use and to better understand how BCIs can be effectively implemented and integrated into the daily lives of children with QCP and their families. Here we present the initial results of an ongoing pilot Home BCI Program as a one-year retrospective. As caregivers play an essential role in the integration of BCIs into daily home activities[124], we captured the experiences of such persons without technical expertise[124], including their perceived expectations, challenges, and opportunities of using BCI at home.

4.3 Methods

4.3.1 Participants

Families of children diagnosed with QCP who met the inclusion criteria for the BCI4Kids clinical research program at the Alberta Children's Hospital (Calgary, Canada; www.bci4kids.com)[133] were invited to take part in the pilot home program. The original in-lab program involved in-person sessions at the BCI Lab where children could try various BCI systems and engage in diverse activities.[103] These included controlling a floating cube, a remote-controlled car, a Sphero SPRK+ robot, an iPad, computer games, and practicing spelling and yes/no communication tasks. The program's objective was to evaluate each child's ability to control a BCI while tailoring the activities to their engagement, interests, and abilities. Families were given equal access to both program formats, although this varied during the pandemic. Inclusion criteria consisted of: (1) age 5 to 18 years; (2) severe QCP diagnosis of any underlying cause; (3) severe physical disability including non-ambulatory status (Gross Motor Function Classification Score Level IV or V)[27] and minimal to no hand use (Manual Ability Classification System Level IV or V)[189]; and (4) perceived ability to understand instructions, as estimated by parents, teachers, and/or clinicians.[103]

4.3.2 Study Design

Parents completed a pre-survey to evaluate their perspectives regarding home BCI use, assessing interest, expectations, benefits, challenges, BCI knowledge, technical readiness, and preferred applications. Upon survey completion, they received a Home BCI Package containing all relevant equipment and training documentation. A home visit to ensure proper child positioning and technology setup supported was completed by an experienced clinician (e.g. the child in their wheelchair or alternate seating system, head positioning to allow EEG-cap placement, position of the screen within the child’s optimal field of view, etc.). A virtual orientation session was then scheduled to review the package contents and equipment setup remotely using zoom. The first participants enrolled in the Home BCI program (P1-P4) were given the choice to use the BCI whenever they wanted or to schedule virtual sessions according to their preferences. However, after observing the advantages of structured virtual sessions, the program shifted to providing all families with scheduled weekly 1-hour virtual BCI sessions. The research team was available for remote support as needed.

4.3.3 Home BCI System

The Home BCI Package consisted of a basic laptop (Microsoft Surface Pro tablet, *Microsoft, USA*) with an Intel Core i5 processor and 8GB of RAM, an Emotiv EPOC X (*Emotiv, USA*) consumer-grade wireless EEG headset with 14 saline-based electrodes (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4, sampling rate of 128Hz), a selection of switch- or keyboard- adapted toys that could be operated using the BCI (i.e., the Sphero SPRK+ toy robot, *Sphero, USA*), a comprehensive training guide, a USB drive of training videos and cleaning/maintenance instructions. The EPOC X headset was chosen for its relative ease of set up in home setting with semi-dry electrodes and low-profile fit, as well as its favorable tolerability in pediatric users[103,104,132,170]. The laptop came with all required software pre-installed, and included: the proprietary Emotiv BCI app (*Emotiv, USA)* for training mental commands; a custom-developed BCI-at-Home software that allowed for the mapping of trained BCI commands to simulated keyboard commands to control computer games and adapted toys[190]; a suite of BCI-enabled games (www.bcigamejam.com)[186]; remote video conferencing (*Zoom Video Communications, USA)* and remote desktop control software (*Remote Utilities, USA)* for running virtual sessions and troubleshooting. Suggested activities included playing skill-enhancing computer games on HelpKidzLearn (www.helpkidzlearn.com); painting using the Sphero; controlling music streaming applications such as YouTube and Spotify; playing BCI Game Jam video games[186]; and controlling switch-adapted appliances and toys. The total cost for the package was approximately $3100CAD.

The BCI system was controlled using a *mental command* paradigm which involved the repeated visualization or imagination of a movement or action (e.g., using brain signals to “pull” as an activation command). Visual feedback was provided to assist the user in performing their mental command (e.g. a cube moving toward user in response to “pulling”) (Fig. 4.1). During training, EEG signals were collected for both active (visualization/imagined movement) and neutral (rest) states and used to train a classifier to discriminate between these states in real time. Up to four different active commands (e.g. pull, push, left, right, etc.) could be trained, depending on how distinct and reproducible a signature a user was able to produce for each visualization. Data were processed and classified through the cloud-based API associated with the Emotiv headset.

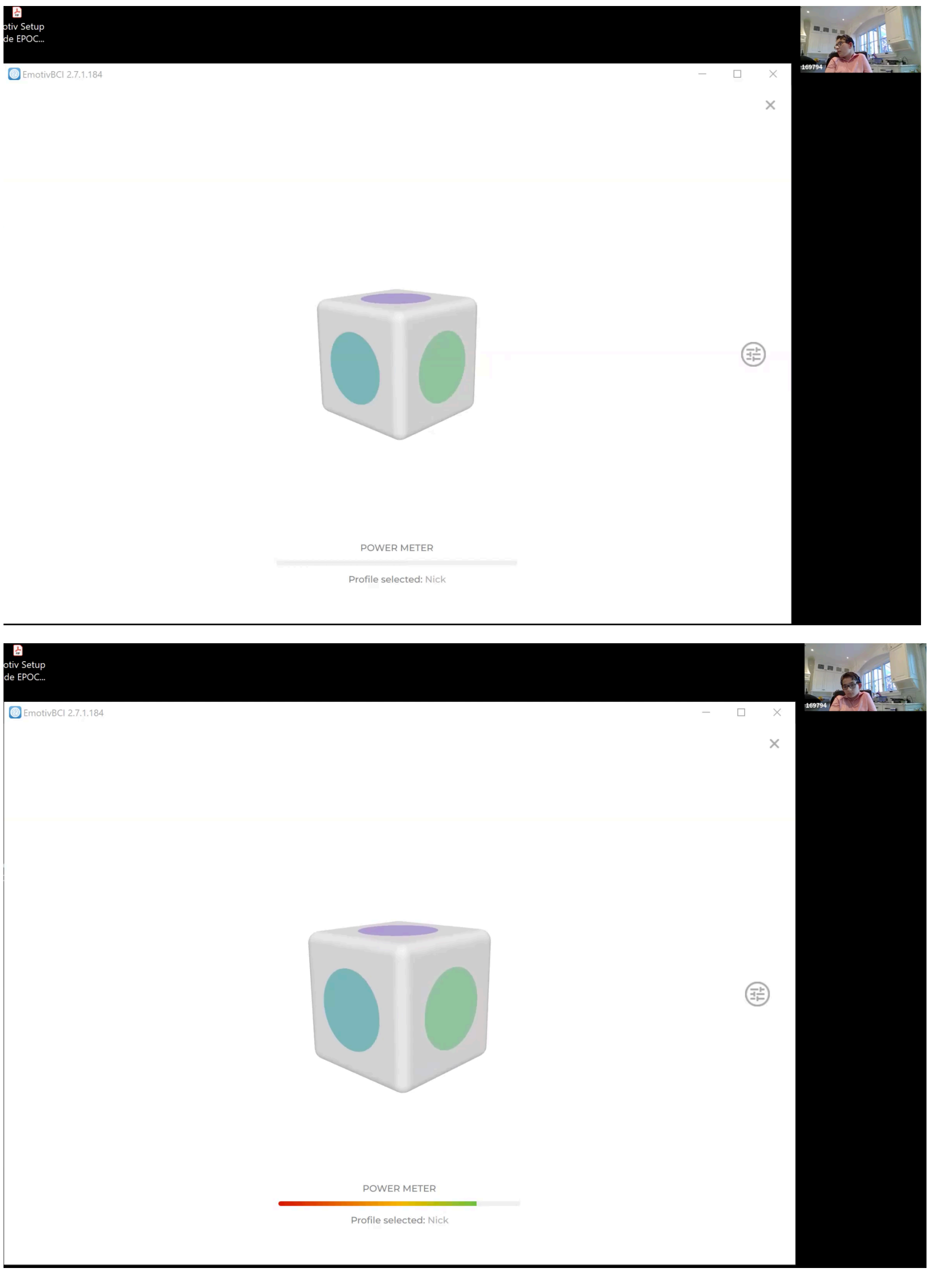


Figure 4.1. A virtual BCI session where the participant is utilizing the *mental command paradigm* offered through the cloud-based API associated with the Emotiv headset. Top: the

participant is practicing their “neutral” command. Bottom: the participant is practicing their “pull” command (the cube “pulls” toward the user and becomes enlarged on the screen).

4.3.4 Home BCI Program

Each virtual session started with setting up the headset and approximately 10 minutes of mental command training. Personalized imagined actions were chosen for each participant (e.g., imagining pushing their wheelchair), as were cues that would help the participant stay calm and relaxed for the neutral state (e.g., counting backward from 10). A symbolic or pictorial visual prompt was also used for some participants to aid in visualization during the mental command training. Each training trial consisted of 8 seconds of imagery or rest which were awarded a consistency score calculated by Emotiv’s proprietary API upon completion. The total number of training trials varied across sessions as trials awarded a low score or trials with excessive movement were rejected and re-attempted up to 3 times to minimize frustration.

4.3.5 Home BCI Questionnaire

To assess the experiences and opinions of families and aides of children with QCP who used BCI at home, we developed and administered a comprehensive questionnaire consisting of short-answer, multiple-choice and Likert-style rating questions. The questionnaire was divided into four sections, each addressing a specific aspect of home-based BCI usage.

Section 1 – Understanding Home-based BCI Adoption and Adaptation: Section 1 included demographic questions about the participant and family, their duration of involvement in the Home BCI Program, expectations, perceived benefits and challenges, factors influencing BCI use, interference with daily routines, related expenses, and involvement of other family members or aides.

Section 2 - Home BCI, Beyond the First Year: Section 2 examined participants' preferences for home or in-lab BCI programs, perceived advantages and disadvantages of each, suggested changes to the program, perceived usefulness of home-based BCI, feelings about weekly virtual sessions, interest in connecting with other participant families, perceptions of future opportunities, and opinions on the translation of BCI to other environments such as school and community settings.

Section 3 – Technical Aspects of Home BCI: Section 3 assessed participants' confidence levels in setting up BCI at home, attitudes towards new technologies, encountered technical challenges, perceived additional support needed, feelings about the Emotiv EPOC X BCI headset, suggested changes to the headset, and any other technical aspects they would change about the Home BCI Program.

Section 4 – Other Assistive Technology: Section 4 inquired about other assistive technologies (AT) and access equipment the children used at home and school, the frequency of AT use, and whether communication devices were used to aid independent play.

4.3.6 Outcome Measures

The outcomes were (1) BCI use, including number of BCI sessions, total duration of use from BCI sessions, use outside of BCI sessions, time between sessions, reason for session cancellations, and preferred activities; (2) effect on the families, including initial expectations, BCI benefits and challenges, factors supporting and hindering use, and effect on daily routine; (3) perceptions of future BCI use in everyday life, and (4) perspectives on the technical aspects of using BCI at home.

4.3.7 Analysis

The collected data was subsequently analyzed to identify trends, preferences, and areas for improvement in the home-based BCI program. Similar short-answer responses were grouped together into common overarching themes. Descriptive statistics were used to summarize the results of the questionnaire. Percentages were computed relative to the total number of responses for each question. The Wilcoxon matched-pairs test was used to compare Likert scale responses. The relationship between the number of BCI sessions and the final BCI setup confidence was modeled using a least-squares fit on an exponential plateau. The equation was: $Y=Y_M -(Y_M-Y_0)*\exp(-k*x)$, where Y_M is the plateau value, Y_0 is the initial value and k is the confidence rate. GraphPad Prism 9.5.1 (GraphPad Inc.) software was used to perform statistical analyses and to generate figures.

4.4 Results

4.4.1 Study population: BCI users & facilitators

Eight children with QCP and their families constituted the study population (mean age=11.8 years, range = 6-16 years, 3 female). Participants had diverse diagnoses, CP syndromes, communication abilities, and previous BCI experience. Their characteristics and BCI experience are summarized in Table 4.1. Eleven parents and aides responded to the questionnaire. Table 4.2 provides a summary of the respondent demographics.

Table 4.1. Participant characteristics and BCI experience.

Participant	Sex	Age (yrs)	Diagnosis	CP Syndrome	Comm. (V=verbal; NV=non-verbal)	Previous in-lab BCI experience	Duration with home kit (mos) up to Oct 2022	Duration of virtual sessions (mos)	Virtual Sessions (#)	Total session use (hrs)	Days between sessions (mean±SD, range)	Use outside of sessions**	Preferred activities
P1	M	14	Extreme prematurity	Dyskinetic QCP	V	5.5 years	21	15	50	58.9	11.9±16.6, 2-117	Low	BCI video games
P2	M	15	Kernicterus	Dyskinetic QCP	NV	1.5 years	24	20	51	82.4	11.0±9.6; 1-64	Low	BCI video games
P3	M	10	Bilateral schizencephaly	QCP	NV	1.5 years	25	13	29	29.6	17.2±21.3; 1-112	Moderate	HelpKidzLearn games, BCI video games, Sphero painting
P4	M	11	Perinatal stroke	Dyskinetic QCP	NV	1 year	26	10	31	33.4	10.8±10.1; 2-52	Moderate	HelpKidzLearn games, BCI video games
P5	M	12	Kernicterus, prematurity	Dystonic QCP	NV	1 session	14.5	8	24	29.2	17.9±36.0; 4-179	Low	HelpKidzLearn games, music applications
P6	F	10	Bilateral schizencephaly	QCP	NV	1 session	11	10	25	28.2	13.4±10.9; 2-42	Low	HelpKidzLearn games, BCI video games
P7	F	6	Hypoxic ischemic encephalopathy	Mixed QCP	NV	1 session	7.5	7.5	20	23.2	11.8±7.25; 2-36	High	HelpKidzLearn games, switch-adapted activities
P8	F	16	Perinatal stroke	Mixed QCP	NV	4 years	1*	1*	2	3.4	3	Low	BCI video games

*Dropped out due to preference for in-person lab-based program

** Frequency of average weekly use outside of BCI sessions: Low = 0-1 times; moderate = 2-3 times; high = 3+ time

Table 4.2. Respondent characteristics.

Respondent	Relation to participant	Occupation	Marital status	# of children	Support worker?	Duration of involvement with home BCI at time of study (mos)	Previous experience with assistive technology
R1	Mother	Engineer	Married	2	After school aide	26	eye-gaze camera, switch, & touch screen
R2	Father	IT communications	Married	2	Fulltime nanny	11	switches &eye-gaze camera
R3	Mother	Engineering consultant	Separated	1	No	24	eye-gaze camera
R4	Mother	Teacher	Married	2	No	25	eye-gaze camera & low-tech communication book
R5	Father	Software developer	Separated	1	No	24	eye-gaze camera
R6	Mother	Stay home parent	Married	3	24 hr nanny	12	custom chin joystick for mouse emulation
R7	Mother	Stay home parent	Married	3	24 hr nanny	11	switches & low tech communication book
R8	Father	Plumber	Married	1	No	1	switches & eye-gaze camera
R9	Aide	Educational aide	-	-	After school aide	12	eye-gaze camera, switch, & low-tech communication book
R10	Aide	Nanny	-	-	Fulltime nanny	9	eye-gaze camera
R11	Mother	Stay home parent	Married	3	No	12	switches & eye-gaze camera

4.4.2 Home BCI Use

The BCI home use periods of the 8 participants varied between 1 to 26 months at the time of data collection (October 2022) (Table 4.1). One family opted out of the Home BCI Program because they preferred attending sessions in-person rather than at home. The duration of participation in virtual sessions ranged from 1 to 20 months. Due to non-use by two out of the first four enrolled families, which was attributed to scheduling difficulties, the research team recommended implementing a weekly format with a designated virtual session time for each family starting in June 2021.

We found the virtual sessions were not only beneficial for facilitating regular BCI use, but also allowed the research team to monitor BCI skill progression and regularly suggest new and challenging activities. Session usage totaled 242 sessions and 288.3 hours (mean duration 72.0±20.0 minutes, range 18-136 minutes). Mean values for time between virtual sessions for each of the 8 participants varied between 10.8 to 17.9 days, with standard deviation values ranging from 7.25 to 35.6, and a mode of 7 days for all participants except for one, who had a mode of 14 days.

Mean cancellation rate among participants was 39.7±8.77%, with a range from 29.9% to 53.9%. Reasons for session cancellations included medical issues, acute illness, equipment not charged, fatigue, emotional dysregulation, busy family life, parents working/busy, family holidays, and weather (i.e., nice day out and would rather go outside than do BCI inside). BCI use outside of virtual sessions was generally low, as 62.5% of families rarely or never used it independently. Only one family exhibited high independent use, engaging almost daily outside of scheduled sessions. Participants engaged in a range of preferred activities, including BCI video games, HelpKidzLearn games, Sphero painting, music applications, and switch-adapted activities. Figure 4.2 shows the activities completed by families using BCI at home.

Figure 4.2. Activities achieved by participants using BCI at home, including Sphero painting (top left, bottom middle right), juicing lemons for their lemonade stand (top middle), playing BCI video games with their siblings and family members (top right, bottom left), playing a Mario computer game (bottom middle left), and controlling disco lights (bottom right).

4.4.3 Effect on families

The responses of 11 parents and aides to section 1 of the questionnaire are summarized in Figure 4.3. For expectations of home BCI use, "Flexibility and Convenience" and "Interactivity/Engagement with Family" were the most common responses, while other expectations were mentioned once. "Increased flexibility and convenience" was the most frequently mentioned benefit, followed by "Increased opportunities for social interaction and engagement with friends and family" and "Increased comfort and safety". The greatest factor supporting home use was support from the research team, followed by accessibility, fun and engagement, and excitement for BCI activities. Most respondents reported that BCI use did not interfere with their daily routine and was incorporated into daily routines.

Challenges with home use included technological problems, setup and connectivity, and scheduling due to fatigue. The most common factor hindering use was technical problems, followed by general fatigue and busy schedules, and incompatibility with the child's physical condition. Some respondents mentioned that other activities would sometimes take priority over BCI use. One respondent indicated that BCI use is flexible and can be rescheduled, if necessary, while another respondent remarked that BCI use can be difficult on weekdays due to school. One respondent pointed out that although BCI doesn’t interfere with regular life, it requires dedicated time.

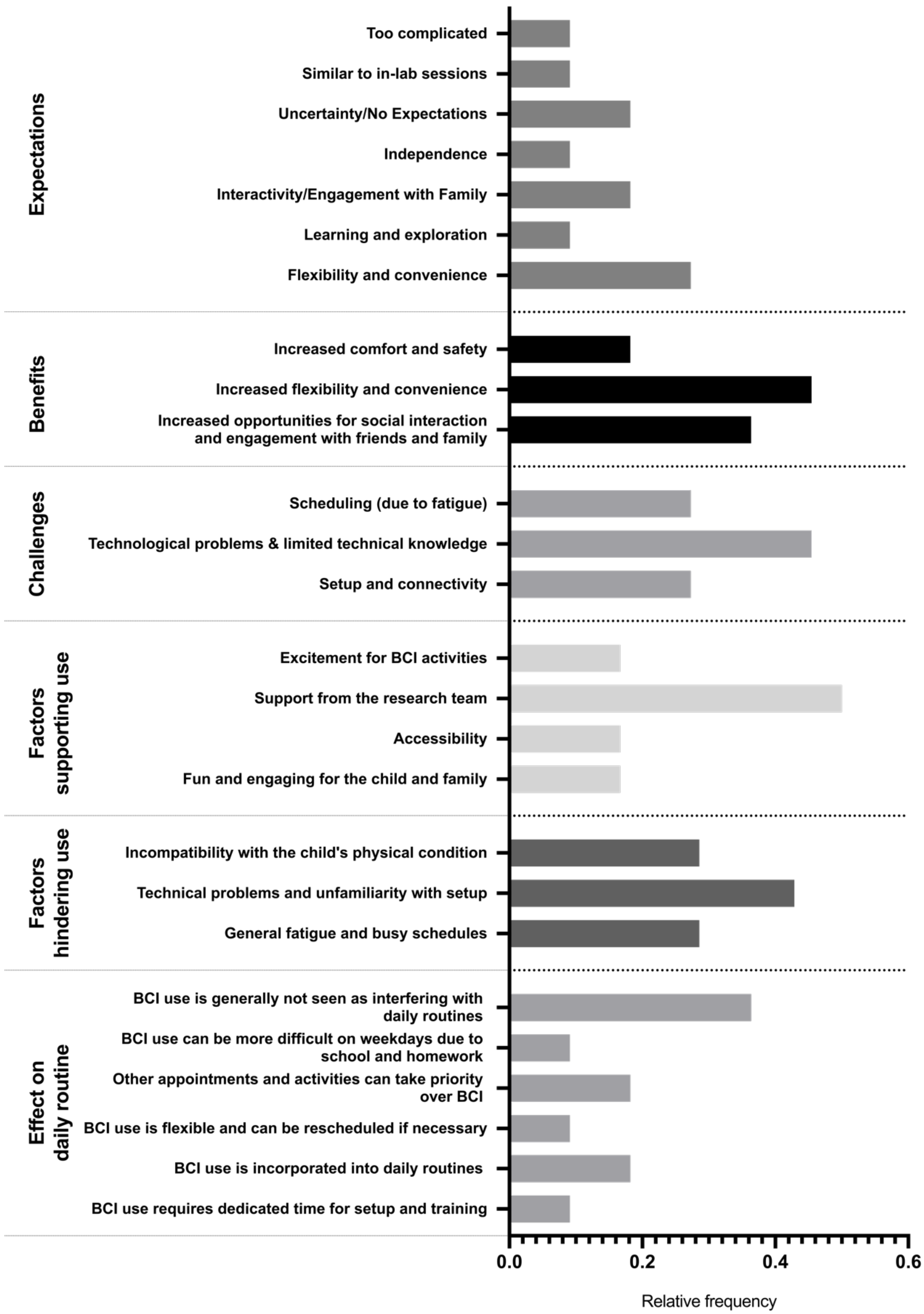


Figure 4.3. Relative frequency of similar responses from questions regarding home BCI use facilitation answered by parents and aides of children with quadriplegic cerebral palsy.

Figure 4.4A shows the relative frequency of responses from parent/aides regarding individuals who were involved with BCI use at home, including setup, monitoring, and engaging with the end-user during home BCI use. Ten out of eleven (91%) of respondents reported that the mother was typically involved, followed by the father and aide (both 55%). Some families cited having siblings, cousins, and grandparents involved with home BCI use.

Figure 4.4B shows the initial and final mean (±SD) BCI setup confidence Likert scores for the 11 parent/aide respondents who were involved with BCI use at home. The average initial score was 3.55±2.51 and the average final score was 8.55±2.30. The results of the Wilcoxon matched-pairs test showed a statistically significant difference between the initial and final BCI setup confidence (W=66, n=11, p=0.001, two-tailed).

Figure 4.4C shows the relationship between the number of BCI sessions and the final BCI setup confidence. BCI session setup by any one person ranged from 2 to 31 sessions and the BCI setup confidence from 2 to 10. The best-fit values obtained from the exponential plateau model were Y_M=9.28, Y_0=-1.62, and k=0.20. The goodness-of-fit measures indicated a strong fit of the model to the data (R^2=0.89). Figure 4.4D displays the children's preferred BCI use locations, with hybrid programs being most common. Likewise, Figure 4.4E shows parent and aide location preferences, also favoring hybrid programs.

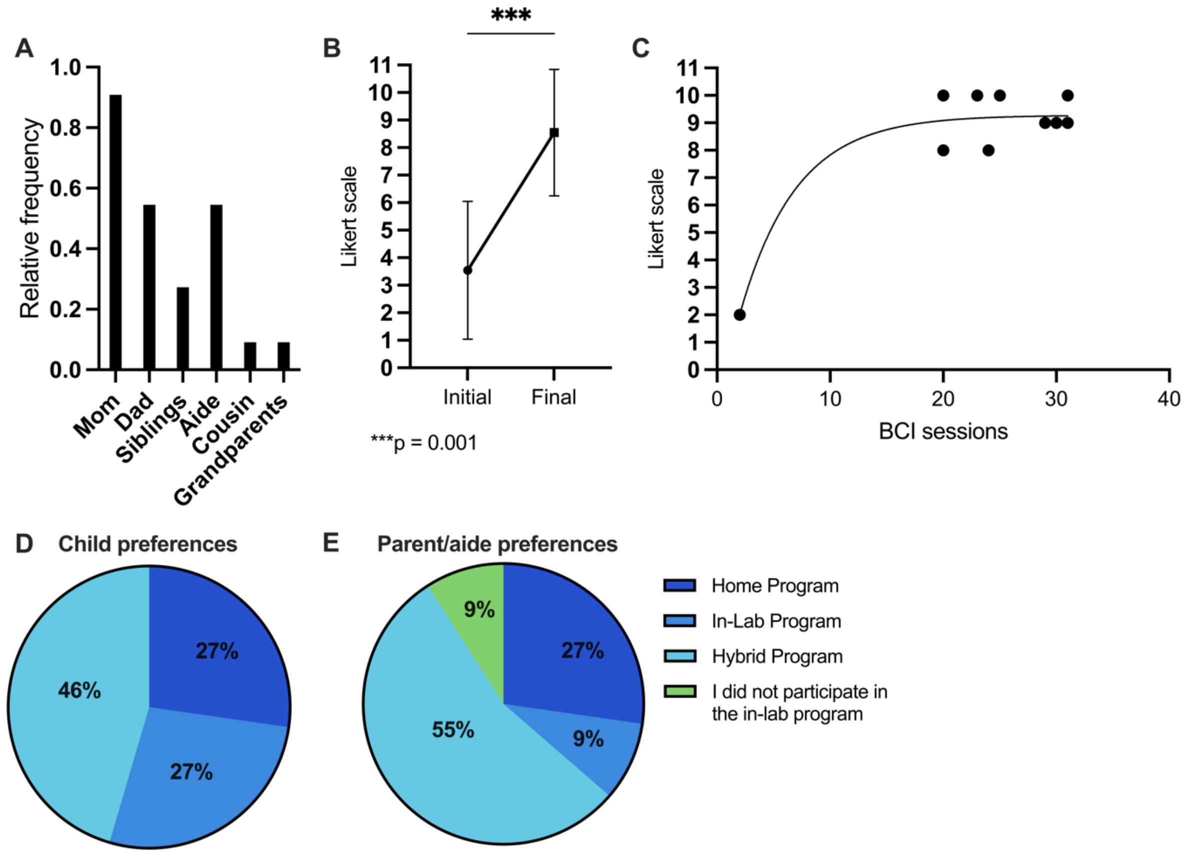


Figure 4.4. Facilitation and preferences of stakeholders for BCI use at home. (A) Relative frequency of responses from parent/aide respondents regarding individuals who are involved with BCI use at home, including setup, monitoring, and engaging with the end-user during home BCI use. (B) Initial and final average (±SD) BCI setup confidence scores for the 11 parent/aide respondents who are involved with BCI use at home. Wilcoxon matched-pairs test was used to compare initial vs final values. (C) Final BCI setup confidence scores as a function of number of BCI sessions that any one respondent was involved in. The line was fit using an exponential plateau model Y=YM−(YM−Y0)*exp(−k*x) where Y_M=9.28; Y_0=-1.62; and k=0.20. (D) Breakdown of respondents' perceptions of the child's preference for location of BCI use. (E) Breakdown of respondents' preferences for location of BCI use. ***p=0.001

4.4.4 Perceptions of future BCI use in everyday life

Table 4.3 presents perceived benefits and challenges of future BCI use in everyday life. Benefits to home use included manageable time investment, convenience, no commute, safe environment, and flexible schedule. Challenges included requirements for technological support and reduced opportunities for in-person interaction with the support team. Moreover, children sometimes reported missing in-lab programs and associated socialization opportunities.

When asked about the prospect of connecting virtually with other BCI families for joint interactive activities, respondents identified potential benefits, including enhanced interaction and socialization, a sense of belonging to a larger BCI community, and increased engagement for participants (Table 4.3). However, some suggested this may pose additional technical and scheduling challenges as well as introduce potential distractions during virtual sessions. Some participants suggested they would prefer to play games in-person with others rather than online.

Translating BCI to school and community settings was identified as a potential benefit for students with complex needs by enabling feedback and demonstration of knowledge, providing opportunities to participate in more school activities, and to increase independence. However, challenges may include training staff, maintaining equipment, ensuring precision and speed, troubleshooting, monitoring multiple children, and finding the best way to utilize BCI in the community.

Table 4.3. Perceived benefits and challenges of future BCI use in everyday life.

Aspect	Benefits	Challenges
Using BCI at home vs. in the lab	More manageable time investment at home Convenience of using the system when desired No need to commute Safe and comfortable environment More flexible schedule Can do BCI anytime Enjoyable to stay at home No car ride, which can upset some children BCI support team is excellent	Less interaction with BCI support team Need for a lot of technological support at home Can't access lab programs like the car which some children enjoy Missing in-person relational aspect of doing in-lab sessions Not having somewhere to go for in-lab sessions Some children are motivated by being with others and enjoy the social aspect of in-lab sessions
Connecting with other BCI families virtually to do interactive BCI activities together (ex. online BCI gaming)	Interaction and socialization with other kids in the program Feeling part of a bigger BCI team More fun and engaging for kids	Potential technical issues Difficulty in scheduling with other families Kids may feel let down or left out if it doesn't happen regularly or if they don't get along More distractions during virtual sessions Some kids may prefer to play live games in a group rather than online

Translation of BCI to other everyday environments, such as school and the community	BCI in school would allow children to interact with their classmates in a way they currently cannot BCI could be integrated into the special education curriculum to enable students with complex needs to give feedback and demonstrate knowledge Using BCI in schools could give children more opportunities to be independent in class BCI could provide more ways for children to participate in school activities Exposing typically developing children to BCI could be beneficial	Challenges of training staff and keeping the equipment organized and maintained The speed and precision required could be a barrier for typical education Troubleshooting and setup Connectivity issues Monitoring all the kids using BCI Community may be tough to figure out the best way to use BCI

4.4.5 Perspectives on the technical aspects of using BCI at home

Figure 4.5 presents the respondents' views on the technical aspects of BCI and other assistive technology at home. The main technical issues included headset connectivity (50%), updates (40%), and setup (10%). Desired BCI headset changes included improved fit and comfort (50%), connection quality and stability (29%), and user-friendliness (21%). Figure 4.5C shows that 27% desired no modifications to the Home BCI Program, while an equal proportion preferred a more reliable headset and faster, more engaging calibration tasks. A minority (9%) suggested home visits and an external webcam would facilitate virtual sessions. Other assistive technologies used by the participants included 57% utilizing eye gaze, 36% employing switches, and 7% not using any additional devices.

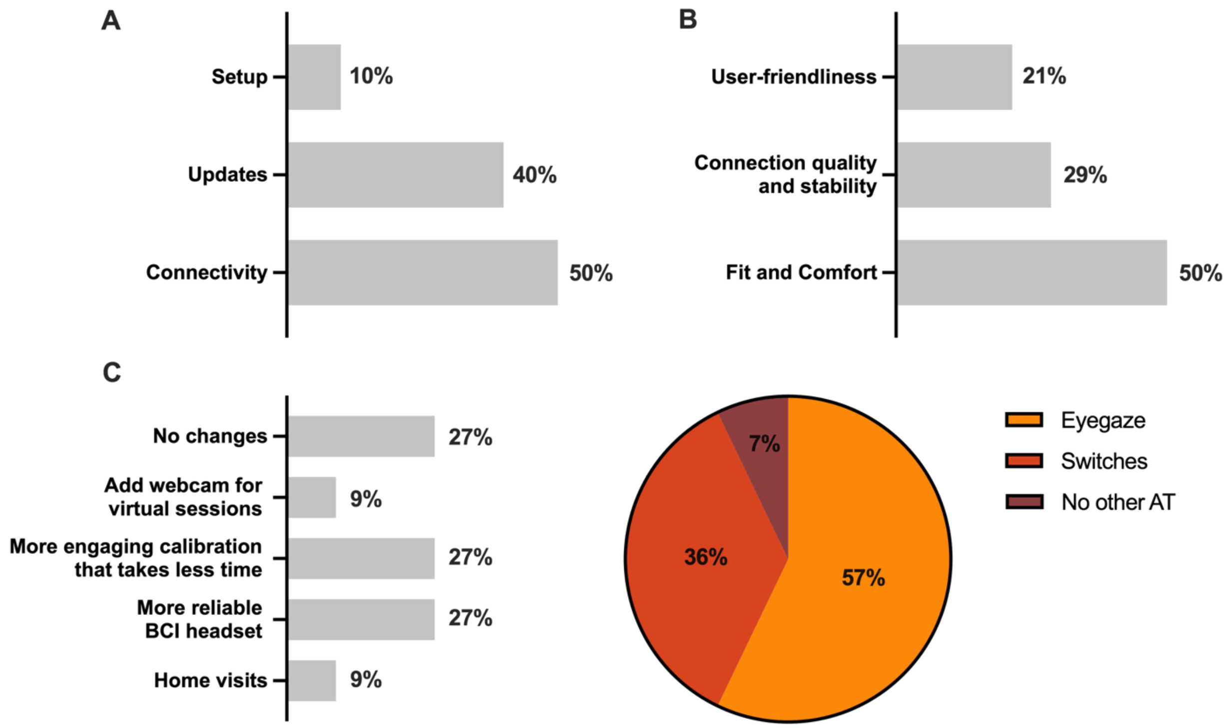


Figure 4.5. Respondents' perspectives about the technical aspects of using BCI and other assistive technology at home. Percentages represent the relative frequency of that response. (A) Common technical issues encountered when using BCI at home. (B) Desired improvements to the BCI headset. (C) Desired changes to the Home BCI Program. The pie chart indicates other assistive technology devices that the child utilizes at home and at school.

4.5 Discussion

This study examined the implementation and impact of a home BCI program for children with QCP from the perspectives of patients, parents, and aides. The program proved feasible and beneficial, with technical issues posing the primary challenge. A key advantage was increased convenience, and scheduled virtual sessions were important for regular use.

Participants in the Home BCI Program were children with QCP with differing abilities, needs, goals, and preferences. Respondents, consisting of parents and aides, also varied in background, occupation, family structure, technical abilities, and support systems. This diversity, further reflected in different types of support workers, family structures, number of children in households, and many other variables suggests that home BCI program may be appealing to a wide range of families with varying levels of support.

Weekly, regularly scheduled appointments were generally achievable with good attendance rates. The mean time between virtual sessions for participants ranged from 10.8 to 17.9 days. The standard deviation for each participant's mean time between sessions varied, ranging from 7.25 to 35.6, indicating considerable variation in the time between sessions for different participants. The mode of 7 days for all participants except for one, who had a mode of 14 days, suggests that most participants were able to attend sessions on a weekly basis. Interestingly, the one family whose mode was double that of other participating families had high BCI usage outside of virtual sessions. This could imply that some families are more self-sufficient in utilizing the BCI technology and may require less frequent support or guidance from the research team. Whether such independence could be instilled over longer training times remains to be determined. The varied attendance due to numerous cancellation reasons and generally low BCI use outside virtual sessions emphasize the unique challenges faced by such diverse families in consistently integrating and utilizing BCIs in their daily lives. Challenges with use are not unique to BCIs but are a universal experience with AT. One of the six most reported barriers to successful AT implementation in children is the time demand from the individual setting up and supporting use.[191] The differences in BCI usage and engagement underscore the flexible approach to scheduling and support the need to accommodate the unique needs and preferences of each family participating in home BCI programs. This aligns with the expectation of "flexibility and convenience" valued across different populations using BCI at home.[89]

In response to these challenges, the implementation of a weekly format with set times for each family proved beneficial. This structured approach not only improved regular BCI usage by allowing families to integrate BCI sessions into their routines but also enabled the research team to closely monitor participants' BCI skill progression, provide tailored support, and suggest engaging activities. Furthermore, consistent virtual sessions fostered collaboration between the research team and families, facilitating prompt issue resolution and a better understanding of individual needs. Understanding the demographics and personalized needs of the families through regular interactions may also help identify any trends or factors affecting BCI use, session attendance, and support needs, which can inform the design of future BCI programs and interventions to better accommodate diverse families.

As with our previous study on lab-based pediatric BCI programs, respondents identified increased social interaction and engagement with friends and family as a key benefit of using home-based BCI technology.[133] This underscores the potential for BCIs to enhance the quality of life and social well-being of children with QCP. As "increased flexibility and convenience" also emerged as a main benefit, it may be that home BCI use allows children to participate in activities in a comfortable, safe environment and the encouragement of more frequent and meaningful interactions.

Technical issues, such as connectivity and setup, were identified as common challenges hindering home BCI use, aligning with findings from previous studies.[85,86,127,128] Continuous support from the research team emerged as a key factor for supporting home use, similar to prior studies which have shown the importance of technical support for successful home use and avoiding abandonment.[89,128,192–194] Addressing these technical challenges and providing continuous support are crucial for fostering confidence in BCI setup among parents and aides. Our results reveal a significant improvement in parent and caregiver confidence in BCI setup over time, supported by the strong fit of the exponential plateau model. This suggests a clear relationship between the number of sessions and BCI setup confidence. Our model indicates that after just 5 sessions, BCI setup confidence may be adequate. This is consistent with Zulauf-Czaja et al. (2021), who found that caregivers felt confident with their BCI system after 4-5 sessions.[195]

Addressing technical challenges and providing adequate training remains a crucial aspect for caregivers managing BCI systems at home. In adult studies of home-based BCI use, similar concerns have been reported, with caregivers emphasizing the need for more technical training on BCI systems, more reliable connections between the headset and computer, and easy-to-clean, preconfigured caps/headsets.[124,125] Furthermore, the time commitments for BCI setup, training, and cleanup have been reported by caregivers to contribute to attrition of home-based BCI.[89] These examples provide obvious targets for industry developers, researchers, and clinicians to advance pediatric specific BCI solutions.

As parents and caregivers become more confident in BCI setup through increased sessions, it is essential to consider their usage preferences in order to optimize user experience. Our study found that many children (46%) and parents/aides (55%) prefer a hybrid approach of in-lab and at home BCI use, consistent with Huggins et al. (2015).[194] Only one family opted out of the Home BCI Program, preferring in-lab sessions due to the social aspects of collaborating with the research team. To enhance social interaction and engagement at home, involving other family members in the BCI process could be beneficial. We found that mothers were most involved in home BCI use (91%). Including additional family members could enhance social aspects, reduce caregiver workload, and prevent burnout. Kleih and Kübler highlight the importance of considering caregivers' mental and emotional state, as frustration and dissatisfaction could hinder successful BCI use and increase the risk of abandonment commonly seen with other AT systems.[126] Encouraging diverse family involvement in home BCI programs may facilitate balanced caregiving dynamics, improved social engagement, and enhanced support for the child, contributing to a more successful and sustainable home BCI experience.

To successfully implement and integrate AT, both usability and accessibility are vital components. Neglecting either of them increases the probability of abandonment.[192] In response to limited access to technology during the COVID-19 pandemic, our goal was to provide access to BCIs. We mitigated delays in access by packaging user-friendly, commercial BCI devices and applications with our custom software, which connected their usage. This approach saved time, with access delays resulting only from software development and shipment of the package components. Similarly, Holz et al. (2015) opted for an easy setup EEG cap with a small number of electrodes, prioritizing usability and accessibility.[128] Although developing a new BCI may have been ideal, our strategy aligns with Kübler, Nijboer, and Kleih's (2020) finding that end-users prioritize urgently needed functionality.[192] Our study focused on understanding the requirements and context of home use, addressing concerns accessing the technology and attempting to provide ongoing support. However, to be adopted and utilized, a device must cater to each user's specific capabilities, requirements, and preferences while outperforming or providing greater convenience than existing alternative access technology (e.g. eye-gaze systems) and low-tech methods (e.g. partner-assisted scanning).[89] Areas that require further attention, are better customization of BCI systems, improved low-cost consumer-grade hardware tolerable for pediatric users with disabilities, continuous support for caregivers/setup assistants, and stronger collaboration between stakeholders to create more effective and user-centered BCI solutions.[192]

4.6 Conclusion

This study evaluated the feasibility and impact of a pilot home BCI program for children with QCP, highlighting the importance of addressing technical challenges, providing continuous support, and understanding the unique needs of diverse families. The program was found to be feasible and beneficial, with participants appreciating the increased convenience and social benefits of home-based BCI use. Our findings emphasize the need for flexible scheduling, tailored support, and involving various family members to create balanced caregiving dynamics and a more successful, sustainable home BCI experience. As we continue to develop and refine BCI technologies, focusing on improved customization, continuous support, and stronger collaboration between stakeholders will be essential to ensure that these systems effectively cater to each user's specific capabilities, requirements, and preferences. By addressing these areas, we can move closer to integrating BCIs seamlessly into the daily lives of children with QCP and their families, ultimately enhancing their quality of life and social well-being.

Chapter 5: Brain-Computer Interface Goals for Children with Quadriplegic Cerebral Palsy

Dion Kelly, Danette Rowley, Erica D. Floreani, Eli Kinney-Lang, Ion Robu, & Adam Kirton

5 Chapter 5

5.1 Abstract

There is a pressing need for alternative access technologies that enable children with severe physical disabilities, as current options often require some degree of controlled movement to be used efficiently. Brain-computer interfaces (BCIs) hold significant potential for improving the lives of children with severe physical disabilities, however research must prioritize user-centered approaches and real-world applications to maximize benefits. This paper examines the integration of home-based BCIs for children with quadriplegic cerebral palsy through user-centered design, focusing on the feasibility, usability, and impact on personal goals and activities of daily living. Seven children aged 6-15 years with quadriplegic cerebral palsy and their families participated in this pilot study, using personalized BCI packages and home-based virtual sessions to help them achieve individualized goals in self-care, productivity, and leisure. We utilized a collaborative goal-setting approach and assessed satisfaction and performance changes using the Canadian Occupational Performance Measure (COPM) and the BCI-adapted Quebec User Evaluation of Satisfaction with Assistive Technology (e-QUEST2.0). Significant improvements in performance and satisfaction were observed in the COPM scores, while parents were most satisfied with professional services and least satisfied with the adjustability of the BCI system, as per the eQuest2.0 questionnaire. Despite no significant improvement in BCI consistency across nine sessions, the intervention positively impacted participants' perceived performance and satisfaction in goal-oriented activities. Future research should focus on enhancing BCI design, comfort, and effectiveness while considering user priorities and feedback for personalized goal achievement.

5.2 Introduction

Children with physical disabilities, such as cerebral palsy (CP), require alternative approaches to facilitate participation and interaction. The last decade has seen many exciting innovations in the space of alternative access technologies, paving the way for new opportunities for children who face access challenges.[196] Such access technology varies from adapted keyboards and touch screens to switch scanning, head-tracking, and eye-tracking. While these approaches show potential, their usage requires a certain degree of controlled movement, often performed in a repeated fashion, leading to fatigue or repetitive strain. In addition, many of these traditional access technologies require optimal positioning of the equipment for the child to effectively interact/control the access method. Unfortunately, these limitations with existing access technologies result in children with the most severe physical disabilities having insufficient opportunities to interact with their environment. This loss of opportunities for life participation infringes upon their fundamental human rights.[40,41] The World Health Organization's International Classification of Functioning, Disability and Health (ICF) defines participation in the context of nine domains, including learning and applying knowledge, communication, mobility, and social life.[36] Though severity of impairment has only been shown to be modestly correlated with quality of life, restricted participation is strongly related.[197] Therefore, to improve the quality of life for children with severe physical disabilities, interventions must focus on increasing activity and participation in daily life.

Brain-computer interfaces (BCIs) offer a promising solution to better enable children with severe physical disabilities to interact and participate without the need for muscular control.[112] Despite their potential, BCI research has largely overlooked children, particularly those with severe physical disabilities who stand to benefit the most.[167] Recent studies have demonstrated that both able-bodied children and those with disabilities can successfully use BCIs to perform various tasks in a controlled laboratory setting.[103,170,186] To truly benefit children with disabilities and ensure that BCIs have a lasting impact on their lives, it is vital to adopt a user-centered approach that actively involves end-users in the development process.[85] This co-creation of solutions is essential to address the unique needs, preferences, and challenges faced by children and their families in their day-to-day lives. By closely collaborating with end-users, we gain valuable insights into their requirements and barriers to technology access. This understanding enables the design of accessible, intuitive BCI systems and applications that enhance children's participation in meaningful activities within their natural environments, such as homes, schools, and communities. Incorporating these user-centered strategies will contribute to positioning BCIs as a lifelong assistive technology for children with disabilities, empowering them to lead more fulfilling and independent lives. We applied principles of user-centered design to investigate the feasibility and usability of home-based BCIs for children with quadriplegic CP to achieve their personal goals. This paper reports on the goals, challenges, successes, and areas for improvement in the integration of BCIs into the daily lives of children with quadriplegic CP and their families.

5.3 Methods

5.3.1 Participant criteria and recruitment process

Children with quadriplegic CP enrolled in the BCI4Kids clinical research program at the Alberta Children's Hospital (Calgary, Canada)[103] were invited to participate in this pilot study. Inclusion criteria for the research program and for this study were: (1) age 5 to 18 years; (2) severe mobility impairment (Gross Motor Function Classification Score Level IV-V)[198] and severe manual dexterity limitations (Manual Ability Classification System Level IV or V)[189]; and (3) evidence from parents, teachers, and/or therapists of at least grade 1 level cognitive capacity. A prerequisite for inclusion in our sample required parents to independently set up the BCI equipment at home successfully on at least three previous distinct instances (verified by BCI researchers) and provide verbal assurance of their comfort and proficiency in the setup process.

5.3.2 Collaborative goal-setting

Potential goals were identified using established applications from the BCI4Kids program combined with the Canadian Occupational Performance Measure (COPM) through semi-structured interviews with the child and family. Reflective questions about daily activities were posed to help identify potential goals. Children with reliable communication methods shared their perceptions. For children with less established communication methods, their parents shared their perceptions on behalf of the child. The child confirmed their parents' answers through a 'yes' or 'no' communication method.

The goals were categorized into three dimensions: self-care, productivity, and leisure. The participants, (or their parent) rated the importance of the goals on a scale of 1 to 10 using their current communication method (outlined in Table 1). Then they chose up to five most important goals and rated the child's perceived performance and satisfaction for each.

5.3.3 Home-based kits to facilitate individual goals

Personalized BCI Packages were delivered to each participating family and included a Microsoft Surface Pro tablet with an Intel Core i5 processor and 8GB of RAM (Microsoft, USA), an Emotiv consumer-grade wireless EEG headset (*Emotiv, USA*), a switch interface, and switch- or keyboard-adapted activities in line with BCI skill acquisition and individual goals.

The laptop was equipped with all necessary pre-installed software, including the Emotiv BCI App (*Emotiv*, USA) for BCI calibration, custom-created BCI-At-Home software for mapping BCI commands to simulated keyboard inputs for computer games and switch-adapted toys/appliances[190], a collection of BCI-compatible games, and remote video conferencing software (*Zoom Video Communications,* USA*)* for conducting virtual sessions and addressing technical issues.

BCI System: Participants received either an Emotiv EPOC X (headband-style) or an Emotiv Flex (cap-style) headset, depending on their individual needs. Both headsets utilized 14 saline-based electrodes placed at the same channel locations (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4). The Emotiv headsets were chosen because of their user-friendliness for parent/caregiver independent set up and suitability for pediatric users.[103,190] The Emotiv headsets employ a *mental command*-based paradigm, where the user repeatedly visualizes a selected action or movement to activate the system. Electrical brain signals (EEG) collected during mental command visualization are compared to signals collected during a resting or neutral state to generate a classifier for decoding these states in real time. EEG data were processed using Emotiv's proprietary Cortex API.

Software: Electrode signal quality verification and BCI calibration was performed using Emotiv's BCI App. Then, once mental commands were trained, they could be streamed via the Cortex API and were integrated into a custom-developed home BCI software to map BCI outputs to useful applications like sending simulated keyboard commands or activating the switch interface.[190] This software was designed to be user friendly, targeting caregivers and parents with limited BCI or technical experience. The software allowed for simultaneous control of up to four different mental commands, each with adjustable sensitivity and activation settings to suit individual user needs. For more details on this home BCI software, refer to Floreani et al. (2022).[190]

Switch Interface: A custom switch interface module was also developed, enabling participants to control any switch-adapted device with the BCI system.[170] The switch interface, through USB connection to the laptop, streamed mental command activations from the Emotiv Cortex API to general purpose input/output pins of a microcontroller, which then directed the activation of four standard 3.5mm mono jack outputs. Custom designed software for the switch interface was used to adjust settings and control of BCI.

Activities: Included in all packages were BCI skill-enhancing activities such as a collection of switch-activated computer games on HelpKidzLearn (www.helpkidzlearn.com) and BCI-enabled games developed for children with complex needs from the BCI Game Jam Series (www.bcigamejam.com). Examples include Sumo Bootle[186], a multiplayer game requiring precise BCI activation for directional control to defeat opponents in a sumo ring, and Alex Runs, an immersive 2D obstacle avoidance game where players must adeptly time activations for their character to jump over obstacles and evade traps.[199] Additionally, participants received relevant switch-adapted devices or a PowerLink™ to provide switch control for electronic appliances (lights, blender, etc.) according to personal goals. The single goal chosen by the participant, or their parent was the main outcome, with options to try other things.

5.3.4 Session Protocol

Families attended nine virtual 1-hour BCI sessions with the BCI researcher over the course of 12 weeks to monitor BCI skill progression and facilitate regular BCI use at home. Prior to beginning the 9 BCI sessions, a home-visit was performed to ensure the child was properly positioned in the preferred seating system to use the BCI technologies and that the equipment was adequately positioned for the child to see the software training screen and/or the switch interface along with all required equipment for the child's personal goal. In addition, personalized mental commands were identified for each participant, including imagined actions (e.g., visualizing pushing their wheelchair) and cues to remain calm and relaxed during the neutral state (e.g., counting backward from 10). Some participants employed visual cues, while others used auditory cues to aid in training personalized mental commands.

At the beginning of each session, parents reported the frequency of their independent BCI usage between sessions. The weekly usage frequency was categorized as high (3+ times), moderate (2-3 times), or low (0-1 times). This was followed by BCI calibration. Calibration trials were each 8 seconds long and users received a "consistency score" via Emotiv's proprietary API. This score was generated after a minimum of trials were trained for a mental command by comparing the most recent trial to the previous ones. The total number of trials varied across sessions, as low-scoring or artefactual trials were rejected and retried up to 3 times to minimize frustration of the participant. A total of 8 acceptable neutral and 8 active command training sets were accepted for each participant in each session.

After training, children selected activities for their session but performed the primary goal activity during the first (baseline), fifth (interim), and ninth (final) sessions to evaluate online BCI performance, which is not reported here.

5.3.5 Outcome measures

Children or parents' self-reported COPM rating of performance and satisfaction were the primary subjective outcomes. The COPM is a personalized tool for setting and assessing individual goals related to everyday challenges that restrict participation in everyday activities.[200] With proven reliability and validity in pediatric settings, the COPM is widely used as a subjective outcome measure in children's clinical trials.[201] COPM assessments were administered before the study (initial) and within a week of study completion (final) to evaluate performance and satisfaction changes in the goal activity. Baseline and reassessment scores were compared, with a 2-point or greater change indicating clinical significance.[201] The BCI-adapted version of the Quebec User Evaluation of Satisfaction with Assistive Technology (e-QUEST2.0) was used to assess parents' satisfaction with various aspects of the BCI system.[87] The e-QUEST2.0 incorporates eight items from the original QUEST 2.0 questionnaire (headset dimensions, weight, adjustment, safety, comfort, ease of use, efficacy, and professional services), which was designed for a wide range of technology, and added four items specifically for BCI evaluation—reliability, speed, learnability, and aesthetic design. Users rated items on a 5-point satisfaction scale (1 = completely unsatisfied, 2 = somewhat unsatisfied, 3 = moderately satisfied, 4 = fairly satisfied, 5 = extremely satisfied). Finally, parents were asked to indicate the three most important components.

5.3.6 Data analysis

Data was analyzed using GraphPad Prism version 9 (GraphPad Inc.). We conducted normality tests (Shapiro-Wilk and Kolmogorov-Smirnov) and proceeded to use parametric or non-parametric tests as appropriate. A one-way repeated measures ANOVA was performed to examine differences in BCI consistency between sessions. Post hoc tests were conducted using Tukey's multiple comparisons test to identify significant differences between specific sessions. To evaluate differences in COPM scores between initial and final assessments, a Wilcoxon matched pairs signed rank test was performed. To examine differences between the various aspects of the BCI system as scored by parents on the eQUEST2.0, a Kruskal-Wallis test was performed. The significance threshold (alpha) was set at 0.05.

5.4 Results

5.4.1 Participant characteristics

Seven children (mean age = 11.1 years, range = 6-15, 2 female) with quadriplegic CP and diverse diagnoses participated in this study (Table 1). Each participant had a unique active command, while 3/7 participants utilized a similar neutral command. The goals and independent use levels varied among participants, with most goals (57%) falling into the "leisure" category, followed by "productivity" (29%) and "self-care" (14%). Participant 1 had moderate independent use and focused on making cards for friends and family. Participants 2, 3, 4, 5, and 6 had low independent use, with goals ranging from turning on lights and playing board games to improving mobility and self-regulation through music control. In contrast, Participant 7 demonstrated high independent use, focusing on food preparation (Fig. 5.1).

Figure 5.1. Participant 7 using the BCI system to control a stand mixer to participate in their goal activity of food preparation.

Table 5.1 Participant Demographics, Commands, Goals, and Activities in BCI Sessions

ID	Sex	Age	Diagnosis	Comm. Method	Neutral command	Active command	Goal (COPM domain)	Activities (frequency)	Independent use*
P1	M	11	Bilateral perinatal stroke	PODD Book	Imagine being Peter Parker (still, calm, no superpowers)	Imagine being Spiderman (superpowers, propelling himself quickly in wheelchair)	Making cards for friends and family (Leisure)	SpinArt (6/9) HKL (2/9) Alex Runs (1/9)	Moderate
P2	M	10	Bilateral schizencephaly	Parent-assisted	Counting backwards	Visualize pushing the Sphero® ball	Turning on the lights (Productivity)	Lights (3/9) HKL (6/9)	Low
P3	M	15	Kernicterus	Eyegaze	Thinking of nothing	Imagine sticking out tongue	Going to the park with cousins (mobility) (Self-care)	Driving (3/9) Alex Runs (3/9) Sumo Bootle (3/9)	Low
P4	M	14	Extreme prematurity	Verbal	Imagine being wrapped in yellow, fuzzy blanket	Imagine puckering lips (to kiss mom)	Playing board games with brothers (Leisure)	Dice roller (3/9) Alex Runs (6/9)	Low
P5	M	12	Kernicterus, prematurity	Parent-assisted	Counting backwards	Imagine "Go"	Music control (for self-regulation) (Leisure)	Spotify (6/9) HKL (3/9)	Low**
P6	F	10	Bilateral schizencephaly	Parent-assisted	Counting backwards	Imagine being pushed on skateboard	Playing games with family (Leisure)	Card shuffler (3/9) HKL (6/9)	Low
P7	F	6	Hypoxic ischemic encephalopathy	Parent-assisted	Singing "Five Little Ducks" song	Imagine racing – "Go"	Food preparation (Productivity)	Blend food (8/9) HKL (1/9)	High

**Frequency of average weekly use outside of BCI sessions: Low = 0-1 times; moderate = 2-3 times; high = 3+ times*

**BCI use low, but practiced neutral and active command without BCI daily

5.4.2 BCI Consistency

Participants exhibited a wide range of BCI consistency scores (Fig. 5.2), with mean minimum scores for one session ranging from 11.83 to 63.83 and mean maximum scores spanning from 79.33 to 99.83. The range of BCI consistency scores also varied among participants from 25.83 to 88.00. The mean BCI consistency scores across the nine sessions ranged between 54.20 and 79.30, indicating diverse levels of performance. Within session variability was large between participants, with standard deviations ranging from 10.29 and 28.31.

The one-way repeated measures ANOVA conducted on the BCI consistency data from nine sessions involving seven participants showed no statistically significant improvement in consistency across the sessions ($F(3.70, 22.2)=1.55$, $p=0.22$). However, there was a significant effect of individual participants on BCI consistency ($F(6, 48)=2.76$, $p=0.02$). Overall, 20.8% of the variability in BCI consistency was explained by the differences across sessions, while 21.5% of the variability was attributed to the differences between participants. The results of the post hoc tests revealed no significant differences in BCI consistency between any of the sessions.

5.4.3 COPM

The initial and final COPM scores for performance and satisfaction showed clinically significant improvements (Fig. 5.3). The initial performance scores ranged from 1.0 to 5.0 with a mean of 1.71, while the final scores ranged from 8.0 to 9.0 with a mean of 8.57. Similarly, initial satisfaction scores ranged from 1.0 to 4.0 with a mean of 1.57, and final scores ranged from 8.0 to 10.0 with a mean of 9.0. The Wilcoxon matched-pairs signed rank test revealed significant differences between initial and final COPM scores for both performance (p=0.0156, median difference of 7.0) and satisfaction (p=0.0156, median difference of 8.0).

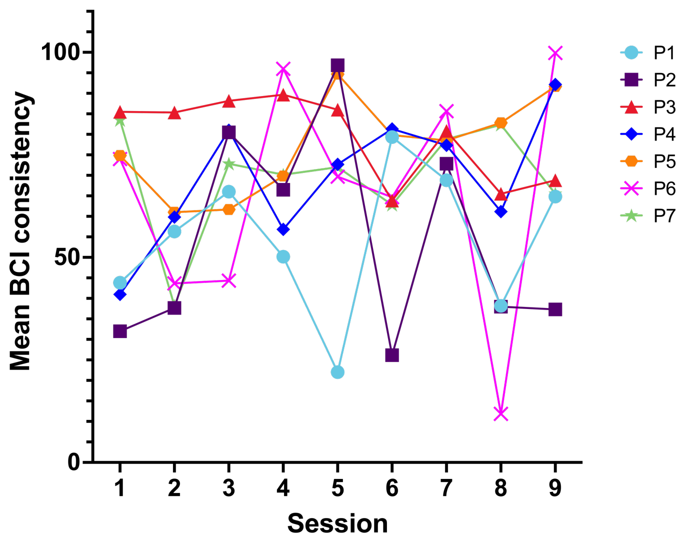


Figure 5.2. Mean BCI consistency for each participant during each BCI session.

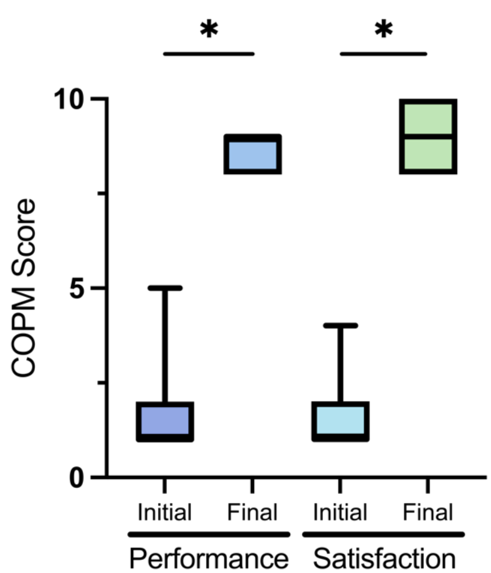


Figure 5.3. Mean COPM Scores for performance and satisfaction before (initial) and after (final) BCI intervention. * $p < 0.05$

5.4.4 eQuest2.0

Parents were most satisfied with the professional services offered by the BCI team, as demonstrated by an average rating of 4.71±0.49, and least satisfied with the adjustability of the BCI system (3.14±1.22) (Fig. 5.4). The average ratings for the other aspects were as follows: headset dimensions (3.86±0.70), weight (4.43±0.79), safety (4.43±0.53), comfort (3.57±0.96), ease of use (4.0±0.82), effectiveness (3.86±1.07), reliability (3.86±0.90), speed (4.0±0.82), learnability (4.0±1.0), and aesthetic (3.29±1.11). The top three most important aspects of the BCI system that were named by parents, were comfort, ease of use, and effectiveness. Scores between each aspect were not significantly different.

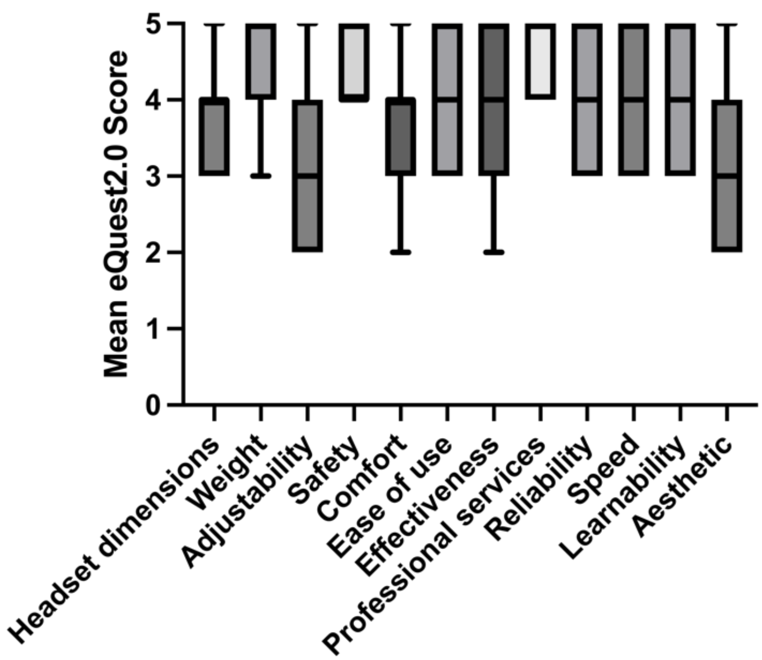


Figure 5.4 Mean score for each item of the eQuest2.0 questionnaire.

5.5 Discussion

This study investigated home-based BCIs in seven children with quadriplegic CP with diverse diagnoses, independent use, and goals. Our results demonstrated significant, clinically relevant improvements in personalized goal achievement and suggest that home-based BCIs hold significant potential for facilitating participation in meaningful life activities for children with quadriplegic CP.

Children with quadriplegic CP used BCI to pursue diverse goals across leisure, productivity, and self-care domains. These goals included creative activities, environmental control, mobility enhancement, family leisure, music-based self-regulation, and food preparation skills. This diversity underscores the importance of versatility in BCI systems for addressing the distinct needs and aspirations of children with severe neurological disabilities. Metzler et al. 2020 reported that leisure-related goals were most prevalent among a large sample of school-aged children with unilateral CP participating in an interventional trial.[202] Goal distributions differed between this and the present study, with children with unilateral CP setting more self-care related goals (45%). These differences suggest that CP type and severity may influence goal prioritization, emphasizing the need for individualized, client-centered intervention planning. The same study also found that the type of goal set did not have a strong influence on goal achievement, suggesting children should be free to set the goals that matter most to them. This principle remains to be validated for BCI but seems reasonable to assume in the design of future studies.

Most participant families exhibited low independent use levels. However, children displaying higher independent use levels, such as practicing mental commands without the BCI (e.g., P5), tended to engage more frequently in their goal activities during BCI sessions. This observation implies varying degrees of motivation and engagement among participants and families. Specifically, children and families who demonstrate greater engagement in their goal activities may experience increased BCI usage both during and outside of sessions. Their motivation for achieving their goals could encourage more practice and ultimately lead to enhanced proficiency in using the BCI system. Future work should focus on strategies to facilitate independent BCI use, such as developing motivational interventions, incorporating gamification elements, and providing personalized feedback to increase user engagement and drive higher levels of practice. By fostering greater motivation and commitment to goal activities, these approaches have the potential to improve BCI proficiency.

BCI consistency scores varied widely among participants, reflecting the challenge of intra- and inter-session variability in BCI implementation.[203] Individual differences among participants significantly affected the consistency scores, and may be attributed to differences in mental commands, cognitive style, and learning processes. Brain topography, influenced by participants' unique injuries and neurodevelopment, may also contribute to this variability, emphasizing the need for personalized BCI approaches.[203] Future research should explore the connection between brain topography, structure, network dynamics, and BCI skill acquisition for this population.

Each participant had a distinct active mental command/visualization, demonstrating individual differences in how they interacted with and controlled the BCI system. However, 3/7 participants shared a similar neutral command, indicating that there might be some commonalities in BCI usage among certain children with quadriplegic CP. This highlights the importance of considering individual differences while also identifying potential shared strategies when designing and implementing BCI systems for children with quadriplegic CP to ensure effectiveness and usability.

On average, participants and families tended to rate their performance and satisfaction in their goal activity as very low prior to this BCI intervention. This is expected (higher rankings would presumably lower the priority as a goal) and in keeping with COPM scoring in other populations of children with CP. Although BCI consistency did not show marked improvements over time, the participants' perception of their performance and satisfaction after BCI intervention improved significantly. This may suggest that the intervention positively impacted their overall experience and self-perceived abilities, as well as the sense of accomplishment in goal-oriented activities. That this was observed in the face of minimal direct BCI "learning" suggests enhancements to achieve higher rates of learning could result in even greater impact. BCI learning over time is only poorly understood in adults and remains virtually unstudied in children. Our home-based study may have lacked sufficient intensity or duration of intervention to demonstrate learning effects, but the convenience of the approach may create future opportunities to increase dose of learning in a patient and family-centered manner.

Overall, parents' satisfaction scores for the BCI system were consistent across aspects. While parents were highly satisfied with the professional services provided by the BCI team, there is room for improvement in the adjustability, comfort, dimensions, and aesthetic of the BCI hardware. The same aspects have also been rated the lowest by adults using BCI at home.[127] Adult participants similarly named effectiveness and comfort as the most important aspects of a BCI system, which was reflected by parent responses in the current study.[127,193] This indicates that priorities for BCI systems are aligned for both pediatric and adult users. While service provision was considered a positive by families, this requires substantial expertise and time so that future efforts to optimize delivery mechanisms and efficiencies will be required. Future research and development should involve refining the BCI design for enhanced adjustability and comfort, optimizing the system's effectiveness, and considering parental feedback to address their priorities and concerns for a more user-centered experience. This may involve conducting user-centered design studies[85], gathering more feedback from users, and iterating on the system design to address these concerns.

5.6 Conclusion

Our study demonstrates that simple BCIs hold significant potential for enhancing participation in meaningful life activities for children with quadriplegic CP and their families. Individual differences and unique goals necessitate personalized approaches to BCI design and implementation. Shared strategies may exist among certain children and families, emphasizing the importance of identifying commonalities and developing programs collectively. Motivation and engagement appear to be key factors in BCI goal-achievement and proficiency, suggesting future work should focus on strategies to facilitate independent use and enhance user commitment. Additionally, user satisfaction highlights the need for refining BCI design and incorporating user-centered feedback to improve comfort, adjustability, and overall effectiveness. By addressing these areas, BCIs can better serve the diverse needs and aspirations of children with quadriplegic CP.

Chapter 6: Separated by distance, united by technology: The psychosocial impact of brain-computer interfaces for online video game play for children with quadriplegic cerebral palsy

Dion Kelly, Erica D. Floreani, Danette Rowley, Adam Kirton, & Eli Kinney-Lang

6 Chapter 6

6.1 Abstract

Thousands of Canadian children are living with profound physical disabilities, rendering them unable to sufficiently participate in play, which is their fundamental right. Video games are played by 90% of children. Unfortunately, commercial gaming systems are inaccessible for children with quadriplegia. Brain-computer interfaces (BCIs) are an emerging technology that could provide a means of access to play video games through the manipulation of electrical brain activity. In this pilot case study, we assessed the feasibility and explored the psychosocial impacts of using BCI as an access method for online video game play for children with quadriplegic cerebral palsy during the COVID-19 pandemic.

Four children with quadriplegic cerebral palsy and their families participated in online remote play of a multiplayer BCI video game from their homes over a six-week period. Various stakeholders responded to user feedback surveys and the Psychosocial Impact of Assistive Devices Scale (PIADS) before and after gaming to explore perceptions and impacts of BCI-facilitated participation in online multiplayer video game play.

Two categories emerged from pre-gaming surveys: 'dependent participation' and 'non-physical interaction/engagement'. 'Ability to participate' and 'sense of connection' emerged as categories from post-gaming surveys. These categories were aligned with the participants' PIADS scores, which increased as a result of BCI-facilitated gaming. Our results demonstrate that BCI-facilitated access of an online, multiplayer video game at home can enhance the quality of life for children with quadriplegia through increased competence, enhanced participation and sense of connection.

6.2 Introduction

Most children with severe quadriplegic cerebral palsy cannot walk, talk, or use their hands. Many are cognitively normal, leaving them locked inside a body that does not work.[31] The United Nations Convention on the Rights of a Child (UNCRC) acknowledges that children with disabilities have the right to enjoy a full and adequate life in environments that ensure dignity, promote independence, and foster their participation in society, including the fundamental right to engage in play.[40] Children with severe physical disabilities are significantly disadvantaged in their ability to exercise these rights, with limited opportunities for participation.

The mental health and psychosocial wellbeing of children with profound physical disabilities is often poor when compared with their able-bodied peers.[204] A study of children with cerebral palsy demonstrated a significantly higher prevalence of mental health problems compared to the broader pediatric population.[205] Health-related quality of life has also been found to be poorer in children with cerebral palsy.[206] A large, multi-study project examining the relationship between quality of life and participation of 1174 children with cerebral palsy (SPARCLE) found that severity of impairment is weakly related to quality of life, but restricted participation is strongly related to quality of life.[37] To improve quality of life for children with significant physical disabilities, interventions must focus on increasing opportunities for participation and play.

The UNCRC states that every child has the right "to engage in play and recreational activities appropriate to the age of the child".[40] A 2020 report published by the Entertainment Software Association of Canada revealed that almost 90% of Canadian children ages 6-17 years play video games.[207] The majority of psychological research on video gaming has focused on the negative outcomes; particularly the impact of violent video games and amplification of aggressive behaviour.[208] However, given the drastic evolution and increasing pervasiveness of video games over the past few decades, a wide variety of games now exist that are more complex, realistic, and social in nature. With this, a growing body of research has demonstrated positive effects of playing video games.[209] Developmental psychologists now suggest that engaging in video game play has valuable benefits for social, cognitive, and emotional development, as well as the potential to enhance mental health and well-being.[210] With growing access to Internet, the number of people playing games online has been steadily increasing. This has proved to be important for maintaining socialization throughout the COVID-19 pandemic, during which social participation had to be redefined. Online multiplayer games encourage interaction and allow players around the world to enjoy a shared activity, resulting in opportunities to collaborate, learn, build friendships, or simply have fun.[211] Unfortunately, commercial gaming systems generally remain inaccessible for children with quadriplegic cerebral palsy because their physical abilities do not match the ability-assumptions of the hardware (e.g., bimanual hand coordination) or the software (e.g., full access to all controller functions).[212] Children with physical disabilities are thus excluded from online multiplayer gaming and the potential psychosocial benefits that it may confer.

Kinetic access methods that require coordinated physical manipulation range from being extremely difficult to impossible for children with quadriplegia to use. Alternative access methods such as eye-gaze systems, which involve tracking movements of the eye around a computer screen[46], can still present critical issues for children with quadriplegic cerebral palsy, whose involuntary and uncontrolled movements restrict the system's ability to accurately and reliably track gaze.[47] Therefore, due to their strong dependence on sufficient motor control and the assistance of others, current assistive technologies may have limited benefit for children with severe cerebral palsy. There is urgent need for novel solutions that bypass the requirement of voluntary, reliable motor control and coordination.

Brain-computer interfaces (BCIs) are an emerging technology that could offer children with quadriplegia new opportunities for social development and participation. BCIs function by non-invasively acquiring, analyzing and translating brain signals into commands to control an external device.[72] This mental control can be achieved by processing patterns in electrical brain activity (referred to as electroencephalography, or EEG) in real time. These patterns can be produced in response to the presentation of external stimuli or they can be generated internally by imagined movements.[72] We have already shown the positive impact that BCIs have for providing new avenues of independence for children with significant disabilities[103,133,170,190,213] and home-based gaming applications could maximize participation while training BCI skills for other practical applications in life.

In November 2019, we organized the first North American BCI Game Jam to motivate the development of BCI-compatible video games specifically for children with severe neuromuscular disabilities (https://bcigamejam.com/).[186] Nine games were submitted by 30 participants across North America. The most popular game was *Sumo Bootle*—a local multiplayer, turn-based, battle-themed game akin to sumo wrestling, where players bump into each other to try to push each other outside of a sumo ring. Given the challenges with socialization and physical interaction amidst the COVID-19 pandemic, especially for children with increased vulnerability, we sought to establish a new outlet for connecting with family members and peers by enabling online peer-to-peer multiplayer of *Sumo Bootle* to allow children to play together from anywhere in the world. In doing so, we provided a unique opportunity for children with severe quadriplegia in separate households to engage in online video game play together via BCI. To demonstrate the feasibility of such an intervention and to explore associated perceptions from various stakeholders, we implemented a pilot survey study with a sample of children and families from our clinical BCI research program, BCI4Kids (www.bci4kids.com). The rest of this paper describes the intervention and an assessment of the potential psychosocial impacts of using BCI to enable multiplayer online gaming.

6.3 Methods

6.3.1 Recruitment

Children with quadriplegic cerebral palsy enrolled in *BCI4Kids* clinical research program at the Alberta Children's Hospital (Calgary, Canada)[103] were invited to participate in this pilot study. Inclusion criteria for the research program and for this study were: (1) age 6 to 18 years; (2) severe physical disability including non-ambulatory (Gross Motor Function Classification Score Level V)[198] and minimal to no hand use (Manual Ability Classification System Level IV or V)[189]; and (3) evidence from parents, teachers, and/or therapists of at least grade 1 level cognitive capacity[103]. Prerequisite criteria to be included in our sample was successful independent setup of the BCI equipment by parents at home on at least three separate occasions (confirmed by BCI researchers) and verbal confirmation by parents of setup competence. This was done to ensure a sufficient level of comfort with the basic BCI hardware and software before introducing new software programs required for online gaming. Participants from the BCI4Kids program were invited to play *Sumo Bootle* against other children from the program, their friends, and their family members, who we refer to as the 'game opponents'. Game opponents and parents were also invited to complete surveys to provide their perspectives on the experience of online multiplayer gaming with BCI. Parents or guardians provided written informed consent. Methods were approved by the University of Calgary Research Ethics Board REB15-2567.

6.3.2 BCI Equipment

Brain signals were acquired using the Emotiv EPOC X headset (Emotiv Systems Inc., USA), which is a low-cost, wireless, consumer-grade EEG headset. The EPOC X has 14 channels (positions AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4) and utilizes saline-based semi-dry electrodes for signal acquisition. The sampling rate of the headset is 128 Hz. The EPOC X was chosen for its relative simplicity to set up and favourable tolerability by participants.[104]

The EPOC X system employs a "mental command" paradigm, where the user engages in repetitive visualization of a mental thought or action with the assistance of visual neurofeedback. The specific mental thought or action to be rehearsed is chosen to be salient to the user (e.g., imagining pushing a large box). During training, a series of trials of the rehearsed mental command are collected and compared to a series of "neutral" or "rest" trials, generating a classifier that can predict the execution of the mental command in real time. The EPOC X system processes and classifies data through Emotiv's proprietary cloud-based service, *Cortex*, so additional details of the data processing pipeline are unknown. We hesitate to refer to this paradigm as traditional motor imagery commonly used in BCI studies, as the EPOC X headset has limited coverage over areas of the brain commonly associated with imagined-movement related desynchronizations in oscillatory activity.[175]

6.3.3 Software & Online Gaming Interface

Acquired brain signals from the EPOC X headset were transmitted a Microsoft Surface Pro laptop computer via Bluetooth and processed through the *Cortex* cloud service. Resulting mental command detections were received at a rate of 8 Hz, downsampled, and translated to simulated keyboard commands for game control at a rate of 2 Hz via a custom-developed desktop application. This desktop application was built using the Electron framework employing the NodeJS runtime. In addition to mapping mental commands to simulated keystroke outputs, the *Emotiv Games Hub* application allowed families to monitor headset contact quality and adjust system parameters. This custom-developed software is described in further detail in Floreani et al. (2022).[190]

Simulated keystrokes from the *Games Hub* application were received by the *Sumo Bootle* video game as input. *Sumo Bootle* is played in turns, with players attempting to time their input (a single button press) to ram their character into other characters around the ring. The input will cause the character to move in the specified direction a predetermined amount. Successfully running into an enemy character will deplete the opponent's health bar and slow movement. Visual feedback for timing movement is provided by an arrow rotating in a limited arc. The game is over once all but one character's health has been reduced to 0. See figure 1 for a detailed visual of the game.

The Steam Gaming Platform (Valve Corporation, USA) was used to configure a solution for playing *Sumo Bootle* online. Steam offers a "Remote Play Together" feature, enabling one of the players to temporarily act as a host server for online gaming with another user, effectively establishing a peer-to-peer connection. Discord, (https://discordapp.com) was used for video conferencing so that children could see each other while playing together (Fig. 6.1). Research personnel utilized Remote Utilities® remote desktop software (https://www.remoteutilities.com/) to connect to participants' computers to assist with software setup and management of technical difficulties. Running the *Games Hub* application, Steam for online play, Discord for video conferencing and Remote Utilities for desktop control sometimes led to high resource consumption on the laptop computer. If this became an issue during the sessions, we switched video conferencing to a mobile platform (e.g., FaceTime, WhatsApp, etc.) to offload resource demands on the computer. Additionally, once the gaming interface was set up, research personnel could log off the remote desktop application and monitor sessions via the video conferencing platform and offer verbal guidance rather than direct control.

6.3.4 Study Procedures

Each participant received a "Home BCI Kit", consisting of an Emotiv EPOC X headset and Surface Pro laptop with all software pre-installed. The Home BCI Kit also came with knowledge guides and video instructions to help families set up the headset and run BCI calibration independently. Immediately after receiving a Home BCI Kit, parents attended a virtual orientation session with research personnel to become familiarized with the contents of the kit, and to do a guided run-through of the equipment and software setup. Personal accounts were created for each participant on Steam and Discord, and other participants were added as friends. Family members external to the BCI4Kids program who participated as game opponents also attended a virtual orientation session so that research personnel could setup their Steam and Discord accounts and download *Sumo Bootle* onto their personal computer. Gaming sessions were scheduled by parents with matching availability and were run virtually with support from the research team. Immediately prior to the gaming session, families independently set up the headset and calibrated the BCI by training their personalized mental command (Table 6.1). Then, families signed on to the Discord video interfacing platform along with the research personnel. At this point, the research personnel logged onto the participant computers via the remote desktop platform to set up the online video game stream via Steam and remained throughout the gaming session to monitor for any technical issues and to offer encouragement to participants. Up to four players could play *Sumo Bootle* at once. When the title screen appeared, players joined the game by executing their trained mental command, which was mapped to a unique key on the keyboard. Players were required to sustain activation of their mental command until their player was 'loaded'. Once all players were loaded, the game began. Players were allocated turns in a clockwise fashion starting with the player at the top of the ring. An arrow circled around the player's Bootle character (Figure 6.1). Players were required to activate their mental command when the arrow was pointing in the direction that they wished to move, with the goal of moving towards and bumping into another player. Players could also run into health boxes to increase their health and longevity in the game. The round continued until there was only one player left in the ring, at which point they were crowned the winner of the round. Players were permitted to play the *Sumo Bootle* game for as long as they liked.

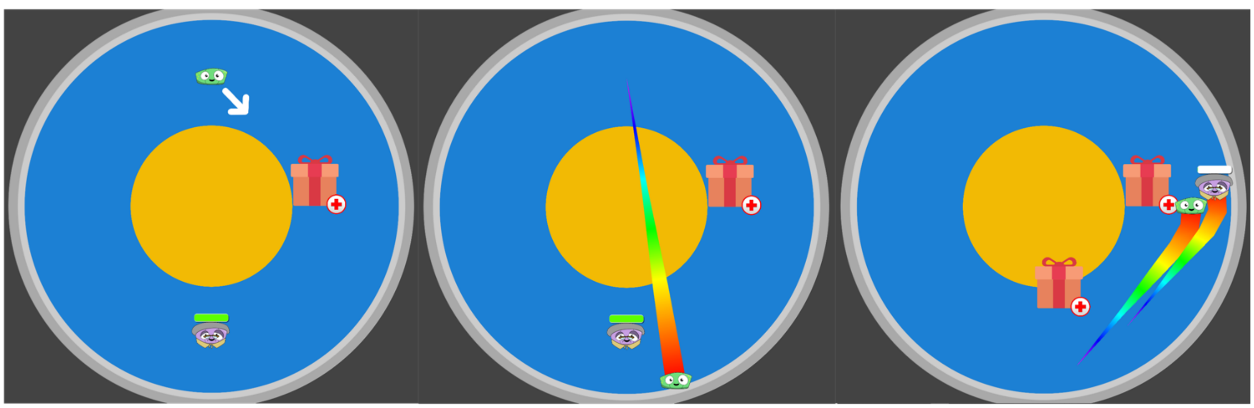


Figure 6.1. Screen shot of Sumo Bootle game play. An arrow circled around the player's Bootle character. Players were required to activate their mental command when the arrow was pointing in the direction that they wished to move, with the goal of moving towards and bumping into another player to knock them out of the ring or towards the health box to increase their health. Participants utilized the "Remote Play Together" feature of Steam to enable online streaming of Sumo Bootle. Participants used Discord to see each other and interact while playing.

6.3.5 User feedback surveys

Surveys were administered before (pre-gaming survey) and after the first BCI gaming session (post-gaming survey) to participants with reliable communication methods. Friends and family members who participated as game opponents and parents/caregivers facilitating BCI set up and use were also surveyed. Pre-gaming surveys with game opponents consisted of four open-ended questions, one yes/no question, and two rating questions to gauge perception of the participants' baseline ability to play with their game opponent both in person and not in-person. Pre-gaming surveys with parents/caregivers consisted of four open-ended questions, two yes/no questions, and two rating questions to gauge their perception of their child's ability to interact and engage in play with friends and family both in person and not in person. Post-gaming surveys consisted of three open-ended questions that allowed game opponents and parents to share their opinions about the online BCI gaming experience and their feelings about future opportunities for the participant.

Survey questions were administered by a designated researcher (DK) via telephone or videoconferencing. The answers were tape recorded and transcribed verbatim (DK). Initial transcriptions were then verified for 100% accuracy by a second researcher (EDF). The transcripts were analyzed using the Grounded Theory approach[214], which best suited the explorative nature of this research. Using a Computer Aided Qualitative Data Analysis Software (CAQDAS) package, Delve (www.delvetool.com) transcripts were broken down into conceptual components using an inductive coding procedure to group similar concepts. Similar concepts were then merged to give rise to two main categories from the pre-gaming surveys (dependent participation and non-physical interaction/emotional engagement) and two categories from the post-gaming surveys (ability to participate and sense of connection).

6.3.6 Assessments

Prior to engaging in the online gaming session, participants responded to 15 statements from the children's version of the Psychosocial Impact of Assistive Devices Scale (PIADS). The questionnaire was repeated following the online gaming intervention. The Children's PIADS is a self-report questionnaire with five-point Likert-type Smiley Face Scale responses ranging from "never" to "always" to assess the impact of assistive devices on quality of life.[215] The items are subdivided into three subscales reflecting key dimensions of quality of life: competence (7 items), adaptability (3 items), and self-esteem (5 items). The highest score for any statement is 5, therefore the highest possible score for the competence subscale is 35 (lowest score=7); for the adaptability subscale is 15 (lowest score=3); and for the self-esteem subscale is 25 (lowest score=5). Higher scores represent better psychosocial impact of the BCI in influencing the child's subjective well-being.

Statements were framed relevant to multiplayer gaming without using a BCI (non-BCI gaming e.g. playing Monopoly) for baseline assessment and when using a BCI (BCI gaming with Sumo Bootle video game) for endline assessment of the intervention. Children responded to statements using their established communication method as detailed in Table 6.1. Statements were read out loud followed by the five potential responses (never, almost never, sometimes, almost always, always). At the same time, the visual of the Smiley Faces was shared via videoconferencing by the researcher or by parents who had a copy. Non-verbal participants indicated 'yes' to the response they wanted to select for each statement. Likert-type Smiley Face answers were converted into numerical responses and scores for items in each subscale were summed to get the subscale score. Overall scores were analyzed to determine the psychosocial impact of using a BCI for multiplayer gaming. Paired sample t-tests were performed to compare the psychosocial impact BCI-facilitated gaming versus gaming without using a BCI.

6.4 Results

6.4.1 Participants

Four participants (P1-P4, median age=10.5 years, all male) from the BCI4Kids clinical research program participated in this pilot study. Participant details, including prior experience in the BCI4Kids program, can be found in Table 6.1. All participants were able to play against at least one other participant at least once, and some also played with other family members (F1, F3) as outlined in Table 6.1. Over a six-week period, participants engaged in 2-5 online gaming sessions with a total of 12.1 hours of game time.

Table 6.1. General information about participants

Participant	Age	Communication Method	Mental Command	Approximate BCI Experience	Game opponents	Gaming sessions	otal game play me
P1	13	Verbal	Imagine "push" with arms	4 years in lab; 3 months at home	P2, F1 (twin brother)	3	3.4 hours
P2	13	Non-verbal: reliable communication with eye movements ('yes' = up/down eye movements; 'no' = left/right eye	Attempt lifting left arm	1.5 years in lab; 3 months at home	P1, P3, P4	5	5.3 hours

		movements), some parent-assisted interpretation with subsequent confirmation by the participant					
P3	8	Non-verbal: yes/no signals through facial expressions, parent-assisted interpretation	Imagine "push" with arms	1.5 years in lab; 7 months at home	P4, F3 (cousin, same age)	2	2.1 hours
P4	9	Non-verbal: uses Pragmatic Organisation Dynamic Display (PODD) book, has some left-hand function that permits pointing to responses, some parent-assisted interpretation	Attempt pushing onto wheelchair pommel	1 year in lab; 7 months at home	P3, P2	2	1.3 hours

* Participant codes (P1-P4) are provided in the left-hand column. When playing against other participants from this cohort, their same code is used in the "Game Opponents" column. If participants also played against family members, they were denoted by the letter "F" followed by the participant's number.

6.4.2 Surveys

Pre-gaming surveys: perceptions of the BCI user's ability to engage in play when in-person and not in-person

P1, P2, F3, and each participant's parent (M1-M4) completed pre-gaming surveys (n=7). P2 communicated their responses with the help of their mother, who interpreted the responses and which they subsequently confirmed using their 'yes/no' communication via eye movements as described in Table 6.1. Two main categories emerged from the analysis of the pre-gaming surveys: dependent participation and non-physical interaction/emotional engagement.

Dependent participation. The concept of 'dependent participation' was referenced by six out of seven respondents, who remarked that participation in play, when both in-person and not in-person, is dependent on factors such as who is around, how much time they have, positioning of the child and their assistive technology, and what assistive technology they have.

Every parent indicated that participation in play when in-person was dependent on help from someone: "Someone is always manipulating for [him]…he can't physically do it. He needs someone else to do it"(M1); "…if the joystick wouldn't have been comfortable to hold, his hand, like my hand over his hand on the joystick, it would have been troublesome"(M2); "it becomes a huge job for [us] to help him engage with others"(M3); "It's just not possible… and realistically, he can't really hold [a controller]. There's just no way to do it. So, I guess we've kind of tried like "okay use his fingers" but then there's just like all these movements that we can't even hold his hands on it really"(M4). Similarly, F3 stated that P3 "needs help to play with us". P2 also conveyed that his participation was dependent on having someone present who could understand him and what he is trying to communicate.

Some parents remarked about their child's in-person participation in play being dependent on the amount of time they have: "When we have time he's actually like willing to, he's actually happy to play a board game whenever we have time"(M2); "In situations where we've tried board games and stuff like that it's like there's a limited amount of time"(M4).

Positioning was commented on by multiple parents, who indicated that it had to be right for the child to participate in play: "If you were to say be building blocks or be wanting to knock down blocks, he can't really do that on his tray because usually where he's building it's out of his reach unless he's laying down on the floor, and then he needs to be in the perfect position to be able to reach them"(M3); "he can't be close to the game board because things go flying"(M4). Positioning of the assistive technology itself was also commented on: "They transferred [the eye gaze device] and now the mount is not in the right position and at home it was not working well"(M2); "…activating a switch is tricky for him. It has to be in the right spot, it can move. For us, it can be frustrating"(M3); "He tends to want to mess with the screen and stuff like that if it's too close to him"(M4).

This category is nicely summarized by a comment by M2 when asked if they felt their child was able to sufficiently interact/engage in play not in-person: "Only when he A) has a device, B) when he's on that device, and C) when somebody else has decided that he needs to go on that device."

Non-physical interaction/emotional engagement. Six out of seven respondents referred to the concept of 'non-physical interaction/emotional engagement' when talking about the participant's ability to engage in play. M1 underlined the importance of this concept in supporting their child's ability to play, despite physical limitations: "[He] can engage by sharing his thoughts but he physically can't do it". Parents and family members voiced that physical participation wasn't necessarily required for their child to participate in play: "I think that what happens with [him] is that mentally, he is in the game. Like emotionally, he is in the game. So, even though he is physically not doing it but he's totally capable of indulging into this emotional and non-physical interaction"(M2); "he is so engaged like emotionally in what other people are doing and he's vocal and he's watching so closely and he will vocalize when people will do certain things that excite him, but I mean in terms of him actually being a part of it, not really, no"(M3); "He'll be around and try to kind of make him feel engaged, but also not doing much in it"(M4). When talking about playing instruments together, F3 stated: "My sister was singing and then [P3] was behind me when I was playing the drums and he really liked it".

Yes/no and rating-type questions. Table 6.2 summarizes responses from each respondent. When asked to rank the participants' ability to engage in play in person on a scale from 1 (not able to play at all) to 10 (no problem whatsoever with playing), scores from game opponents (P1, P2, F1, F3) ranged from 1 to 8 and scores from parents (M1-M4) ranged from 1 to 10. When asked if they feel as though the participant is able to sufficiently interact/engage in play with friends and family in-person, three parents said 'no'. The one parent who responded 'yes' (M1) added that their child's ability is sufficient, but would be better with the ability to physically engage in play rather than sit on the sidelines. Half the game opponents indicated 'yes' as well. When asked to rate the participant's ability to engage in play when not together in-person, three game opponents reported a 1 and one reported a 3. Parents on the other hand reported some higher scores, with two parents reporting a 1, one parent reporting a 4, and one parent reporting a 6. When asked if they feel as though the participant can sufficiently interact/engage in play with friends and family when not in-person, all the parents said 'no'. In response to this question, all the game opponents indicated that they had never played together when not face-to-face, thus all responses were recorded as 'no'.

Table 6.2. Pre-gaming survey yes/no and rating-type question responses from parents and game opponents

Respondent	Associated participant	Ability to engage in play in-person	Is in-person ability to engage in play sufficient?	Ability to engage in play when not in-person	Is ability to engage in play when not in-person sufficient?
Parents					
M1	P1	9	Yes	1	No
M2	P2	10	Yes	1	No
M3	P3	3	No	4	No
M4	P4	1	No	6	No

Game Opponents					
P1	P2	1	No	1	No
P2	P1	5	No	1	No
F1	P1	8	Yes	3	No
F3	P3	5	Yes	1	No

Post-gaming surveys: opinions about the online BCI gaming experience

P1, P2, F1, F3, and each participant's parent (M1-M4) completed post-gaming surveys (n=8). Two categories were identified regarding the online BCI gaming experience: ability to participate and sense of connection.

Ability to participate. This category was evident in all eight transcripts. Parents in particular demonstrated excitement in underlining this category: "activities are so limited, so it's something that he loves and enjoys and interacts"(M1); "it was the first time to play with another teenager!"(M2); "it's just cool seeing him thriving at something"(M3); "I know that sometimes when we're on calls with cousins and stuff like that, they kind of get their own thing, and everybody starts a side conversation and it's just kind of chaos. But with this, that was kind of the point of it and the two of them were doing it through BCI and I think that was a really cool part"(M4). Similarly, game opponents also expressed enjoyment about the participant's ability to play with them: "[I feel] happy, because I get to play games with other friends"(P1); "With [P1] being able to play, as he normally can't with other games, I think it's a really good experience"(F1); "I really want to do it again with [P3]"(F3). P2 also conveyed to his parents that it "feels great to play with [P1]".

Sense of connection. This category was prevalent across all eight transcripts, with a sense of connection felt between both game opponents and parents. When talking about his experience playing with P2, P1 said: "I get to smile. And we both love video games!". P2 also expressed to his parents that he "liked that he had to compete with [P1]" and conveyed his excitement at the prospect of competing against more people in the future through a gaming tournament. When asked about further comments or suggestions, P2 expressed via his parents that if there were more people playing, then some people could work as a team together. P1's twin brother, F1, said: "I think that if we play video games more often, we'll be able to connect and socialize easier because we'll have something to talk about…like how he beat me because he can actually play the game".

Parents commented on the sense of connection they felt having their child participate in play with another child who has similar challenges as them: "It's one thing having [P3] playing against his cousin, but it's super cool with him seeing someone who's like him. I always think that's super powerful when people get to see someone who's similar to them"(M3); "I actually really did like the element that the friend has special needs because I think that while it would probably be fun with cousins, I think having another kid with similar kind of challenges… I don't think that they get to experience [that] in this kind of a setting"(M4). Additionally, M3 remarked how the experience allowed their child to be seen and be heard: "For him specifically, it's super awesome because people are engaging with him instead of talking to everyone else besides him because he can't respond, you know, it's a way for him to engage with other people". Along the same lines, M1 expressed optimism for connecting with new people in the future: "it will open up opportunities for him to meet people that he normally wouldn't meet and interact with and play with".

Parents also referred to the sense of connection they felt between each other through the experience: "I think it was also really cool meeting another BCI family in that setting because we've kind of run into people before in some of the other things, but we've never really played with them. It's kind of similar issues that we run into…which kind of makes it more "okay, it's not just us"…I think there was something else that [M3] mentioned that I was like "Oh we get that too""(M4). The sentiment was reiterated by M3, who said: "It was just so nice to have the boys play each other and it's even nice just to chat with [M4]. She's so lovely as well".

6.4.3 Children's PIADS

All four participants completed the children's PIADS questionnaire before using BCI for gaming and after their first gaming session (Fig. 6.2). The highest scores that could be obtained for the competence, adaptability, and self-esteem subscales were 35, 15, and 25, respectively. Scores for the competence, adaptability, and self-esteem subscales with respect to multiplayer gaming without BCI ranged from 14-27, 6-15, and 13-25, respectively. After engaging in online video game play of *Sumo Bootle* using BCI scores ranged from 27-35 for competence, 11-15 for adaptability, and 16-25 for self esteem. The largest change in score was observed in the competence subscale for P4, with an increase of 15 points. Modest changes were observed for the adaptability and self-esteem subscales, with the largest change in score for adaptability observed with P4 (5 points) and for self esteem with P3 (6 points). Comparison of PIADS scores in the competence construct between BCI-facilitated gaming and non-BCI gaming showed a significant increase in competence when using a BCI for gaming ($M=8.25$, $SD=4.99$; $t(3)=3.31$, $p=0.045$).

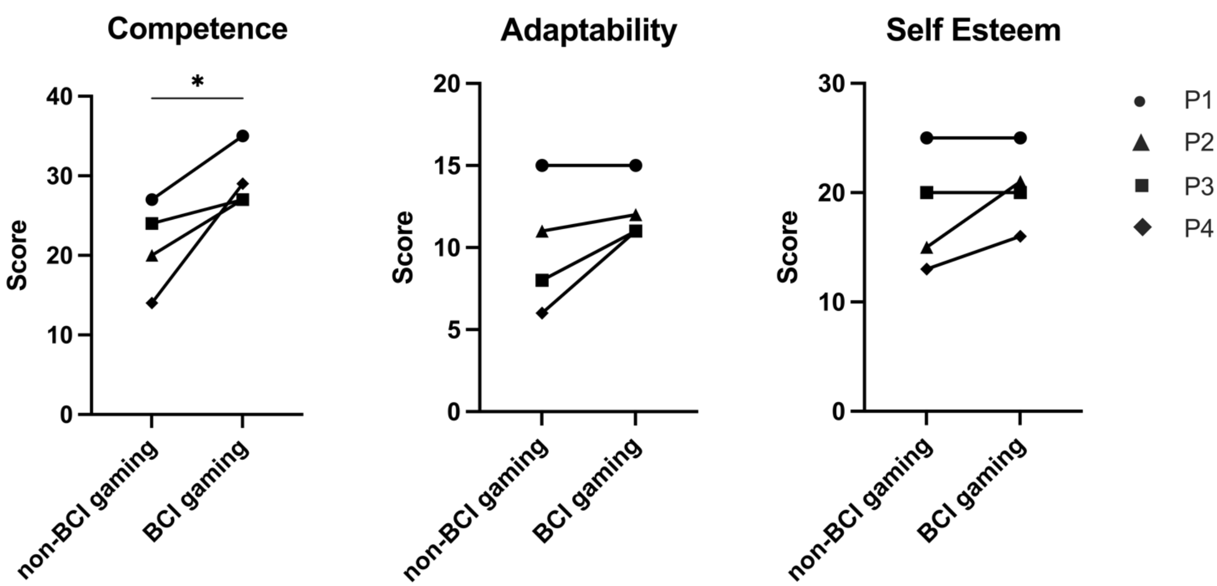


Figure 6.2. Psychosocial Impact of Assistive Devices Scale (PIADS) construct scores for non-BCI gaming and BCI gaming for the four participants. Each participant is represented by a different shaped data point. The highest (and lowest) possible scores were 35(7), 15(3), and 25(5) for competence, adaptability, and self-esteem, respectively. Statistically significant differences are shown with two-tailed paired sample t-tests, * $p < 0.05$.

6.5 Discussion

The results obtained from this study present an important view of the perspectives of children with quadriplegic cerebral palsy and their family members on participation in play when in-person and when not in-person. Over a six-week period, participants engaged in an average of 2 hours of online BCI gaming per week. In comparison, the average Canadian boy aged 6-12 years spends approximately 12 hours per week playing video games, 60% of which is online.[207] Canadian boys aged 13-17 years spend the most time playing video games out of any other demographic, with an average game play time of 21 hours per week and 78% accounting for online game play.[207] The disparities in game play time can also be observed in our small sample, with the older boys playing approximately two times longer than the younger boys. However, it is important to note that game play time for this population is also limited by parents' schedules, comfort with setting up all required hardware and software, and the cooperation of the lineup of interfacing software necessary for the realization of online BCI video game play. In multiple instances, participants signed on for their scheduled gaming session and an update was required on one or more of the interfacing software applications, which obstructed access and caused delays, leading to reduced time for game play. Understandably, these setbacks at the start of a gaming session, when children were expecting to be able to play right away, caused annoyance and wore thin on their patience. For able-bodied users, hardware and software are typically neatly integrated, readily accessible, and easy to use. These examples highlight key factors outlined in the Social Model of Disability, which describes disability as a product of societal barriers that "takes no or little account of people who have physical impairments and thus excludes them from participation in the mainstream of social activities".[216] This is echoed by the results of the pre-gaming interviews, where parents and family described the participant as having limited participation and independence when engaging in play in person, and essentially no ability to engage in play when not in person. All categories extracted from post-gaming surveys with game opponents and parents reflected positive experiences from online BCI video game play. Of note, the sense of connection, a basic human need, was not only felt by participants, but also their parents, who expressed excitement for meeting other "BCI families" and playing together. Parents of children with special needs often face more challenges, which can increase the risk of stress and depression, and feelings of loneliness, isolation, and disconnect, especially when practical and emotional resources are stretched.[217] Generating a sense of social connectedness by befriending other families who share similar challenges can influence parents' ability to cope, improve their sense of well-being, and foster a more positive outlook for themselves and their child.[217] Additionally, the reported sense of connection that is stimulated by engaging in play with children with similar challenges, is directly related to participation and can lead to feelings of support, inclusion, and capability–all of which have been described as the primary constituents of well-being by children with disabilities.[218] In particular, sports participation is an activity that facilitates the development of peer relationships, however, due to their severe motor impairments, children with quadriplegia are generally unable to participate in traditional sports and are therefore deprived of the opportunity to learn social sportsmanship skills.[219] By participating in multiplayer video game play, children with quadriplegia have the opportunity to learn sportsmanship skills, including complimenting, turn taking, being a good sport, and importantly, building friendships.[219] All interviewees expressed optimistic outlooks on the potential of online video game play for enhancing participation and social interaction, demonstrating the capacity for the activity to improve the well-being of children with quadriplegia.

Five out of eight respondents rated the participants in-person ability to engage in play as a 5 out of 10 or higher. However, when asked to rate their ability to engage in play when not in-person, seven out of eight respondents reported scores below 5, with five respondents rating the participants ability to play as a 1 out of 10 when not in-person. Every respondent expressed that the participants ability to engage in play when not in-person was not sufficient. These ratings are supported by the sentiments that were elucidated by the qualitative analysis, specifically the category of "non-physical interaction/emotional engagement". The International Classification of Functioning, Disability, and Health for Children and Youth (ICF-CY) defines participation as involvement in life situations, which reflect the interaction of the person, the activity and the environment.[220] Working from this definition, Palisano and colleagues conceptualized a model of optimal participation for children with physical disabilities as a "dynamic interaction of determinants (attributes of the child, family, and environment) and dimensions (physical, social, and self engagement) of participation.[221] The results of the pre-gaming surveys reinforce this model, as respondents articulated that physical engagement, defined as "what the child is actually doing" was poor, however the social and self engagement, respectively defined as the "interpersonal interactions that occur during the activity" and the "child's enjoyment in the moment" were consistently positively referenced and reflected in the rating-type questions. Specifically, respondents indicated that social and self engagement dimensions were just as valuable to participation as the physical dimension.

All themes extracted from interviews with game opponents and parents are evident in the participants' PIADS scores, which indicate a positive impact of BCI use on participants' quality of life. The highest possible summed score that could be obtained in the competence subscale, which evaluates feelings of competency and efficacy, was 35, while the lowest possible score was 7. The psychosocial impact scores in this construct can be interpreted by range: 7-13 is considered poor; 14-20 is considered fair; 21-27 is considered good; and 28-35 is considered excellent. With respect to non-BCI gaming, two of four participants scores were “fair”, while one was “good”, and one was “excellent”.[215] BCI-facilitated gaming increased participant scores by at least 3 and up to 15 points, bringing all scores up to the “good” and “excellent” range. The highest possible score for the adaptability subscale, which indicates a willingness to take risks and socially participate, was 15 and for the self-esteem subscale, which indicates feelings of emotional health and happiness, was 25.[215] Interpretation of scores in the adaptability construct are as follows: 3-5 is poor; 6-8 is fair; 9-11 is good; and 12-15 is excellent. For the self-esteem construct, 5-9 is considered poor; 10-14 is fair; 15-19 is good; and 20-25 is excellent. Similar to the competence construct, two participants reported scores in the “fair” range for adaptability, with the other two scores falling into the “good” and “excellent” ranges. Following BCI-facilitated gaming, scores increased to the “good” and “excellent” ranges. For the self-esteem subscale, only scores for two participants changed, with one score increasing from “fair” to “good” and the other score increasing from “good” to “excellent”. Similar to our results, a case study with an adult patient with amyotrophic lateral sclerosis using EEG-BCI at home for recreational applications reported using BCI 1-3 times per week and reported an “excellent” psychosocial impact of BCI in all three subscales.[128] Unfortunately, additional studies evaluating the psychosocial impact of BCI use are limited since non-invasive BCI has not yet transitioned from the laboratory and into end-users’ homes.

Although the impact of online multiplayer video game play using BCI has very promising potential, the primary limitations lie with the amount of software programs required to enable its use. Interfacing many applications causes an exaggerated use of the computer’s resources and lagged performance speed in the multiple software programs, including the game itself. Unsurprisingly, this leads to frustration due to waiting for the applications to catch up and compromised game control accuracy. The trade-off between cost and resources is obvious: using a more powerful computer would be more expensive and could make BCI-enabled video gaming less accessible. For BCI systems to effectively be used in the home for applications like online gaming, software interfaces must be able to run on low- to mid-range computers found in the average home. Further, a chain of several software applications introduces many opportunities for technical failures. To optimize this new outlet for individuals with severe motor disabilities, next steps must include streamlining software, improving processing speed and applying solutions that are both cost-effective and efficient.

6.6 Conclusion

The results of this pilot study highlight a simple, home-based strategy to enhance participation and independence of children with severe physical disabilities who are significantly disadvantaged in their abilities to participate in daily life. We demonstrate that something as simple as playing video games online, something that most able-bodied individuals take for granted, can have a profound psychosocial impact on children who have not had the opportunity to participate in this activity before due to inaccessibility of their physical environment. Though larger studies are needed to generalize the results, our modest sample demonstrates the potential of BCI-facilitated video gaming for enhancing the quality of life for children with severe neurological disabilities through increased participation and group play. We hope this study will motivate more groups to build on this research and develop more accessible technologies and systems for children who cannot access traditional assistive technologies, in addition to serving as a catalyst for future research on psychosocial impacts of BCI technology.

7 Chapter 7: Conclusion & Discussion

7.1 Discussion

Despite more than 3 decades of BCI development, leading to an extensive body of literature with over 10,000 publications, a major gap remains in our understanding of how BCIs can be tailored to help children. Research focused on pediatric applications accounts for less than 2% of all BCI studies. However, children, particularly those with severe disabilities, arguably stand to benefit most from BCI technology. Ignoring this population not only limits our understanding of BCI's full potential but also deprives these children of potential benefits. The scarcity of research in this area underscores the urgent need for more concerted efforts towards understanding and developing BCIs for children. In this regard, my research aimed to help fill this void and lay foundations on multiple fundamental issues relevant to pediatric BCI. The ultimate goal is to enable the widespread adoption and use of BCIs by children with severe neurological disabilities, for whom this technology could provide immediate, lifelong improvements in quality of life.

Each chapter represents a unique contribution to pediatric BCI, filling significant gaps in the existing body of research. In *Chapter 2*, we provided the first comprehensive description of baseline BCI performance in typically developing children across different age groups and BCI paradigms.[169] We demonstrated that children can competently control BCIs using a suite of paradigms with an accuracy comparable to adults. We also found an effect of age in the auditory P300 paradigm, but not other paradigms, demonstrating that performance in different BCI paradigms may not be equally affected by age. This research not only refutes previous assumptions about children's BCI capabilities but also set a new, promising trajectory for the development of BCI training programs, specifically designed for children. Therefore, in *Chapter 3*, we aimed to improve performance in P300 and SMR paradigms through child-friendly calibration paradigms that incorporated elements of gamification, such as engaging characters, quests, and rewards. Our findings validated our previous results, confirming the abilities of children to control P300-based BCIs with high levels of accuracy. They also revealed that, while gamification may enhance BCI performance for some, it doesn't serve all children equally, challenging the 'one-size-fits-all' approach to BCI calibration. This study further elucidated potential avenues for the development of strategies to optimize BCI performance in children and emphasized the need for personalized approaches.

To extend the practical application of our previous work, *Chapters 4-6* focused on transitioning BCI technology from the lab to the homes of children with QCP and their families. *Chapter 4* outlined the development of our pioneering, patient- and family-centred Home BCI Program designed specifically for children with QCP and their families. This study explored both the feasibility of this program and its impact on parents and caregivers. We found that the program was feasible and offered benefits such as enhanced flexibility, convenience, and social interaction. Our results emphasized the importance of flexible scheduling, personalized support, and the active participation of various family members as key elements for maintaining balanced caregiving and ensuring a successful, sustainable at-home BCI experience. The insights gained from this work underscore the need for further research and development in this direction to ensure that BCI technology truly meets the unique needs and preferences of children with QCP and their families.

In *Chapter 5* we utilized principles of user-centred design to evaluate the feasibility of using BCIs as an access technology for children with QCP to achieve personal goals in their home. Employing a collaborative goal-setting approach using the COPM, we assessed satisfaction and performance changes relating to goals, as well as satisfaction with the BCI system using the e-QUEST2.0, after nine weeks of use. We observed significant improvements in both performance and satisfaction and identified the top three most important aspects of BCI systems for parents for use by their child: comfort, ease of use, and effectiveness. This research demonstrated the potential of BCIs as an empowering tool for children with QCP in their home environment and emphasized the importance of user-centered design and goal setting in enhancing the usability and satisfaction of these systems.

Chapter 6 further expanded our understanding of how BCIs can bring about meaningful changes in the lives of these children. Through surveys and psychosocial questionnaires, we demonstrated that being able to participate in an activity that many take for granted, such as online video gaming, can dramatically enhance the quality of life for children who have previously been sidelined due to the inaccessibility of their environments. This study adds a new dimension to pediatric BCI research by highlighting the tangible enhancements in the quality of life brought about by such inclusive activities. Importantly, our study underscores the transformative potential of BCIs beyond their clinical or therapeutic applications and encourages further research into expanding the scope of BCI use in the everyday lives of children with disabilities.

Altogether, this body of work brings us a step closer to realizing the full potential of BCI technology in improving the quality of life for these children with severe disabilities and their families. The insights and findings from these studies not only contribute to our understanding of pediatric BCI but also highlight critical areas for future research and development. The subsequent sections will consider some of the lessons learned along this journey and discuss potential implications for future studies. As we move forward, these insights will guide our approach to further optimize BCI technology and make it a more integrated, effective, and enriching tool in the lives of children with disabilities and their families.

7.1.1 Development of pediatric-specific BCI paradigms and training programs

7.1.1.1 Understanding Age-Related BCI Performance in Children

The review by Orlandi et al. found that across the twelve pediatric-focused BCI studies, only four participants were less than 9 years old.[66] Across all studies in this body of work alone, we recruited over 20 participants under the age of 9 years old, and demonstrated that children as young as 5 years old can use BCIs for various applications, including entertainment and communication. This represents a significant advancement in pediatric BCI research, addressing the considerable gap in the existing literature as highlighted by Orlandi et al. Furthermore, our research demonstrated that performance in different BCI paradigms may not be equally affected by age. Only one other study has investigated the effect of age on BCI performance in children, demonstrating age-related differences in online accuracy using an SMR-based BCI.[104] Both our baseline study (*Chapter 2*) and our gamification study (*Chapter 3*) investigated age effects, finding only significant differences in P300 paradigms, but not SMR paradigms. Our research offers novel insight into the relationship between age and BCI performance, revealing that different BCI paradigms may not be uniformly impacted by age. This finding opens up an avenue for future studies to explore how different factors like cognitive development, brain plasticity, and maturation affect children's BCI performance. It can also guide the design of age appropriate BCI training programs and applications.

7.1.1.2 Gamification and Personalized Approaches

The 'one-size-fits-all' approach may not be effective for all children. Our research suggests that gamification may enhance BCI performance for some children but not for others. Such outcomes have been described in adult BCI studies[184] and pediatric rehabilitation studies[222], with a collective agreement that while gamification may enhance the user experience, it does not guarantee the intended therapeutic effects of the intervention. Our validation of the nuanced nature of gamification underscores the importance of individualizing BCI interventions. There is a need for further research to understand how to optimize the use of gamification in BCI training and therapies for different individuals. One such avenue for personalized gamification could be through BCI Game Jams.[186] These events, forged by our lab, serve as a unique platform where global game developers, children, and their families come together to co-design BCI-compatible games. This approach could significantly enhance the relevance and effectiveness of BCI technology for children, offering a powerful example of user-centered design in action.

7.1.2 Transitioning BCI Technology from Lab to Home for Children with Severe Neurological Disabilities

Our research demonstrated the feasibility and benefits of a home-based BCI program for children with QCP and their families, underlining the significance of user-centered design and goal setting in enhancing the usability and satisfaction of BCI systems. This marks a notable first in pediatric studies and aligns with previous work done in adult populations[64,65,89,124,127], substantiating the relevance and applicability of home-based BCI programs across the lifespan. These investigations yield crucial insights on improving the usability and satisfaction of BCI systems for children with severe neurological disabilities, laying a robust foundation for future research. Our work illuminates the transformative potential of home-based BCI technology, capable of significantly improving the quality of life for children with QCP and their families. Looking ahead, continued exploration into enhancing the feasibility and acceptance of home-based BCI programs is identified as a top priority. Future research should consider factors such as personalized support, flexible scheduling, and active family involvement, ensuring BCI systems are meticulously designed to cater to these needs. Our work underscores the importance of these considerations, setting a promising trajectory for the further development and refinement of home-based BCI interventions.

7.1.3 BCI as an empowering tool for enhancing quality of life and inclusion for children with disabilities

Our work has demonstrated that BCIs can be a powerful tool for children with QCP, enhancing their ability to achieve personal goals and participate in everyday activities that they have been typically excluded from, such as playing games with siblings, preparing meals, and creating art (Figs. 7.1 & 7.2). Remarkably, these advancements have been achieved using a single personalized mental command. This one command has allowed these children to exercise control over their environment, contributing to a sense of independence and accomplishment. It has empowered them to interact with the world around them in ways that were previously inaccessible, thus fostering a greater sense of inclusivity and participation.

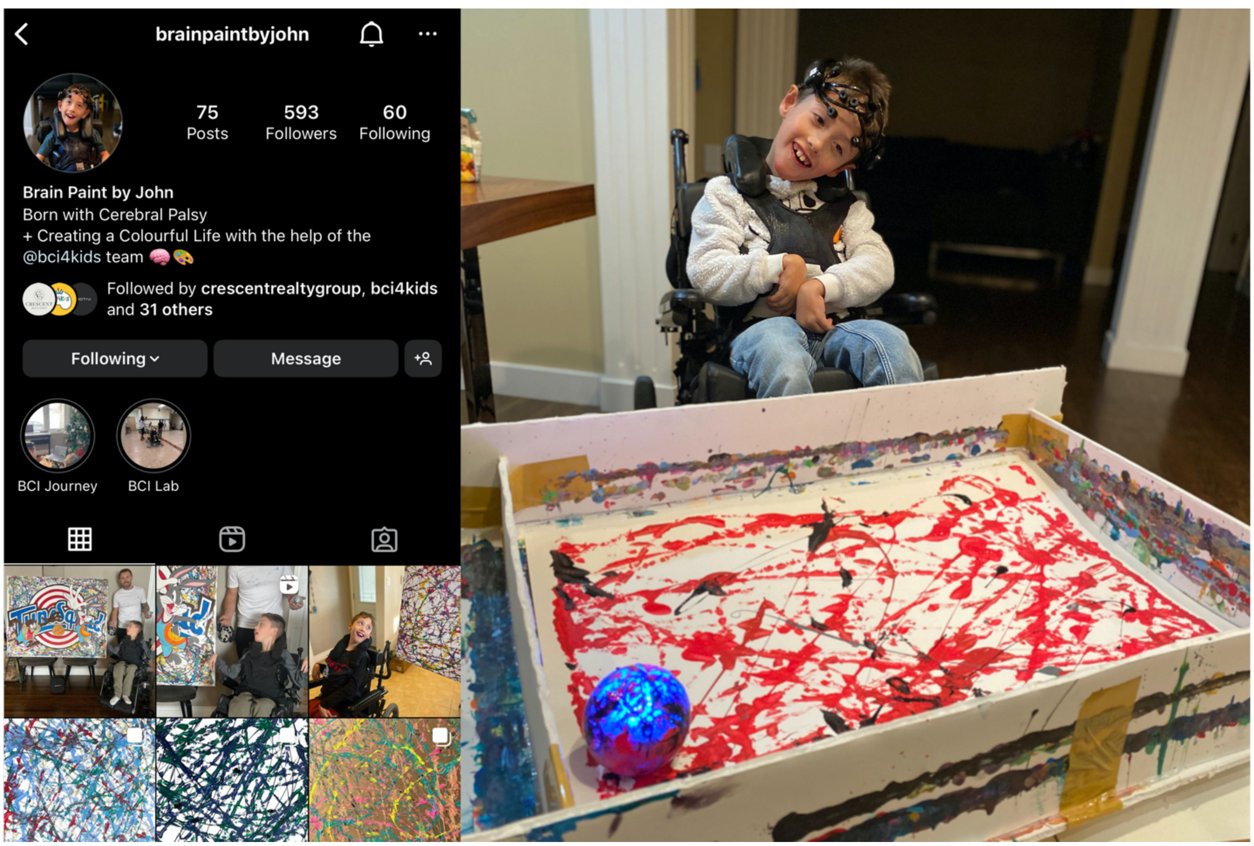


Figure 7.1. Empowerment through BCI case example. John is a 10-year-old boy thriving as an artist despite his QCP, a result of bilateral schizencephaly. John's unique talent lies in creating "brain paintings" using BCI technology, a skill that has earned him global recognition with artworks sold around the world. He showcases his exceptional abilities on his Instagram page @brainpaintbyjohn. John's talent has won him several awards, including the "Make Your Mark"

art contest by the Canadian CP Network for World CP Day. Among his many admirers is Olivier Oullier, the former President of Emotiv.

Figure 7.2. Examples of BCI use at home leading to a multitude of previously unimaginable activities. This collage of photos showcases a sample of the variety of activities achieved by Claire, a 7-year-old girl with mixed QCP secondary to HIE, who started in our BCI program at the young age of 5. Beginning with lab-based activities, such as controlling a remote-controlled car, Claire can now manipulate various devices in her environment through the BCI, facilitating participation in a range of activities. These include activities typical for children her age, like painting, as well as those specific to her family, such as their annual pickling tradition.

Yet, this is just the beginning. The potential for expanding this technology to incorporate multiple personalized mental commands holds immense promise. By providing these children with the ability to execute a broader range of commands, we can further enhance their interaction with their environment. Whether it is controlling a wider array of devices, performing more complex tasks, or engaging in more intricate forms of communication, the possibilities are vast and exciting. Moreover, the use of multiple commands could also bring about subtler, yet equally important benefits. It could allow these children to express their thoughts and intentions more accurately, contribute to their sense of autonomy and complexity of thought, and further strengthen their self-confidence and self-esteem.

In essence, our work underscores the transformative potential of BCIs as a medium for empowerment and inclusion for children with QCP. As we continue to expand and refine this technology, incorporating multiple mental commands, we open up a world of possibilities for enhancing their quality of life and participation in society, upholding their fundamental human rights to autonomy, inclusion, and full participation in all aspects of life.[38,40,41] The future of BCI technology is bright, and its potential to bring about meaningful change in the lives of children with QCP is immense.

8 Bibliography

1. Picton, T. W. & Taylor, M. J. Electrophysiological Evaluation of Human Brain Development. *Developmental Neuropsychology* **31**, 249–278 (2007).
2. Ismail, F. Y., Fatemi, A. & Johnston, M. V. Cerebral plasticity: Windows of opportunity in the developing brain. *European Journal of Paediatric Neurology* **21**, 23–48 (2017).
3. Mattson, M. P., Gleichmann, M. & Cheng, A. Mitochondria in Neuroplasticity and Neurological Disorders. *Neuron* **60**, 748–766 (2008).
4. Meredith, R. M. Sensitive and critical periods during neurotypical and aberrant neurodevelopment: A framework for neurodevelopmental disorders. *Neuroscience & Biobehavioral Reviews* **50**, 180–188 (2015).
5. Graham, A. M., Marr, M., Buss, C., Sullivan, E. L. & Fair, D. A. Understanding Vulnerability and Adaptation in Early Brain Development using Network Neuroscience. *Trends in Neurosciences* **44**, 276–288 (2021).
6. Martin, J. H. The Corticospinal System: From Development to Motor Control. *Neuroscientist* **11**, 161–173 (2005).
7. Haines, D. E. & Mihailoff, G. A. *Fundamental Neuroscience for Basic and Clinical Applications E-Book*. (Philadelphia: Elsevier, 2017).
8. Dunbar, M. & Kirton, A. Perinatal Stroke. *Semin Pediatr Neurol* **32**, 100767 (2019).
9. Zuccoli, G., Yannes, M. P., Nardone, R., Bailey, A. & Goldstein, A. Bilateral symmetrical basal ganglia and thalamic lesions in children: an update (2015). *Neuroradiology* **57**, 973–989 (2015).
10. Graham, H. K. *et al.* Cerebral palsy. *Nat Rev Dis Primers* **2**, 15082 (2016).
11. Schuetze, M. *et al.* Morphological Alterations in the Thalamus, Striatum, and Pallidum in Autism Spectrum Disorder. *Neuropsychopharmacol* **41**, 2627–2637 (2016).

12. Jan, M. M. S. Cerebral Palsy: Comprehensive Review and Update. *Annals of Saudi Medicine* **26**, 123–132 (2006).

13. Bai, D. *et al.* Association of Genetic and Environmental Factors With Autism in a 5-Country Cohort. *JAMA Psychiatry* **76**, 1035–1043 (2019).

14. Colledge, N. *A Guide to Cerebral Palsy*. 28 https://www.ofcp.ca/pdf/Web-Guide-To-CP.pdf (2011).

15. Cans, C., De-la-Cruz, J. & Mermet, M.-A. Epidemiology of cerebral palsy. *Paediatrics and Child Health* **18**, 393–398 (2008).

16. Daly, I. *et al.* Exploration of the neural correlates of cerebral palsy for sensorimotor BCI control. *Front Neuroeng* **7**, 20 (2014).

17. Krigger, K. Cerebral Palsy: An Overview. *American Family Physician* **73**, 91–100 (2006).

18. Rosenbaum, P. *et al.* A report: the definition and classification of cerebral palsy April 2006. *Dev Med Child Neurol Suppl* **109**, 8–14 (2007).

19. Moreno-De-Luca, A., Ledbetter, D. H. & Martin, C. L. Genomic insights into the etiology and classification of the cerebral palsies. *Lancet Neurol* **11**, 283–292 (2012).

20. Novak, I. *et al.* Early, Accurate Diagnosis and Early Intervention in Cerebral Palsy: Advances in Diagnosis and Treatment. *JAMA Pediatr* **171**, 897–907 (2017).

21. Springer, A. *et al.* Profile of children with cerebral palsy spectrum disorder and a normal MRI study. *Neurology* (2019) doi:10.1212/WNL.0000000000007726.

22. Benini, R., Dagenais, L. & Shevell, M. I. Normal Imaging in Patients with Cerebral Palsy: What Does It Tell Us? *The Journal of Pediatrics* **162**, 369-374.e1 (2013).

23. Minocha, P., Sitaraman, S. & Sachdeva, P. Clinical Spectrum, Comorbidities, and Risk Factor Profile of Cerebral Palsy Children: A Prospective Study. *J Pediatr Neurosci* **12**, 15–18 (2017).
24. Monbaliu, E. *et al.* Clinical presentation and management of dyskinetic cerebral palsy. *The Lancet Neurology* **16**, 741–749 (2017).
25. te Velde, A., Morgan, C., Novak, I., Tantsis, E. & Badawi, N. Early Diagnosis and Classification of Cerebral Palsy: An Historical Perspective and Barriers to an Early Diagnosis. *J Clin Med* **8**, (2019).
26. Palisano, R. *et al.* Development and reliability of a system to classify gross motor function in children with cerebral palsy. *Dev Med Child Neurol* **39**, 214–223 (1997).
27. Rosenbaum, P. L., Palisano, R. J., Bartlett, D. J., Galuppi, B. E. & Russell, D. J. Development of the Gross Motor Function Classification System for cerebral palsy. *Dev.Med.Child Neurol* (2008).
28. Johnston, M. V. Chapter 616: Encephalopathies. in *Nelson Textbook of Pediatrics* 3168-3186.e1 (Elsevier Inc., 2020).
29. Himmelmann, K. *et al.* Dyskinetic cerebral palsy in Europe: trends in prevalence and severity. *Archives of Disease in Childhood* **94**, 921–926 (2009).
30. Monbaliu, E. *et al.* Functional outcomes in children and young people with dyskinetic cerebral palsy. *Developmental Medicine & Child Neurology* **59**, 634–640 (2017).
31. Bruno, M.-A. *et al.* Locked-In Syndrome in Children: Report of Five Cases and Review of the Literature. *Pediatric Neurology* **41**, 237–246 (2009).
32. Stadskleiv, K. Cognitive functioning in children with cerebral palsy. *Developmental Medicine & Child Neurology* **62**, 283–289 (2020).

33. Stadskleiv, K., Jahnsen, R., Andersen, G. L. & von Tetzchner, S. Neuropsychological profiles of children with cerebral palsy. *Developmental Neurorehabilitation* **21**, 108–120 (2018).

34. Ballester-Plané, J. *et al.* Cognitive functioning in dyskinetic cerebral palsy: Its relation to motor function, communication and epilepsy. *European Journal of Paediatric Neurology* **22**, 102–112 (2018).

35. Shapiro, S. M. Definition of the Clinical Spectrum of Kernicterus and Bilirubin-Induced Neurologic Dysfunction (BIND). *J Perinatol* **25**, 54–59 (2005).

36. World Health Organization. *International classification of functioning, disability and health (ICF)*. (World Health Organization, 2001).

37. Colver, A. F. & Dickinson, H. O. Study Protocol: Determinants of participation and quality of life of adolescents with cerebral palsy: a longitudinal study (SPARCLE2). *BMC Public Health* 10 (2010).

38. United Nations General Assembly. Universal Declaration of Human Rights. (1948).

39. United Nations Children's Fund (UNICEF). What is the Convention on the Rights of the Child? https://www.unicef.org/child-rights-convention/what-is-the-convention.

40. United Nations General Assembly. *Convention on the Rights of the Child*. (1989).

41. United Nations General Assembly. Convention on the Rights of Persons with Disabilities. (2007).

42. Shevell, M. I., Majnemer, A., Poulin, C. & Law, M. Stability of motor impairment in children with cerebral palsy. *Developmental Medicine & Child Neurology* **50**, 211–215 (2008).

43. Light, J., McNaughton, D. & Caron, J. New and emerging AAC technology supports for children with complex communication needs and their communication partners: State of the science and future research directions. *Augmentative and Alternative Communication* **35**, 26–41 (2019).

44. Jeffords, J. M. S.2432 - 105th Congress (1997-1998): Assistive Technology Act of 1998. https://www.congress.gov/bill/105th-congress/senate-bill/2432 (1998).

45. Nisbet, P. Alternative Access Technologies. in *Handbook of Electronic Assistive Technology* 105–148 (Elsevier, 2019). doi:10.1016/B978-0-12-812487-1.00005-3.

46. Myrden, A., Schudlo, L., Weyand, S., Zeyl, T. & Chau, T. Trends in communicative access solutions for children with cerebral palsy. *J. Child Neurol.* **29**, 1108–1118 (2014).

47. Borgestig, M., Sandqvist, J., Parsons, R., Falkmer, T. & Hemmingsson, H. Eye gaze performance for children with severe physical impairments using gaze-based assistive technology—A longitudinal study. *Assist Technol* **28**, 93–102 (2016).

48. Mak, J. N. & Wolpaw, J. R. Clinical Applications of Brain-Computer Interfaces: Current State and Future Prospects. *IEEE Rev Biomed Eng* **2**, 187–199 (2009).

49. Wolpaw, J. R., Birbaumer, N., McFarland, D. J., Pfurtscheller, G. & Vaughan, T. M. Brain–computer interfaces for communication and control. *Clinical Neurophysiology* 25 (2002).

50. Oxley, T. J. *et al.* Minimally invasive endovascular stent-electrode array for high-fidelity, chronic recordings of cortical neural activity. *Nat Biotechnol* **34**, 320–327 (2016).

51. Wang, Y., Yang, X., Zhang, X., Wang, Y. & Pei, W. Implantable intracortical microelectrodes: reviewing the present with a focus on the future. *Microsyst Nanoeng* **9**, 1–17 (2023).

52. Hochberg, L. R. *et al.* Neuronal ensemble control of prosthetic devices by a human with tetraplegia. *Nature* **442**, 164–171 (2006).

53. Hochberg, L. R. *et al.* Reach and grasp by people with tetraplegia using a neurally controlled robotic arm. *Nature* **485**, 372–375 (2012).

54. Downey, J. E. *et al.* Motor cortical activity changes during neuroprosthetic-controlled object interaction. *Sci Rep* **7**, 16947 (2017).

55. Collinger, J. L. *et al.* High-performance neuroprosthetic control by an individual with tetraplegia. *The Lancet* **381**, 557–564 (2013).

56. Chaudhary, U. *et al.* Spelling interface using intracortical signals in a completely locked-in patient enabled via auditory neurofeedback training. *Nat Commun* **13**, 1236 (2022).

57. Ramadan, R. A. & Vasilakos, A. V. Brain computer interface: control signals review. *Neurocomputing* **223**, 26–44 (2017).

58. Vansteensel, M. J. & Jarosiewicz, B. Brain-computer interfaces for communication. *Handb Clin Neurol* **168**, 67–85 (2020).

59. Ramsey, N. F. Chapter 1 - Human brain function and brain-computer interfaces. in *Handbook of Clinical Neurology* (eds. Ramsey, N. F. & Millán, J. del R.) vol. 168 1–13 (Elsevier, 2020).

60. Schalk, G. & Leuthardt, E. C. Brain-computer interfaces using electrocorticographic signals. *IEEE Reviews in Biomedical Engineering* **4**, 140–154 (2011).

61. Vansteensel, M. J. *et al.* Brain-computer interfacing based on cognitive control. *Ann Neurol* **67**, 809–816 (2010).

62. Brunner, P., Ritaccio, A., Emrich, J., Bischof, H. & Schalk, G. Rapid Communication with a “P300” Matrix Speller Using Electrocorticographic Signals (ECoG). *Frontiers in Neuroscience* **5**, (2011).

63. Benabid, A. L. *et al.* An exoskeleton controlled by an epidural wireless brain–machine interface in a tetraplegic patient: a proof-of-concept demonstration. *The Lancet Neurology* **18**, 1112–1122 (2019).

64. Vansteensel, M. J. *et al.* Fully Implanted Brain-Computer Interface in a Locked-In Patient with ALS. *N. Engl. J. Med.* **375**, 2060–2066 (2016).

65. Pels, E. G. M. *et al.* Stability of a chronic implanted brain-computer interface in late-stage amyotrophic lateral sclerosis. *Clinical Neurophysiology* **130**, 1798–1803 (2019).

66. Orlandi, S., House, S. C., Karlsson, P., Saab, R. & Chau, T. Brain-Computer Interfaces for Children With Complex Communication Needs and Limited Mobility: A Systematic Review. *Frontiers in Human Neuroscience* **15**, 643294 (2021).

67. Kübler, A. The history of BCI: From a vision for the future to real support for personhood in people with locked-in syndrome. *Neuroethics* **13**, 163–180 (2020).

68. Gevins, A. & Smith, M. E. Electroencephalography (EEG) in Neuroergonomics. in *Neuroergonomics: The brain at work* (eds. Parasuraman, R. & Rizzo, M.) 0 (Oxford University Press, 2006). doi:10.1093/acprof:oso/9780195177619.003.0002.

69. Müller-Putz, G. R. Electroencephalography. *Handb Clin Neurol* **168**, 249–262 (2020).

70. Jackson, A. F. & Bolger, D. J. The neurophysiological bases of EEG and EEG measurement: a review for the rest of us. *Psychophysiology* **51**, 1061–1071 (2014).

71. Ng, C. R. *et al.* Multi-Center Evaluation of Gel-Based and Dry Multipin EEG Caps. *Sensors* **22**, 8079 (2022).

72. Chaudhary, U., Birbaumer, N. & Ramos-Murguialday, A. Brain-computer interfaces for communication and rehabilitation. *Nat Rev Neurol* **12**, 513–525 (2016).

73. Abiri, R., Borhani, S., Sellers, E. W., Jiang, Y. & Zhao, X. A comprehensive review of EEG-based brain–computer interface paradigms. *J. Neural Eng.* **16**, 011001 (2019).

74. Sooley, M., Cressman, E. K. & Martini, R. Movement imagery as a predictor of online control in typically developing children. *Developmental Neuropsychology* **43**, 508–523 (2018).

75. Kosmyna, N., Lindgren, J. T. & Lécuyer, A. Attending to Visual Stimuli versus Performing Visual Imagery as a Control Strategy for EEG-based Brain-Computer Interfaces. *Sci Rep* **8**, 13222 (2018).

76. Stojic, F. & Chau, T. Nonspecific Visuospatial Imagery as a Novel Mental Task for Online EEG-Based BCI Control. *Int. J. Neural Syst.* (2020) doi:10.1142/s0129065720500264.

77. Wolpaw, J. R. & Wolpaw, E. W. Brain–Computer Interfaces: Something New under the Sun. in (2012). doi:10.1093/ACPROF:OSO/9780195388855.003.0001.

78. Kübler, A., Neumann, N., Wilhelm, B., Hinterberger, T. & Birbaumer, N. Predictability of Brain-Computer Communication. *Journal of Psychophysiology* **18**, 121–129 (2004).

79. Scherer, R. *et al.* Brain–computer interfacing: more than the sum of its parts. *Soft Comput* **17**, 317–331 (2013).

80. Chaudhary, U., Mrachacz-Kersting, N. & Birbaumer, N. Neuropsychological and neurophysiological aspects of brain-computer-interface (BCI) control in paralysis. *The Journal of Physiology* **n/a**,.

81. Allison, B. Z., Kübler, A. & Jin, J. 30+ years of P300 brain–computer interfaces. *Psychophysiology* **57**, e13569 (2020).

82. Letourneau, S. *et al.* Clinician awareness of brain computer interfaces: a Canadian national survey. *J Neuroeng Rehabil* **17**, 2 (2020).

83. Sellers, E. W., Ryan, D. B. & Hauser, C. K. Noninvasive brain-computer interface enables communication after brainstem stroke. *Sci Transl Med* **6**, 257re7 (2014).

84. Müller-Putz, G. R., Daly, I. & Kaiser, V. Motor imagery-induced EEG patterns in individuals with spinal cord injury and their impact on brain–computer interface accuracy. *J. Neural Eng.* **11**, 035011 (2014).

85. Kübler, A. *et al.* The User-Centered Design as Novel Perspective for Evaluating the Usability of BCI-Controlled Applications. *PLOS ONE* **9**, e112392 (2014).

86. Botrel, L., Holz, E. M. & Kübler, A. Brain Painting V2: evaluation of P300-based brain-computer interface for creative expression by an end-user following the user-centered design. *Brain-Computer Interfaces* **2**, 135–149 (2015).

87. Zickler, C. *et al.* A Brain-Computer Interface as Input Channel for a Standard Assistive Technology Software. *Clin EEG Neurosci* **42**, 236–244 (2011).

88. Scherer, R., Schwarz, A., Müller-Putz, G. R., Pammer-Schindler, V. & Garcia, M. L. Game-Based BCI Training: Interactive Design for Individuals with Cerebral Palsy. in *2015 IEEE International Conference on Systems, Man, and Cybernetics* 3175–3180 (2015). doi:10.1109/SMC.2015.551.

89. Wolpaw, J. R. *et al.* Independent home use of a brain-computer interface by people with amyotrophic lateral sclerosis. *Neurology* **91**, e258–e267 (2018).

90. Combaz, A. *et al.* A Comparison of Two Spelling Brain-Computer Interfaces Based on Visual P3 and SSVEP in Locked-In Syndrome. *PLOS ONE* **8**, e73691 (2013).

91. Höhne, J. *et al.* Motor Imagery for Severely Motor-Impaired Patients: Evidence for Brain-Computer Interfacing as Superior Control Solution. *PLOS ONE* **9**, e104854 (2014).

92. Bai, O., Lin, P., Huang, D., Fei, D.-Y. & Floeter, M. K. Towards a User-Friendly Brain-Computer Interface: Initial Tests in ALS and PLS Patients. *Clin Neurophysiol* **121**, 1293–1303 (2010).

93. Scherer, R. *et al.* Thought-based row-column scanning communication board for individuals with cerebral palsy. *Ann Phys Rehabil Med* **58**, 14–22 (2015).

94. Daly, I. *et al.* On the control of brain-computer interfaces by users with cerebral palsy. *Clinical Neurophysiology* **124**, 1787–1797 (2013).

95. Breshears, J. D. *et al.* Decoding motor signals from the pediatric cortex: implications for brain-computer interfaces in children. *Pediatrics* **128**, e160-168 (2011).

96. Ehlers, J., Valbuena, D., Stiller, A. & Gräser, A. Age-Specific Mechanisms in an SSVEP-Based BCI Scenario: Evidences from Spontaneous Rhythms and Neuronal Oscillators. *Computational Intelligence and Neuroscience* vol. 2012 e967305 https://www.hindawi.com/journals/cin/2012/967305/ (2012).

97. Norton, J. J. S., Mullins, J., Alitz, B. E. & Bretl, T. The performance of 9-11-year-old children using an SSVEP-based BCI for target selection. *J Neural Eng* **15**, 056012 (2018).

98. Giedd, J. N. *et al.* Brain development during childhood and adolescence: a longitudinal MRI study. *Nat Neurosci* **2**, 861–863 (1999).

99. Neuper, C., Scherer, R., Reiner, M. & Pfurtscheller, G. Imagery of motor actions: Differential effects of kinesthetic and visual–motor mode of imagery in single-trial EEG. *Cognitive Brain Research* **25**, 668–677 (2005).

100. Lust, J. M., Wilson, P. H. & Steenbergen, B. Motor imagery difficulties in children with Cerebral Palsy: A specific or general deficit? *Res Dev Disabil* **57**, 102–111 (2016).

101. Gabis, L. V., Tsubary, N. M., Leon, O., Ashkenasi, A. & Shefer, S. Assessment of Abilities and Comorbidities in Children With Cerebral Palsy. *J Child Neurol* **30**, 1640–1645 (2015).

102. Cowan, N., Fristoe, N. M., Elliott, E. M., Brunner, R. P. & Saults, J. S. Scope of attention, control of attention, and intelligence in children and adults. *Memory & Cognition* **34**, 1754–1768 (2006).

103. Jadavji, Z. *et al.* Establishing a Clinical Brain-Computer Interface Program for Children With Severe Neurological Disabilities. *Cureus* **14**, (2022).

104. Zhang, J., Jadavji, Z., Zewdie, E. & Kirton, A. Evaluating If Children Can Use Simple Brain Computer Interfaces. *Front Hum Neurosci* **13**, 24 (2019).

105. Jadavji, Z., Zhang, J., Zewdie, E. & Kirton, A. Brain computer interface control in hemiparetic children with perinatal stroke. in *ICNC 2018* (2018).

106. Blair, E., Langdon, K., McIntyre, S., Lawrence, D. & Watson, L. Survival and mortality in cerebral palsy: observations to the sixth decade from a data linkage study of a total population register and National Death Index. *BMC Neurol* **19**, 111 (2019).

107. Allison, B. Z. & Neuper, C. Could Anyone Use a BCI? in *Brain-Computer Interfaces: Applying our Minds to Human-Computer Interaction* (eds. Tan, D. S. & Nijholt, A.) 35–54 (Springer, 2010). doi:10.1007/978-1-84996-272-8_3.

108. Shu, X. *et al.* Fast Recognition of BCI-Inefficient Users Using Physiological Features from EEG Signals: A Screening Study of Stroke Patients. *Front Neurosci* **12**, (2018).

109. Jeunet, C., N'Kaoua, B. & Lotte, F. Advances in user-training for mental-imagery-based BCI control: Psychological and cognitive factors and their neural correlates. *Prog. Brain Res.* **228**, 3–35 (2016).

110. Kübler, A., Holz, E., Kaufmann, T. & Zickler, C. A User Centred Approach for Bringing BCI Controlled Applications to End-Users. *Brain-Computer Interface Systems - Recent Progress and Future Prospects* (2013) doi:10.5772/55802.

111. Kübler, A., Blankertz, B., Müller, K.-R. & Neuper, C. A Model of BCI Control. in (eds. Müller-Putz, G. R. et al.) 100–103 (2011).

112. Mikołajewska, E. & Mikołajewski, D. The prospects of brain — computer interface applications in children. *cent.eur.j.med* **9**, 74–79 (2014).

113. Jeunet, C., Jahanpour, E. & Lotte, F. Why standard brain-computer interface (BCI) training protocols should be changed: an experimental study. *J Neural Eng* **13**, 036024 (2016).

114. Lotte, F., Larrue, F. & Mühl, C. Flaws in current human training protocols for spontaneous Brain-Computer Interfaces: lessons learned from instructional design. *Front Hum Neurosci* **7**, (2013).

115. Su, C.-H. & Cheng, C.-H. A mobile gamification learning system for improving the learning motivation and achievements. *Journal of Computer Assisted Learning* **31**, 268–286 (2015).

116. Brewer, R. *et al.* Using Gamification to Motivate Children to Complete Empirical Studies in Lab Environments. in (2013). doi:10.1145/2485760.2485816.

117. Dovis, S., Van der Oord, S., Wiers, R. W. & Prins, P. J. M. Can Motivation Normalize Working Memory and Task Persistence in Children with Attention-Deficit/Hyperactivity

Disorder? The Effects of Money and Computer-Gaming. *J Abnorm Child Psychol* **40**, 669–681 (2012).

118. Gooch, D., Vasalou, A., Benton, L. & Khaled, R. Using Gamification to Motivate Students with Dyslexia. in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* 969–980 (ACM, 2016). doi:10.1145/2858036.2858231.
119. Pramana, G. *et al.* Using Mobile Health Gamification to Facilitate Cognitive Behavioral Therapy Skills Practice in Child Anxiety Treatment: Open Clinical Trial. *JMIR Serious Games* **6**, e9 (2018).
120. Brown, D. *et al.* Engaging Students with Intellectual Disabilities through Games Based Learning and Related Technologies. in *Universal Access in Human-Computer Interaction. Applications and Services for Quality of Life* (eds. Stephanidis, C. & Antona, M.) 573–582 (Springer Berlin Heidelberg, 2013).
121. McFarland, D. J. & Wolpaw, J. R. Brain–computer interface use is a skill that user and system acquire together. *PLOS Biology* **16**, e2006719 (2018).
122. Rashid, M. *et al.* Current Status, Challenges, and Possible Solutions of EEG-Based Brain-Computer Interface: A Comprehensive Review. *Front Neurorobot* **14**, (2020).
123. Chavarriaga, R., Fried-Oken, M., Kleih, S., Lotte, F. & Scherer, R. Heading for new shores! Overcoming pitfalls in BCI design. *Brain-Computer Interfaces* **4**, 60–73 (2017).
124. Miralles, F. *et al.* Brain–Computer Interfaces on Track to Home: Results of the Evaluation at Disabled End-Users' Homes and Lessons Learnt. *Front. ICT* **2**, (2015).
125. Leeb, R. *et al.* Transferring brain–computer interfaces beyond the laboratory: Successful application control for motor-disabled users. *Artificial Intelligence in Medicine* **59**, 121–132 (2013).

126. Kleih, S. C. & Kübler, A. Psychological Factors Influencing Brain-Computer Interface (BCI) Performance. in *2015 IEEE International Conference on Systems, Man, and Cybernetics* 3192–3196 (2015). doi:10.1109/SMC.2015.554.

127. Käthner, I. *et al.* A Multifunctional Brain-Computer Interface Intended for Home Use: An Evaluation with Healthy Participants and Potential End Users with Dry and Gel-Based Electrodes. *Front Neurosci* **11**, (2017).

128. Holz, E. M., Botrel, L., Kaufmann, T. & Kübler, A. Long-term independent brain-computer interface home use improves quality of life of a patient in the locked-in state: a case study. *Arch Phys Med Rehabil* **96**, S16-26 (2015).

129. Guastello, S. & Jay, K. Improving the patient experience through a comprehensive performance framework to evaluate excellence in person-centred care. *BMJ Open Qual* **8**, e000737 (2019).

130. King, G., Williams, L. & Hahn Goldberg, S. Family-oriented services in pediatric rehabilitation: a scoping review and framework to promote parent and family wellness. *Child: Care, Health and Development* **43**, 334–347 (2017).

131. Gaglio, B. *et al.* Methodological standards for qualitative and mixed methods patient centered outcomes research. *BMJ* **371**, m4435 (2020).

132. Jadavji, Z., Zhang, J., Paffrath, B., Zewdie, E. & Kirton, A. Can Children With Perinatal Stroke Use a Simple Brain Computer Interface? *Stroke* **52**, 2363–2370 (2021).

133. Jadavji, Z. *et al.* A Pediatric BCI Program for Children With Severe Neurological Disabilities: Thematic Analysis of Family Perspectives. Preprint at https://doi.org/10.21203/rs.3.rs-951419/v1 (2022).

134. Letourneau, S., Zewdie, E., Burkholder, L., Andersen, J. & Kirton, A. Clinician Awareness of Brain Computer Interfaces and Eligible Pediatric Populations. in *ANNALS OF NEUROLOGY* vol. 84 S416–S416 (WILEY 111 RIVER ST, HOBOKEN 07030-5774, NJ USA, 2018).

135. Amankwah, N. *et al.* Cerebral palsy in Canada, 2011-2031: results of a microsimulation modelling study of epidemiological and cost impacts. *Health Promot Chronic Dis Prev Can* **40**, 25–37 (2020).

136. Arnfield, E., Guzzetta, A. & Boyd, R. Relationship between brain structure on magnetic resonance imaging and motor outcomes in children with cerebral palsy: A systematic review. *Research in Developmental Disabilities* **34**, 2234–2250 (2013).

137. Myrden, A. & Chau, T. Effects of user mental state on EEG-BCI performance. *Frontiers in Human Neuroscience | www.frontiersin.org* **1**, 308 (2015).

138. Guger, C. *et al.* Complete Locked-in and Locked-in Patients: Command Following Assessment and Communication with Vibro-Tactile P300 and Motor Imagery Brain-Computer Interface Tools. *Front Neurosci* **11**, 251 (2017).

139. Wolpaw, J. R., Birbaumer, N., McFarland, D. J., Pfurtscheller, G. & Vaughan, T. M. Brain-computer interfaces for communication and control. *Clin Neurophysiol* **113**, 767–791 (2002).

140. Orlandi, S., House, S. C., Karlsson, P., Saab, R. & Chau, T. Brain-Computer Interfaces for Children With Complex Communication Needs and Limited Mobility: A Systematic Review. *Frontiers in Human Neuroscience* **15**, (2021).

141. Jadavji, Z., Zhang, J., Paffrath, B., Zewdie, E. & Kirton, A. Can Children With Perinatal Stroke Use a Simple Brain Computer Interface? *Stroke* STROKEAHA120030596 (2021) doi:10.1161/STROKEAHA.120.030596.

142. Taherian, S., Selitskiy, D., Pau, J. & Claire Davies, T. Are we there yet? Evaluating commercial grade brain–computer interface for control of computer applications by individuals with cerebral palsy. *Disability and Rehabilitation: Assistive Technology* **12**, 165–174 (2017).

143. Kübler, A., Kotchoubey, B., Kaiser, J., Wolpaw, J. R. & Birbaumer, N. Brain-computer communication: unlocking the locked in. *Psychol Bull* **127**, 358–375 (2001).

144. Paediatrics HICCUP| University of Calgary |. http://www.ucalgary.ca/paediatrics_hiccup/.

145. Guger, C. *et al.* MindBEAGLE — A new system for the assessment and communication with patients with disorders of consciousness and complete locked-in syndrom. in *2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* 3008–3013 (2017). doi:10.1109/SMC.2017.8123086.

146. Guger, C. *et al.* How many people are able to control a P300-based brain–computer interface (BCI)? *Neuroscience Letters* **462**, 94–98 (2009).

147. Guger, C., Krausz, G. & Edlinger, G. Brain-computer interface control with dry EEG electrodes. (2011).

148. Guger, C., Ortner, R., Dimov, S. & Allison, B. A comparison of face speller approaches for P300 BCIs. in *2016 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* 004809–004812 (2016). doi:10.1109/SMC.2016.7844989.

149. Guger, C. *et al.* How many people are able to control a P300-based brain–computer interface (BCI)? *Neuroscience Letters* **462**, 94–98 (2009).

150. Guger, C., Krausz, G., Allison, B. Z. & Edlinger, G. Comparison of Dry and Gel Based Electrodes for P300 Brain–Computer Interfaces. *Front. Neurosci.* **6**, (2012).

151. Allison, B. Z. *et al.* Validation of a Brain-Computer Interface (BCI) System Designed for Patients with Disorders of Consciousness (DOC): Regular and Sham Testing with Healthy Participants. in *Augmented Cognition. Enhancing Cognition and Behavior in Complex Human Environments* (eds. Schmorrow, D. D. & Fidopiastis, C. M.) 253–265 (Springer International Publishing, 2017).

152. Oldfield, R. C. The assessment and analysis of handedness: The Edinburgh inventory. *Neuropsychologia* **9**, 97–113 (1971).

153. Tatla, S. K., Jarus, T., Virji-Babul, N. & Holsti, L. The development of the Pediatric Motivation Scale for rehabilitation: Le développement de la « Pediatric Motivation Scale » en réhabilitation. *Can J Occup Ther* **82**, 93–105 (2015).

154. Garvey, M. A., Kaczynski, K. J., Becker, D. A. & Bartko, J. J. Subjective Reactions of Children to Single-Pulse Transcranial Magnetic Stimulation. *Journal of Child Neurology* **16**, 891–894 (2001).

155. Ehlers, J., Valbuena, D., Stiller, A. & Gräser, A. Age-specific mechanisms in an SSVEP-based BCI scenario: evidences from spontaneous rhythms and neuronal oscillators. *Comput Intell Neurosci* **2012**, 967305 (2012).

156. Norton, J., Mullins, J., Alitz, B. & Bretl, T. The performance of 9-11-year-old children using an SSVEP-based BCI for target selection. *Journal of Neural Engineering* **15**, (2018).

157. Beveridge, R., Wilson, S. & Coyle, D. Can teenagers control a 3D racing game using motion-onset visual evoked potentials? *Brain-Computer Interfaces* **4**, 102–113 (2017).

158. Beveridge, R., Wilson, S., Callaghan, M. & Coyle, D. Neurogaming With Motion-Onset Visual Evoked Potentials (mVEPs): Adults Versus Teenagers. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* **27**, 572–581 (2019).
159. Vařeka, L. Evaluation of convolutional neural networks using a large multi-subject P300 dataset. *Biomedical Signal Processing and Control* **58**, 101837 (2020).
160. Niso, G., Romero, E., Moreau, J. T., Araujo, A. & Krol, L. R. Wireless EEG: A survey of systems and studies. *NeuroImage* **269**, 119774 (2023).
161. Chatelle, C., Spencer, C. A., Cash, S. S., Hochberg, L. R. & Edlow, B. L. Feasibility of an EEG-based brain-computer interface in the intensive care unit. *Clin Neurophysiol* **129**, 1519–1525 (2018).
162. Albrecht, R., Suchodoletz, W. v & Uwer, R. The development of auditory evoked dipole source activity from childhood to adulthood. *Clinical Neurophysiology* **111**, 2268–2276 (2000).
163. Overbye, K., Huster, R. J., Walhovd, K. B., Fjell, A. M. & Tamnes, C. K. Development of the P300 from childhood to adulthood: a multimodal EEG and MRI study. *Brain Struct Funct* **223**, 4337–4349 (2018).
164. Souto, D. O., Cruz, T. K. F., Fontes, P. L. B., Batista, R. C. & Haase, V. G. Motor Imagery Development in Children: Changes in Speed and Accuracy With Increasing Age. *Front. Pediatr.* **8**, (2020).
165. Blakemore, S.-J. & Choudhury, S. Development of the adolescent brain: implications for executive function and social cognition. *Journal of Child Psychology and Psychiatry* **47**, 296–312 (2006).

166. Weyand, S. & Chau, T. Challenges of implementing a personalized mental task near-infrared spectroscopy brain–computer interface for a non-verbal young adult with motor impairments. *Developmental Neurorehabilitation* **20**, 99–107 (2017).

167. Kirton, A. Trapped Children: A Moral Imperative to Advance Pediatric Brain Computer Interfaces. *JAMA pediatrics* (in press).

168. Sailer, M., Hense, J., Mandl, H. & Klevers, M. Psychological Perspectives on Motivation through Gamification. *IxD&A* 28–37 (2013) doi:10.55612/s-5002-019-002.

169. Kelly, D., Zewdie, E., Carlson, H. & Kirton, A. *Brain-Computer Interfaces for Children: A Comparative Study of Five Common EEG-based Paradigms*. https://www.researchsquare.com/article/rs-2836229/v1 (2023) doi:10.21203/rs.3.rs-2836229/v1.

170. Floreani, E. D., Rowley, D., Kelly, D., Kinney-Lang, E. & Kirton, A. On the feasibility of simple brain-computer interface systems for enabling children with severe physical disabilities to explore independent movement. *Front. Hum. Neurosci.* **16**, 1007199 (2022).

171. Kelly, D., Kinney-Lang, E., Murji, S. & Kirton, A. Can Gamified Brain-Computer Interface Training Paradigms Improve Performance on a P300 Spelling Task? in *Proceedings of the 8th International BCI Meeting 2021 (vBCI)* 24 (2021).

172. Gualtieri, C. T. & Johnson, L. G. Reliability and validity of a computerized neurocognitive test battery, CNS Vital Signs. *Arch.Clin.Neuropsychol.* **21**, 623–643 (2006).

173. Rezeika, A. *et al.* Brain–Computer Interface Spellers: A Review. *Brain Sciences* **8**, 57 (2018).

174. Kaufmann, T., Schulz, S. M., Grünzinger, C. & Kübler, A. Flashing characters with famous faces improves ERP-based brain–computer interface performance. *J. Neural Eng.* **8**, 056016 (2011).

175. Pfurtscheller, G. & Neuper, C. Motor imagery and direct brain-computer communication. *Proceedings of the IEEE* **89**, 1123–1134 (2001).

176. Lotte, F. *et al.* A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update. *J. Neural Eng.* **15**, 031005 (2018).

177. Barachant, A., Bonnet, S., Congedo, M. & Jutten, C. Multiclass Brain–Computer Interface Classification by Riemannian Geometry. *IEEE Trans. Biomed. Eng.* **59**, 920–928 (2012).

178. Laurie-Rose, C., Frey, M., Ennis, A. & Zamary, A. Measuring Perceived Mental Workload in Children. *The American Journal of Psychology* **127**, 107–125 (2014).

179. Müller-Putz, G., Scherer, R., Brunner, C., Leeb, R. & Pfurtscheller, G. Better than Random? A closer look on BCI results. *International Journal of Bioelektromagnetism* **10**, 52–55 (2008).

180. McCullagh, P., Lightbody, G., Zygierewicz, J. & Kernohan, W. G. Ethical Challenges Associated with the Development and Deployment of Brain Computer Interface Technology. *Neuroethics* **7**, 109–122 (2014).

181. Klein, E. Chapter 24 - Ethics and the emergence of brain-computer interface medicine. in *Handbook of Clinical Neurology* (eds. Ramsey, N. F. & Millán, J. del R.) vol. 168 329–339 (Elsevier, 2020).

182. Nijboer, F., Birbaumer, N. & Kubler, A. The Influence of Psychological State and Motivation on Brain–Computer Interface Performance in Patients with Amyotrophic Lateral Sclerosis – a Longitudinal Study. *Frontiers in Neuroscience* **4**, (2010).

183. Edlinger, G., Allison, B. Z. & Guger, C. How Many People Can Use a BCI System? in *Clinical Systems Neuroscience* (eds. Kansaku, K., Cohen, L. G. & Birbaumer, N.) 33–66 (Springer Japan, 2015). doi:10.1007/978-4-431-55037-2_3.

184. de Castro-Cros, M. *et al.* Effects of Gamification in BCI Functional Rehabilitation. *Frontiers in Neuroscience* **14**, (2020).

185. Riggins, T. & Scott, L. S. P300 development from infancy to adolescence. *Psychophysiology* **57**, e13346 (2020).

186. Kelly, D. *et al.* A Child's Right to Play: Results from the Brain-Computer Interface Game Jam 2019 (Calgary Competition). *Annu Int Conf IEEE Eng Med Biol Soc* **2020**, 6099–6102 (2020).

187. Kinney-Lang, E. *et al.* Advancing Brain-Computer Interface Applications for Severely Disabled Children Through a Multidisciplinary National Network: Summary of the Inaugural Pediatric BCI Canada Meeting. *Front Hum Neurosci* **14**, 593883 (2020).

188. Kelly, D., Zewdie, E. & Kirton, A. Pediatric brain-computer interface competency: a pilot study. in *Proceedings of the 8th Graz Brain-Computer Interface Conference 2019* (eds. Müller-Putz, G. R., Ditz, J. C. & Wriessnegger, S. C.) 151–156 (Verlag der Technischen Universität Graz, 2019). doi:10.3217/978-3-85125-682-6-28.

189. Eliasson, A. C. *et al.* The Manual Ability Classification System (MACS) for children with cerebral palsy: scale development and evidence of validity and reliability. *Developmental Medicine & Child Neurology* **48**, 549–554 (2006).

190. Floreani, E. D. *et al.* Iterative Development of a Software to Facilitate Independent Home Use of BCI Technologies for Children with Quadriplegic Cerebral Palsy. in *2022 44th*

Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) 3361–3364 (2022). doi:10.1109/EMBC48229.2022.9871105.

191. Copley, J. & Ziviani, J. Barriers to the use of assistive technology for children with multiple disabilities. *Occupational Therapy International* **11**, 229–243 (2004).
192. Kübler, A., Nijboer, F. & Kleih, S. Chapter 26 - Hearing the needs of clinical users. in *Handbook of Clinical Neurology* (eds. Ramsey, N. F. & Millán, J. del R.) vol. 168 353–368 (Elsevier, 2020).
193. Miralles, F. *et al.* Brain Computer Interface on Track to Home. *ScientificWorldJournal* **2015**, 623896 (2015).
194. Huggins, J. E., Moinuddin, A. A., Chiodo, A. E. & Wren, P. A. What Would Brain-Computer Interface Users Want: Opinions and Priorities of Potential Users With Spinal Cord Injury. *Archives of Physical Medicine and Rehabilitation* **96**, S38-S45.e5 (2015).
195. Zulauf-Czaja, A. *et al.* On the way home: a BCI-FES hand therapy self-managed by sub-acute SCI participants and their caregivers: a usability study. *Journal of NeuroEngineering and Rehabilitation* **18**, 44 (2021).
196. Koch Fager, S., Fried-Oken, M., Jakobs, T. & Beukelman, D. R. New and emerging access technologies for adults with complex communication needs and severe motor impairments: State of the science. *Augment Altern Commun* **35**, 13–25 (2019).
197. Colver, A. Quality of life and participation. *Developmental Medicine & Child Neurology* **51**, 656–659 (2009).
198. Palisano, R. J., Rosenbaum, P., Bartlett, D. & Livingston, M. H. Content validity of the expanded and revised Gross Motor Function Classification System. *Developmental Medicine & Child Neurology* **50**, 744–750 (2008).

199. Orlandi, S. Alex Runs.

200. McColl, M. A., Paterson, M., Davies, D., Doubt, L. & Law, M. Validity and community utility of the Canadian Occupational Performance Measure. *Canadian Journal of Occupational Therapy - Revue Canadienne d Ergotherapie* **67**, 22–30 (2000).

201. Cusick, A., Lannin, N. A. & Lowe, K. Adapting the Canadian Occupational Performance Measure for use in a paediatric clinical trial. *Disability & Rehabilitation* **29**, 761–766 (2007).

202. Metzler, M. J. *et al.* Goals of children with unilateral cerebral palsy in a brain stimulation arm rehabilitation trial. *Dev Med Child Neurol* (2020) doi:10.1111/dmcn.14763.

203. Saha, S. & Baumert, M. Intra- and Inter-subject Variability in EEG-Based Sensorimotor Brain Computer Interface: A Review. *Front Comput Neurosci* **13**, 87 (2020).

204. Parkes, J. *et al.* Psychological problems in children with cerebral palsy: a cross-sectional European study. *J Child Psychol Psychiatry* **49**, 405–413 (2008).

205. Bjorgaas, H. M., Elgen, I., Boe, T. & Hysing, M. Mental health in children with cerebral palsy: does screening capture the complexity? *ScientificWorldJournal* **2013**, 468402 (2013).

206. Ramstad, K., Jahnsen, R., Skjeldal, O. H. & Diseth, T. H. Parent-reported participation in children with cerebral palsy: the contribution of recurrent musculoskeletal pain and child mental health problems. *Dev Med Child Neurol* **54**, 829–835 (2012).

207. Entertainment Software Association of Canada. *Real Canadian Gamer Essential Facts*. (2020).

208. Prescott, A. T., Sargent, J. D. & Hull, J. G. Metaanalysis of the relationship between violent video game play and physical aggression over time. *Proceedings of the National Academy of Sciences of the United States of America* **115**, 9882 (2018).

209. Halbrook, Y. J., O'Donnell, A. T. & Msetfi, R. M. When and How Video Games Can Be Good: A Review of the Positive Effects of Video Games on Well-Being. *Perspect Psychol Sci* **14**, 1096–1104 (2019).

210. Granic, I., Lobel, A. & Engels, R. C. M. E. The benefits of playing video games. *American Psychologist* **69**, 66–78 (2014).

211. UNICEF. *Child Rights and Online Gaming: Opportunities and Challenges for Children and the Industry*. https://www.unicef-irc.org/files/upload/documents/UNICEF_CRBDigitalWorldSeriesOnline_Gaming.pdf (2019).

212. Hobbs, D., Henschke, M., Wilkinson, B. & Reynolds, K. Game on! Accessible gaming for children with disabilities. in (2012).

213. Floreani, E. D. *et al.* Unlocking Independence: Exploring Movement with Brain-Computer Interface for Children with Severe Physical Disabilities. *Annu Int Conf IEEE Eng Med Biol Soc* **2021**, 5864–5867 (2021).

214. Charmaz, K. *Constructing Grounded Theory*. (SAGE, 2014).

215. Cunningham, R. Children's Psychosocial Impact of Assistive Devices (PIADS). (2018).

216. Oliver, M. *Understanding disability: From theory to practice*. x, 192 (St Martin's Press, 1996). doi:10.1007/978-1-349-24269-6.

217. Blake, L., Bray, L. & Carter, B. “It’s a lifeline”: Generating a sense of social connectedness through befriending parents of disabled children or children with additional need. *Patient Education and Counseling* **102**, 2279–2285 (2019).

218. Foley, K.-R. *et al.* To Feel Belonged: The Voices of Children and Youth with Disabilities on the Meaning of Wellbeing. *Child Ind Res* **5**, 375–391 (2012).

219. Ferguson, B. R., Gillis, J. M. & Sevlever, M. A Brief Group Intervention Using Video Games to Teach Sportsmanship Skills to Children With Autism Spectrum Disorders. *Child & Family Behavior Therapy* **35**, 293–306 (2013).

220. World Health Organization. *International classification of functioning, disability and health: children and youth version: ICF-CY*. https://apps.who.int/iris/handle/10665/43737 (2007).

221. Palisano, R. J. *et al.* Participation-based therapy for children with physical disabilities. *Disability and Rehabilitation* **34**, 1041–1052 (2012).

222. Janssen, J. *et al.* Gamification in Physical Therapy: More Than Using Games. *Pediatric Physical Therapy* **29**, 95 (2017).